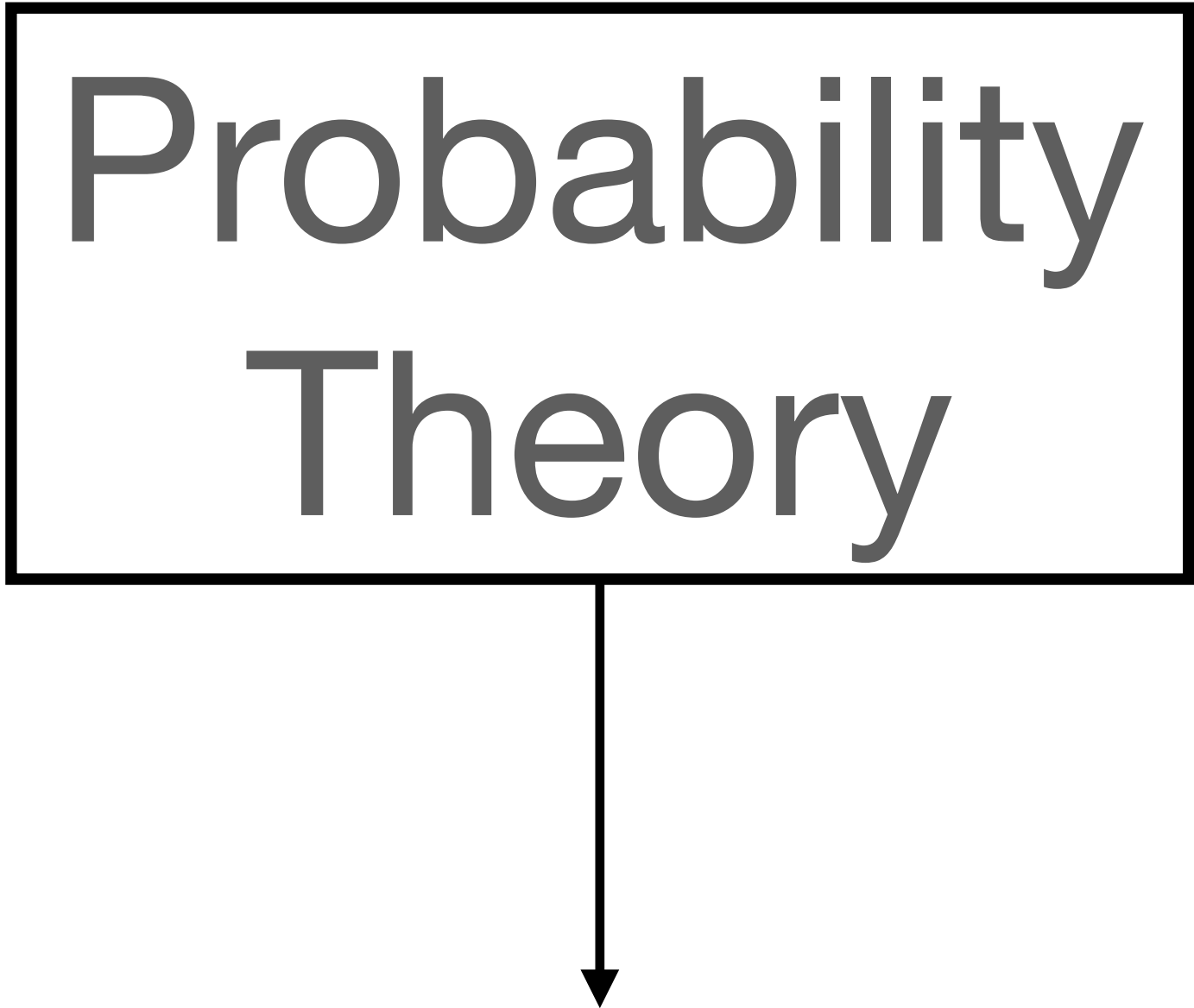


Linear Models



The two sides of statistics

Probability
Theory

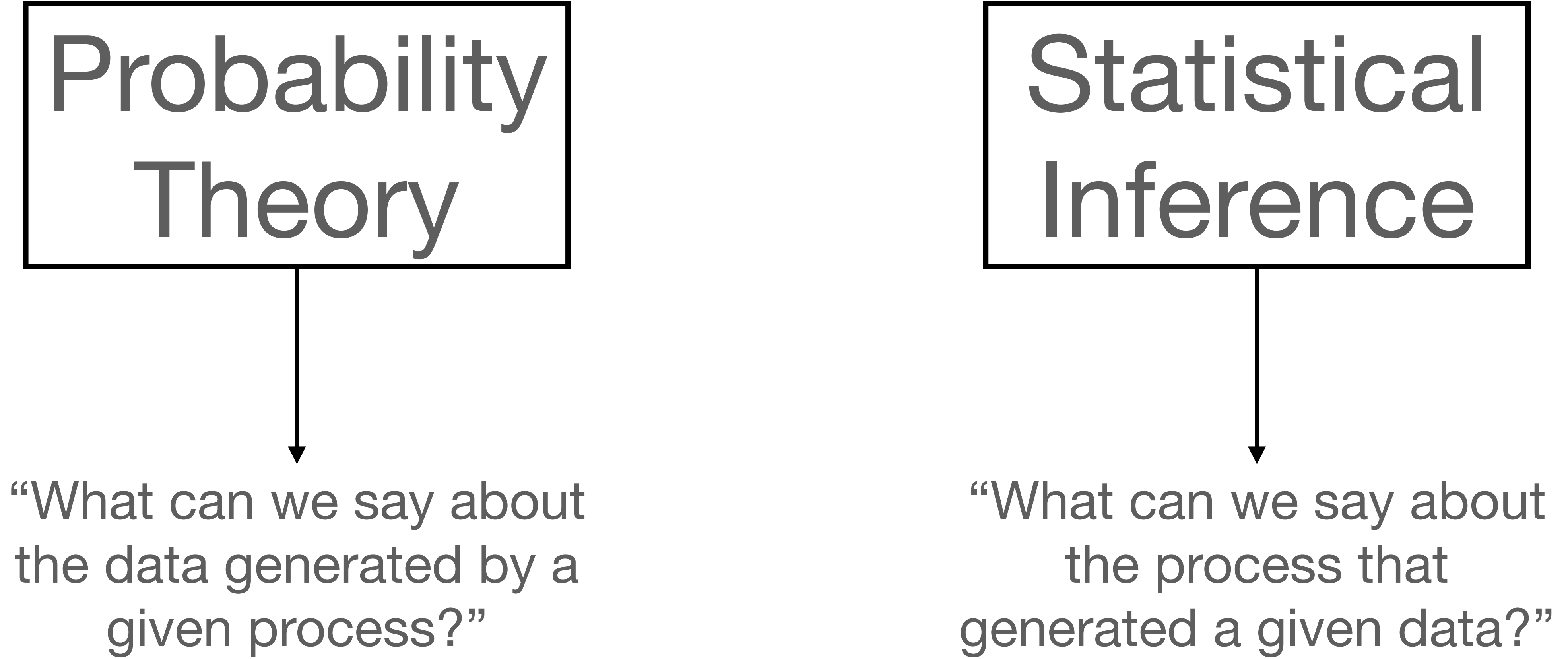


```
graph TD; A[Probability Theory] --> B["What can we say about the data generated by a given process?"]
```

“What can we say about
the data generated by a
given process?”

The two sides of statistics

Probability
Theory



```
graph TD; A[Probability Theory] --> B["What can we say about the data generated by a given process?"]; C[Statistical Inference] --> D["What can we say about the process that generated a given data?"];
```

“What can we say about
the data generated by a
given process?”

Statistical
Inference

“What can we say about
the process that
generated a given data?”

The two sides of statistics

Probability
Theory



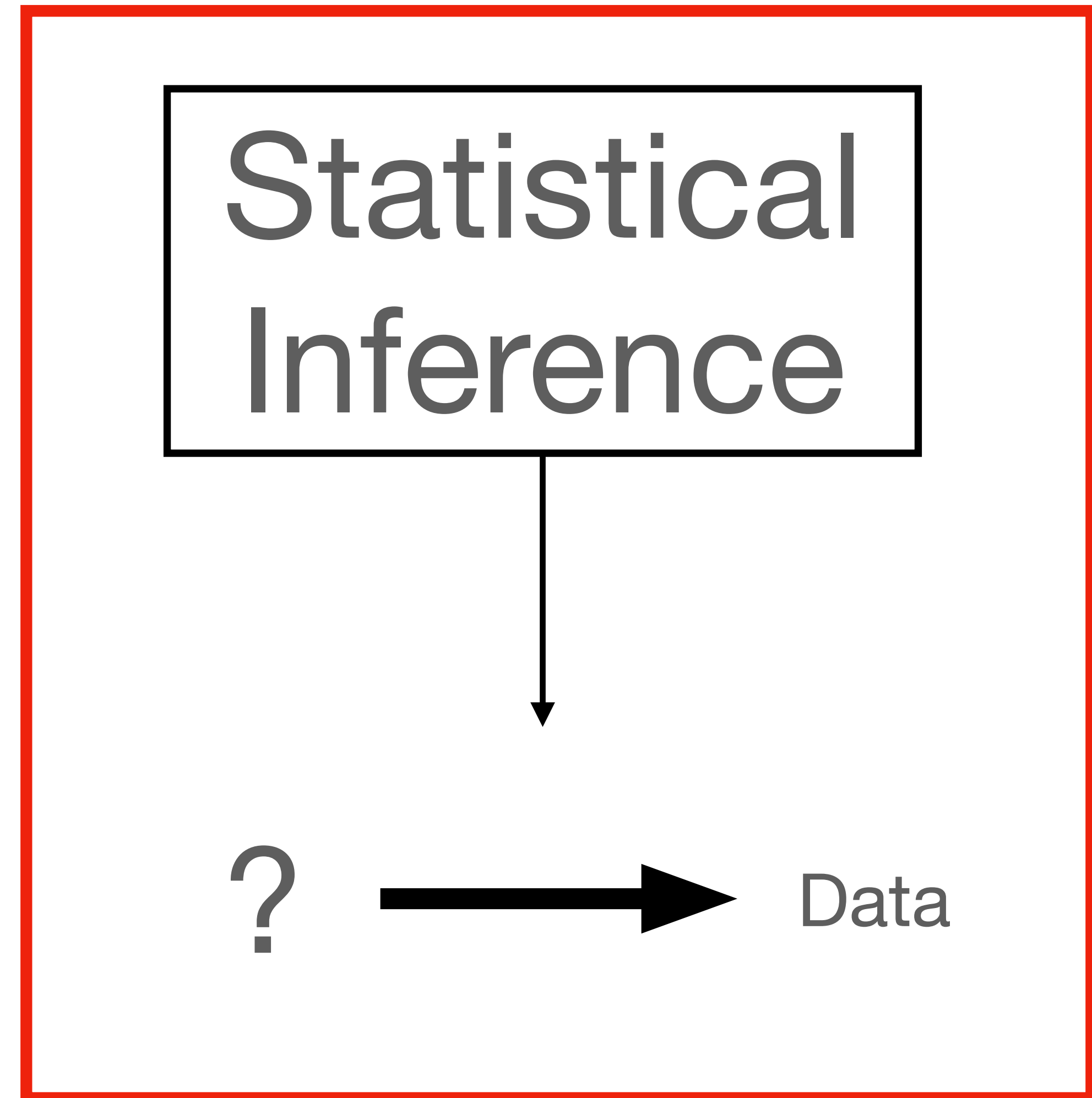
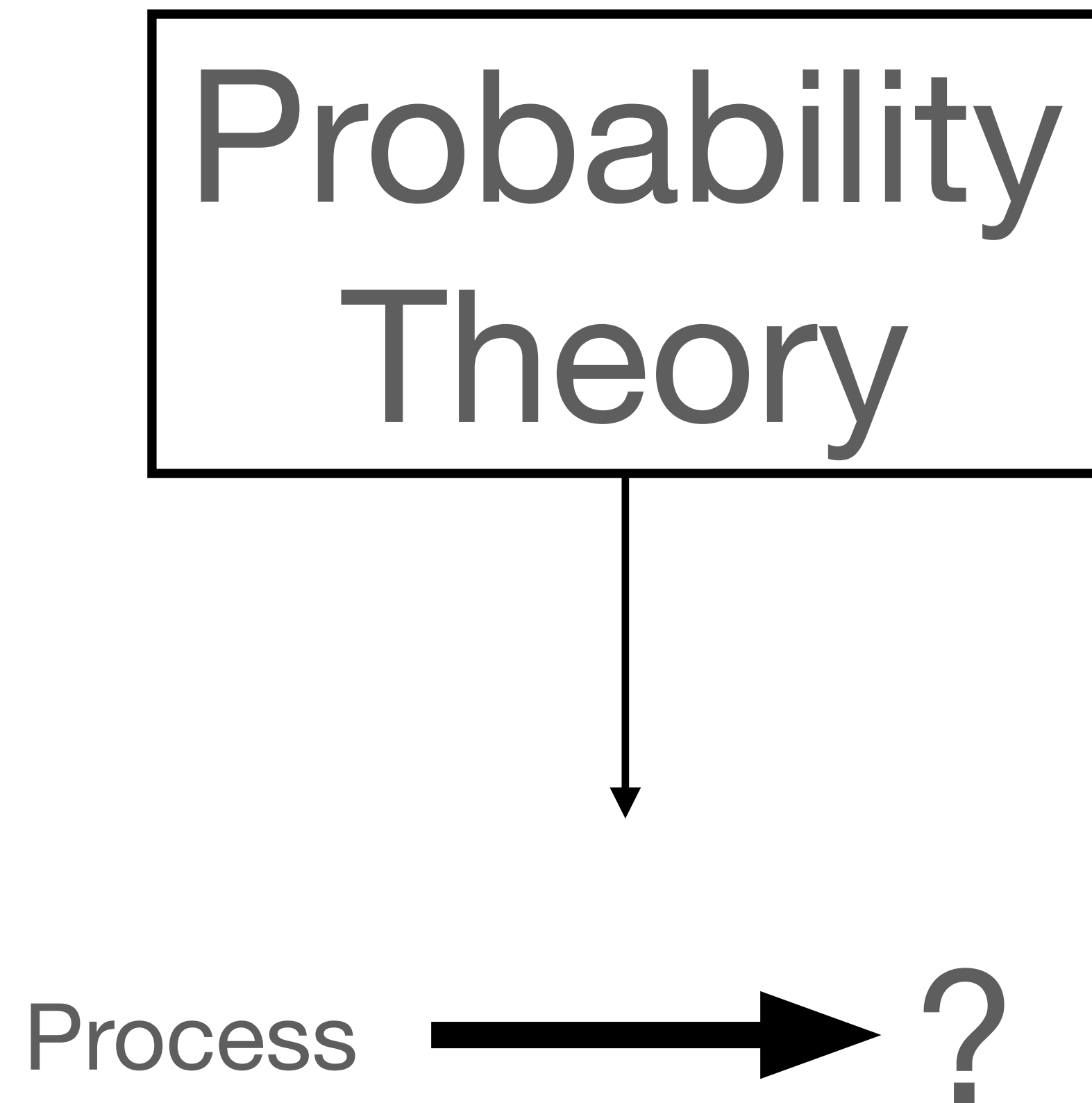
Process → ?

Statistical
Inference



? → Data

The two sides of statistics



Statistical inference: two “flavors”

Supervised learning: **today**

Predictor variables

Response variables



Statistical inference: two “flavors”

Supervised learning: **today**

Predictor variables

Response variables



Unsupervised learning: **tomorrow**

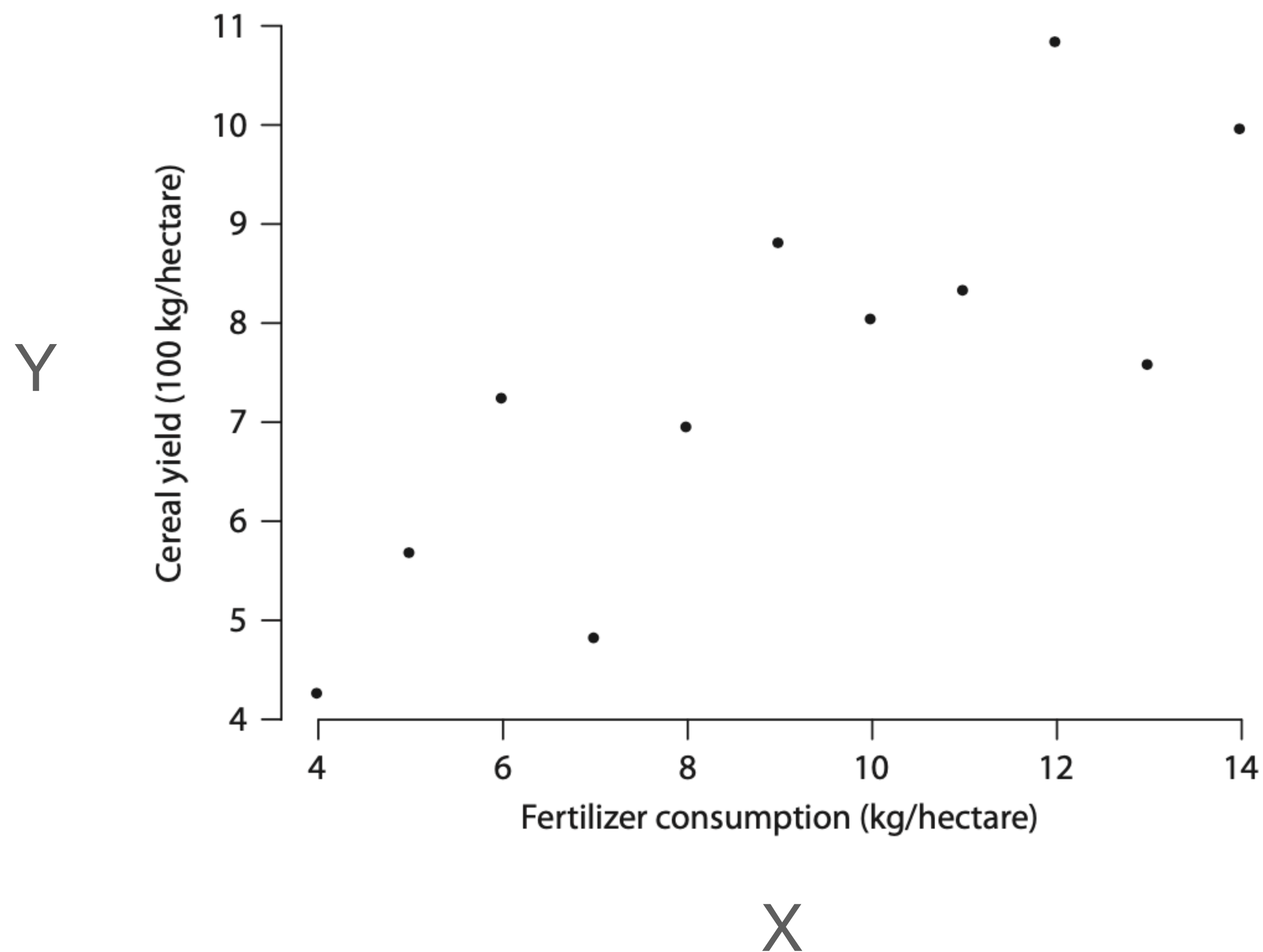
Variables



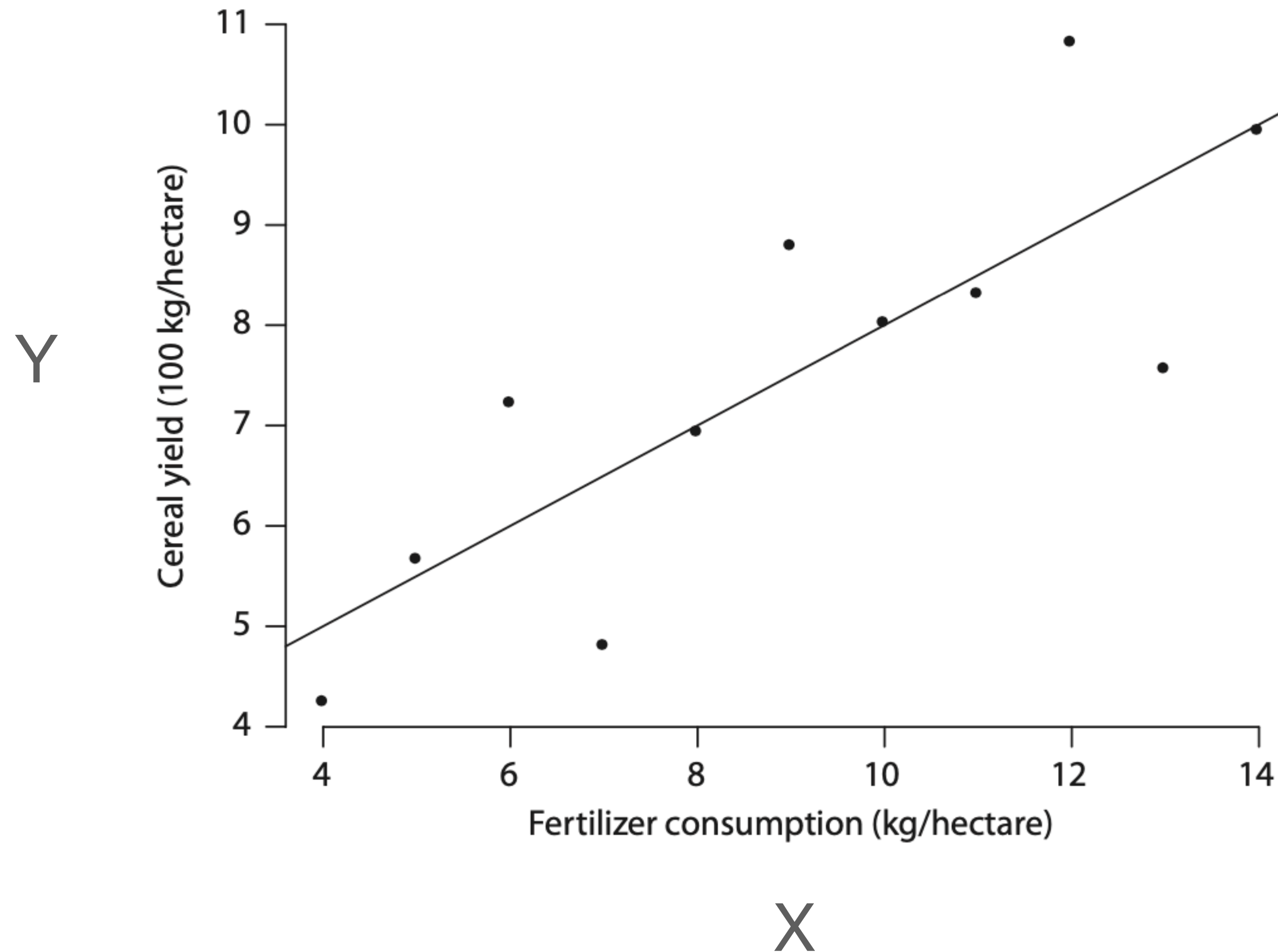
Linear Regression

- Simplest model in supervised learning
- Jumping-off point for more complex models
- Many statistical models are extensions or generalizations of the linear model

Linear Regression



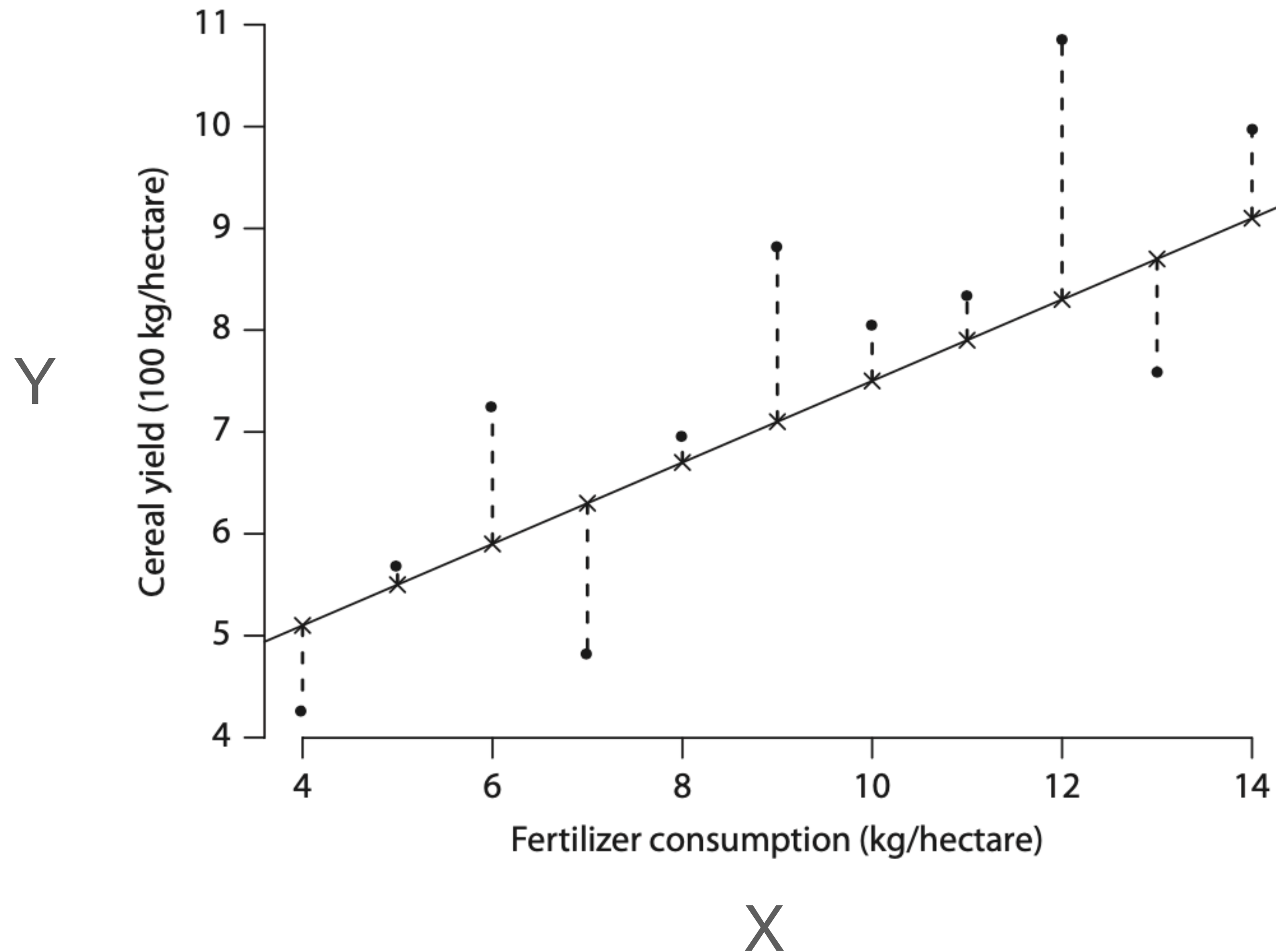
Linear Regression: “a line of best fit”



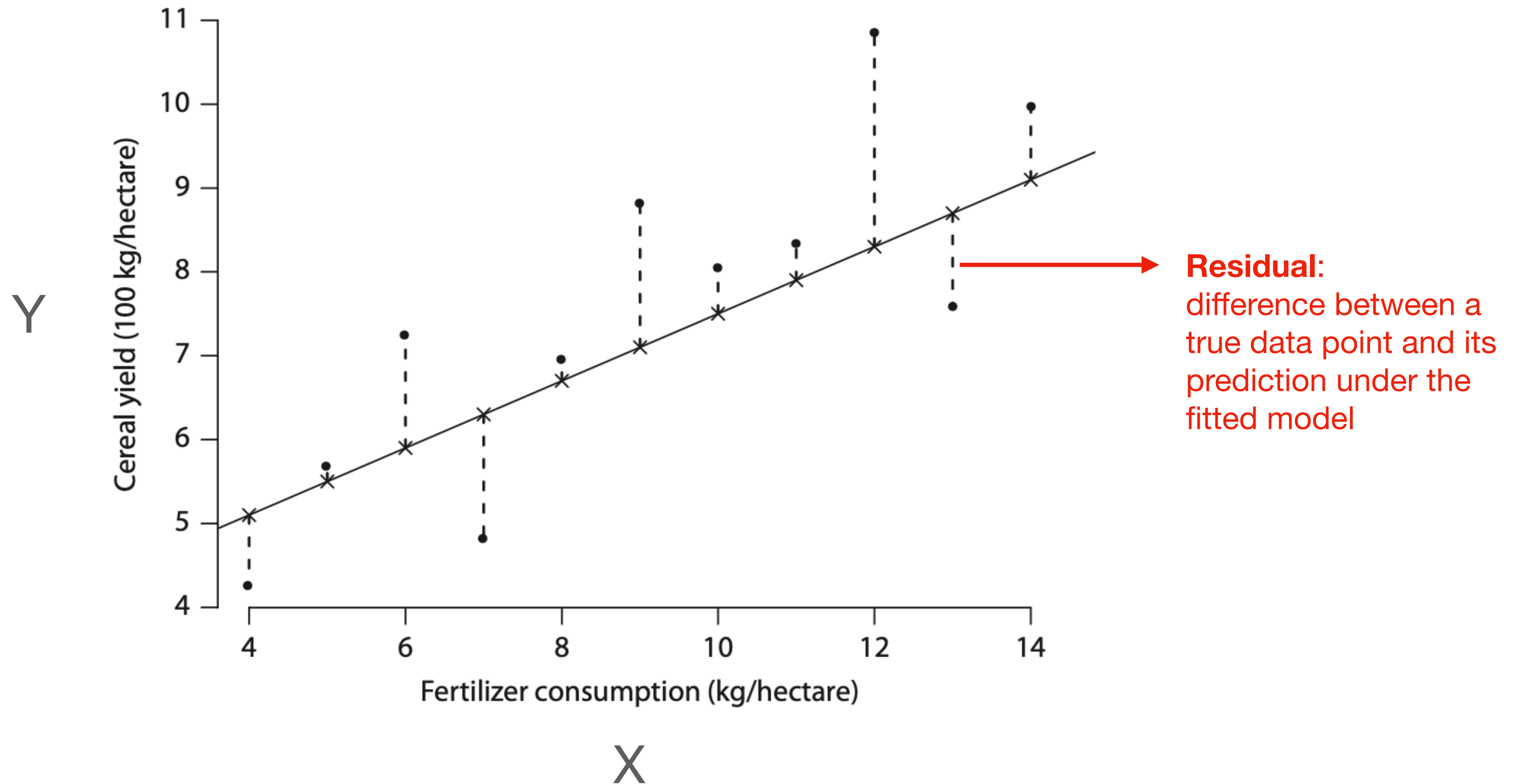
Linear Regression: types of questions

- Is there a relation between variable X and variable Y ?
- How strong is the relation between variable X and variable Y ?
- How well can we predict variable Y from the values of variable X ?
- Is the relationship between variable X and Y linear?
- Which variables contribute to variable Y ?

Linear Regression: “a line of best fit”



Linear Regression: “a line of best fit”



Simple linear regression

- **1** predictor variable (x)
- **1** response variable (y)

Simple linear regression

- **1** predictor variable (x)
- **1** response variable (y)

$$y = f(x)$$

Variable **y** is a **function** of **x**

Simple linear regression

- **1** predictor variable (x)
- **1** response variable (y)

$$y = f(x)$$

Variable **y** is a **function** of **x**

$$y \approx \beta_0 + \beta_1 x$$

Variable **y** is a **linear**
function of variable **x**

Simple linear regression

- **1** predictor variable (x)
- **1** response variable (y)

$$y = f(x)$$

Variable **y** is a **function** of **x**

$$y \approx \beta_0 + \beta_1 x$$

Variable **y** is a **linear**
function of variable **x**

$$y = \beta_0 + \beta_1 x + \epsilon$$

Variable **y** is a linear function
of **x**, plus some **noise**

Simple linear regression

$$y = \beta_0 + \beta_1 x + \epsilon$$

Variable **y** is a linear function
of **x**, plus some **noise**

Simple linear regression

$$y = \beta_0 + \beta_1 x + \epsilon$$

Variable **y** is a linear function of **x**, plus some **noise**

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Each value y_i has a specific predictor value x_i and noise value ϵ_i

Simple linear regression

$$y = \beta_0 + \beta_1 x + \epsilon$$

Variable **y** is a linear function
of **x**, plus some **noise**

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Each value y_i has a specific
predictor value x_i and noise
value ϵ_i

Individual values are represented with subscripts

Simple linear regression

Vectors of values (variables) are represented in bold

$$\mathbf{y} = \beta_0 + \beta_1 \mathbf{x} + \epsilon$$

Variable **y** is a linear function of **x**, plus some **noise**

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Each value y_i has a specific predictor value x_i and noise value ϵ_i

Individual values are represented with subscripts

Parameters vs. Estimates

$$y = \beta_0 + \beta_1 x + \epsilon$$

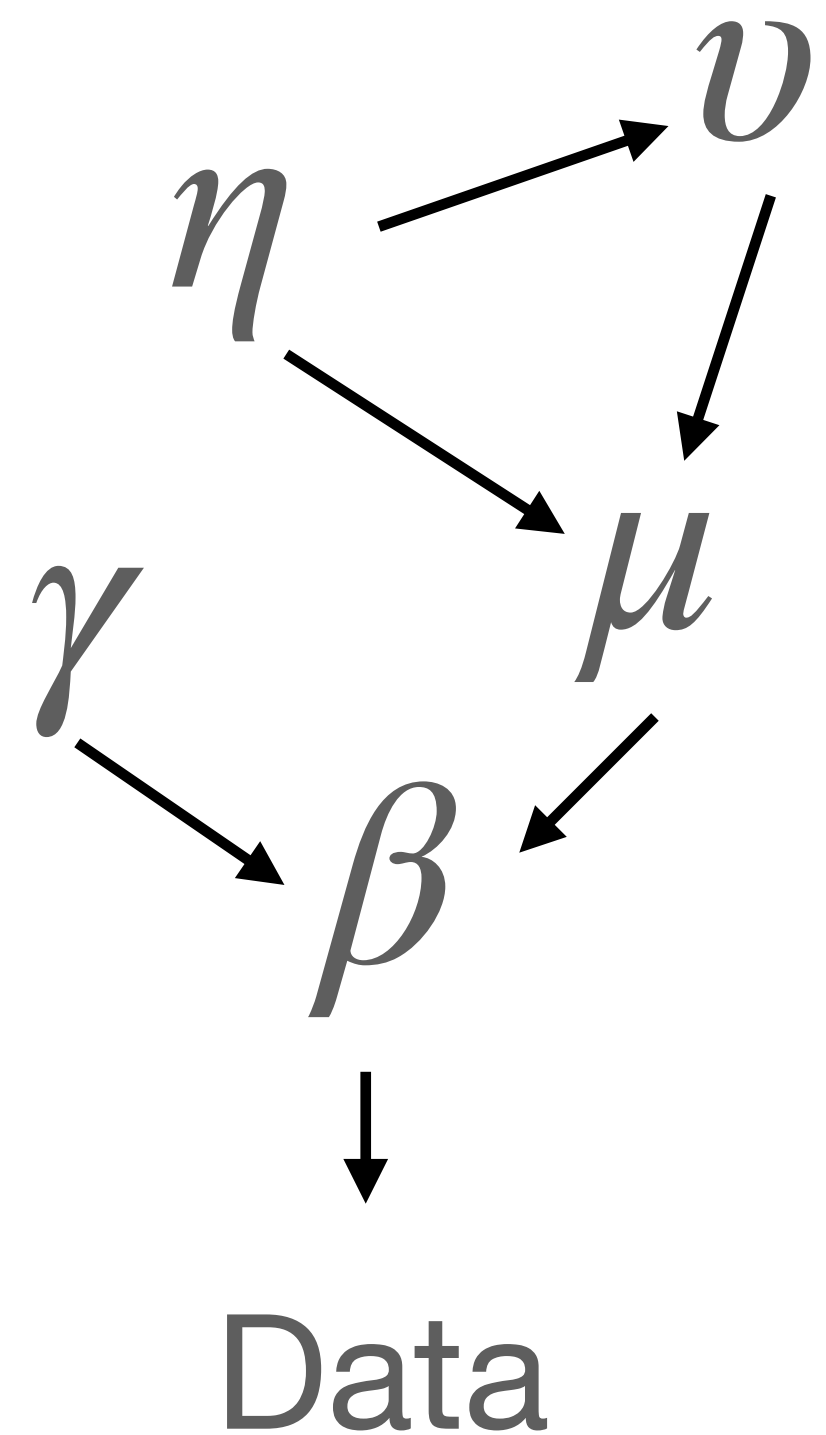
β_0 and β_1 are unknown parameters in our model: **we do not know their value**

$$y = \hat{\beta}_0 + \hat{\beta}_1 x + \epsilon$$

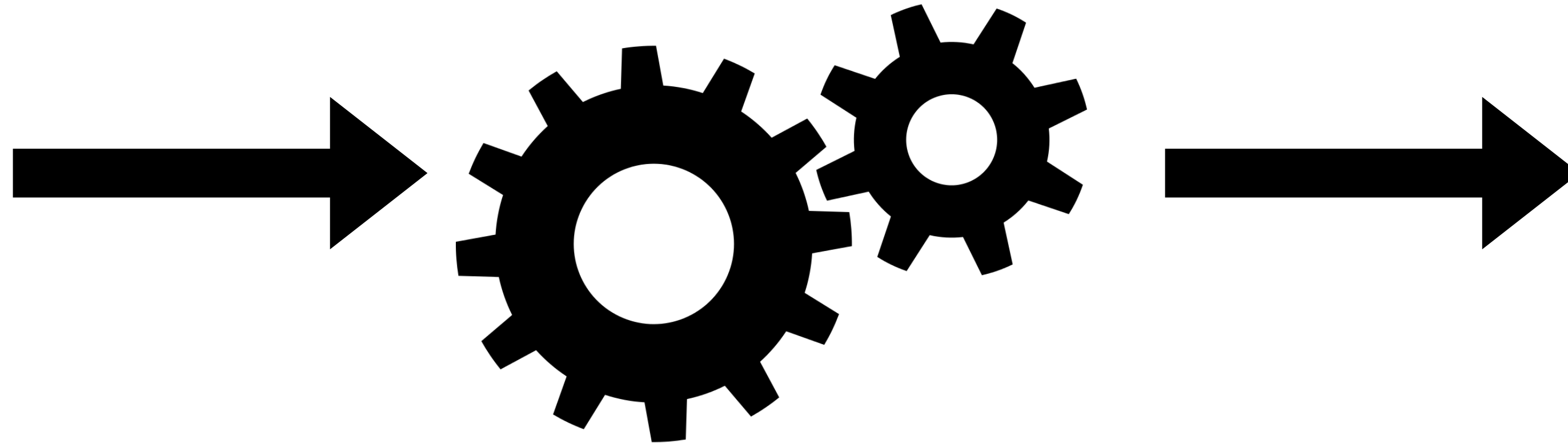
$\hat{\beta}_0$ and $\hat{\beta}_1$ are our **best estimates** of the above parameters

Model vs. Inference Method

Model



Inference method



Parameter estimates

$$\hat{\alpha} = 0.56$$

$$\hat{\beta} = 3.2$$

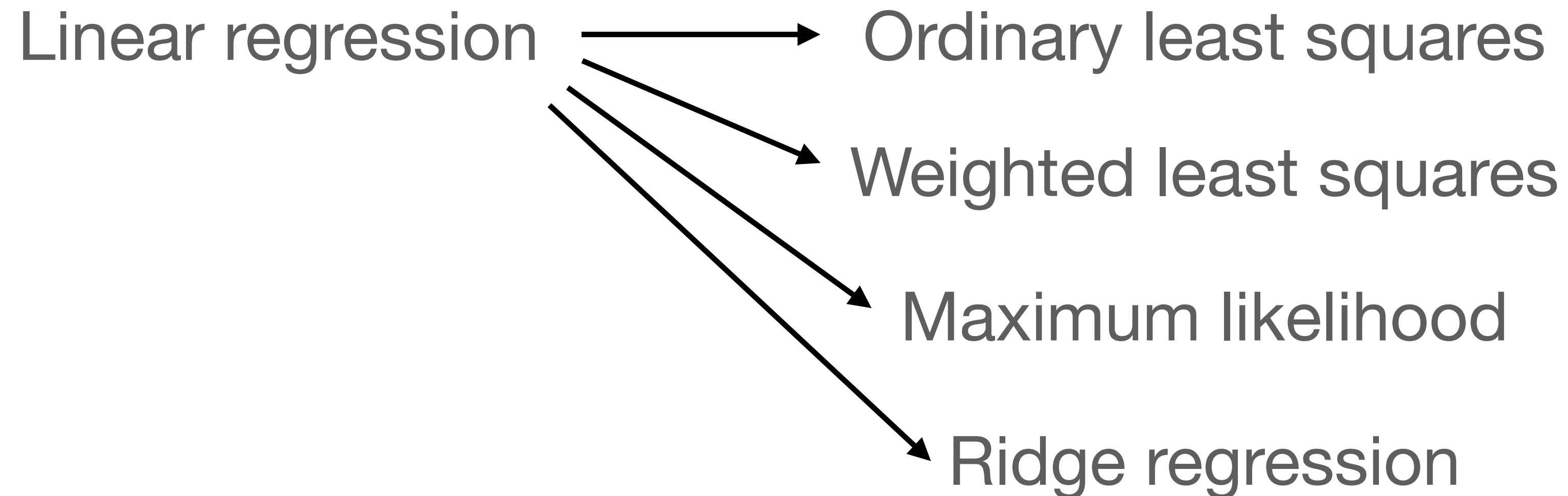
$$\hat{\mu} = -2$$

...

Model vs. Inference Method

Model

Inference method

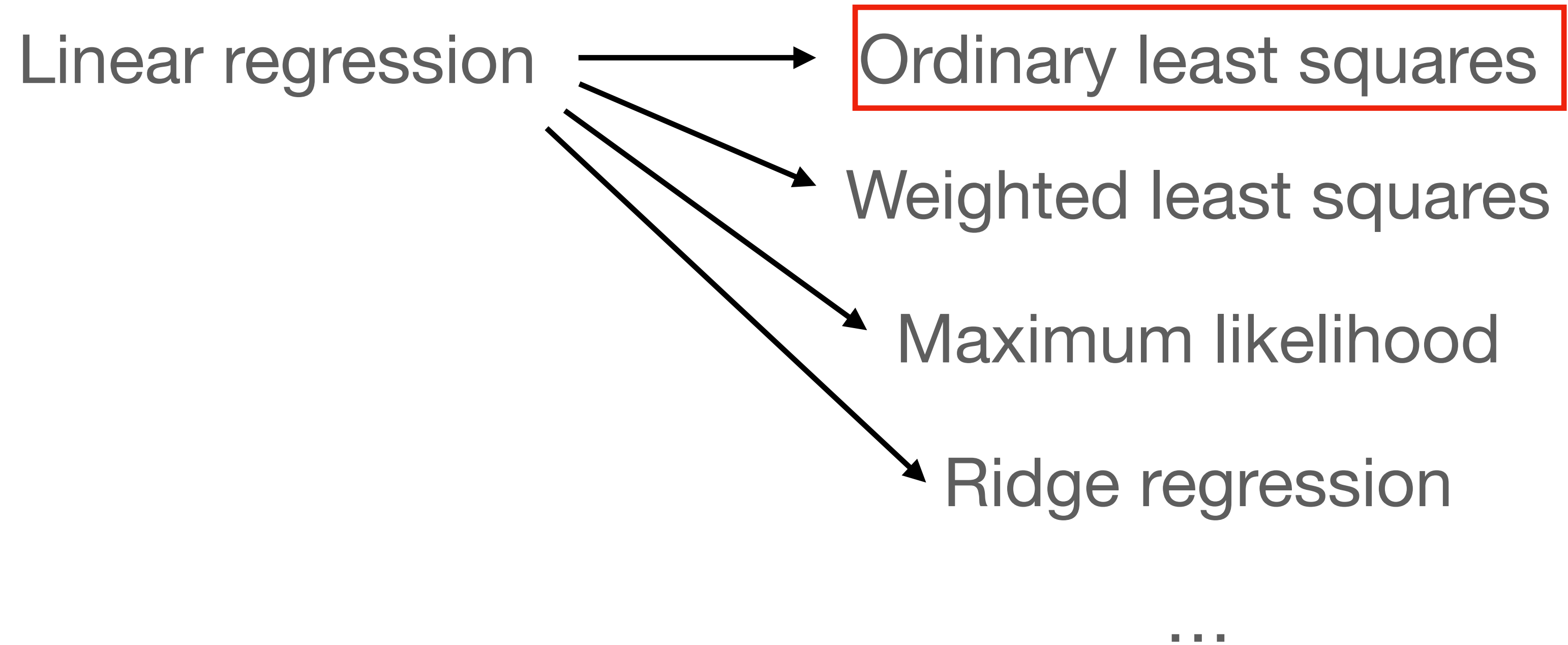


...

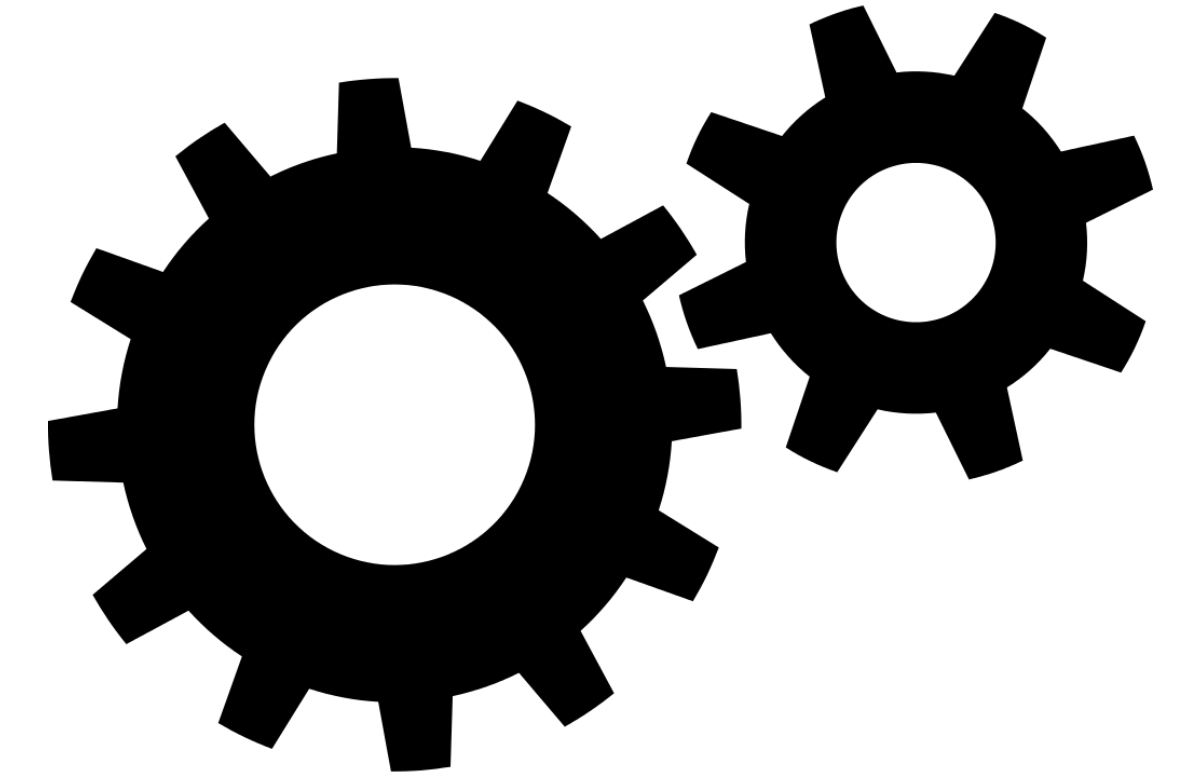
Model vs. Inference Method

Model

Inference method

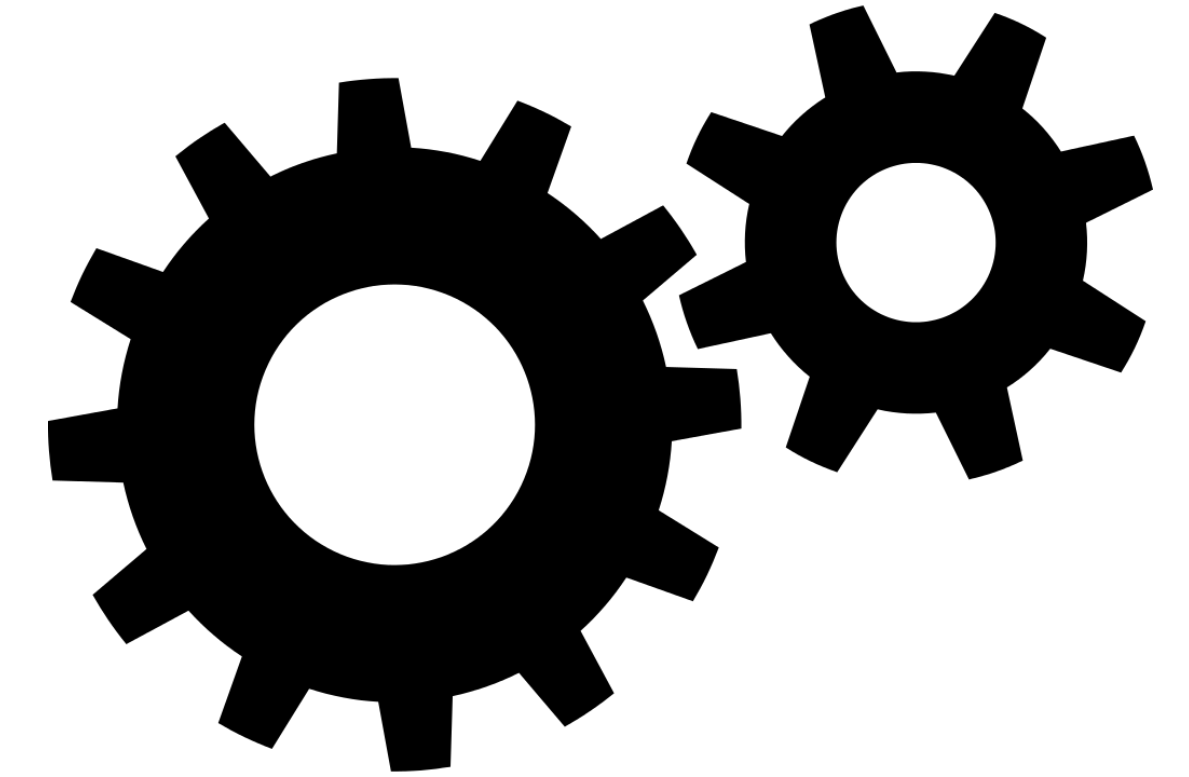


Ordinary least squares



“Find estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ of the parameters β_0 and β_1 by minimizing the **Sum of Squared Residuals**”

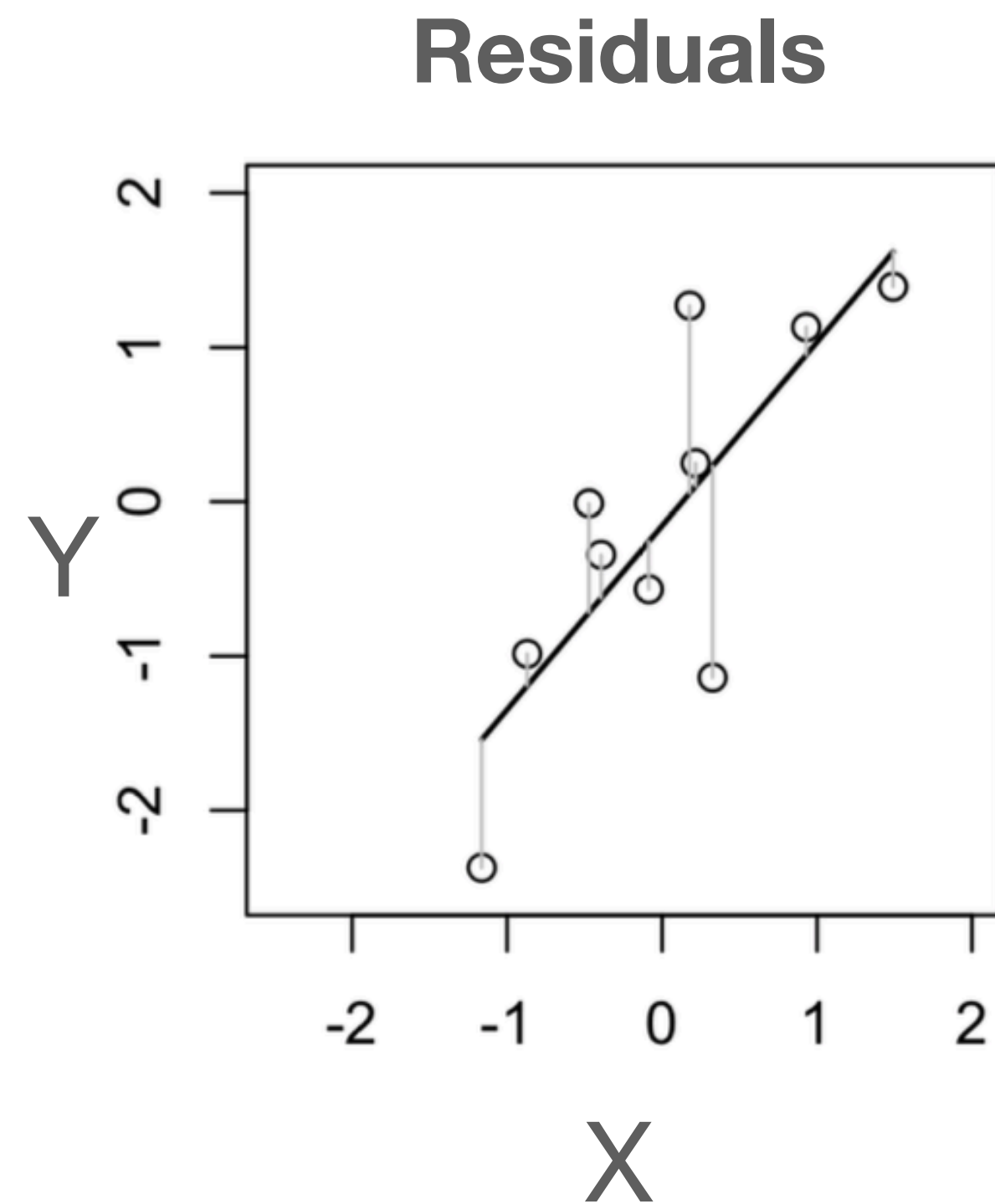
Ordinary least squares



“Find estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ of the parameters β_0 and β_1 by minimizing the **Sum of Squared Residuals**”



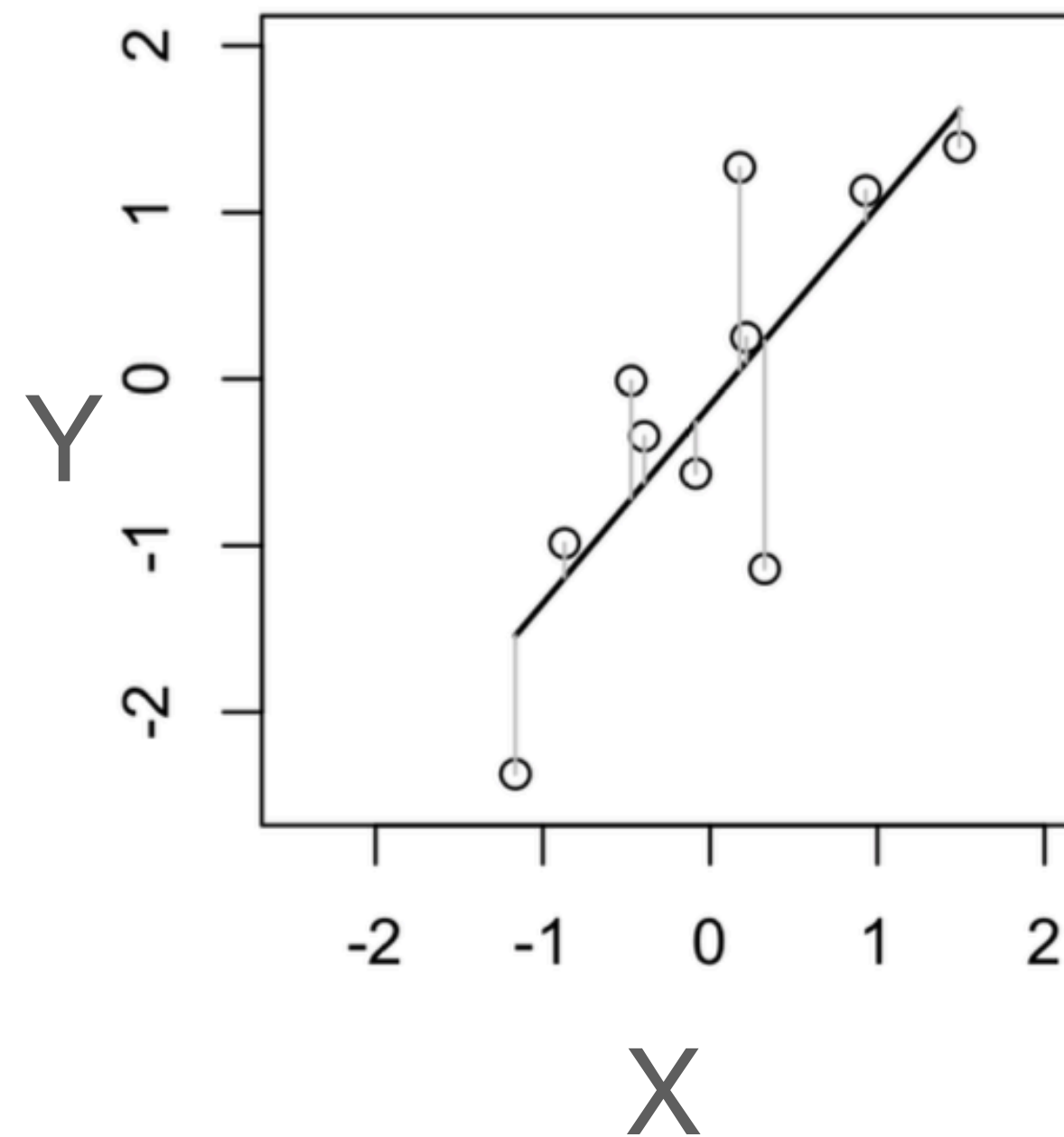
Ordinary least squares: what does this mean?



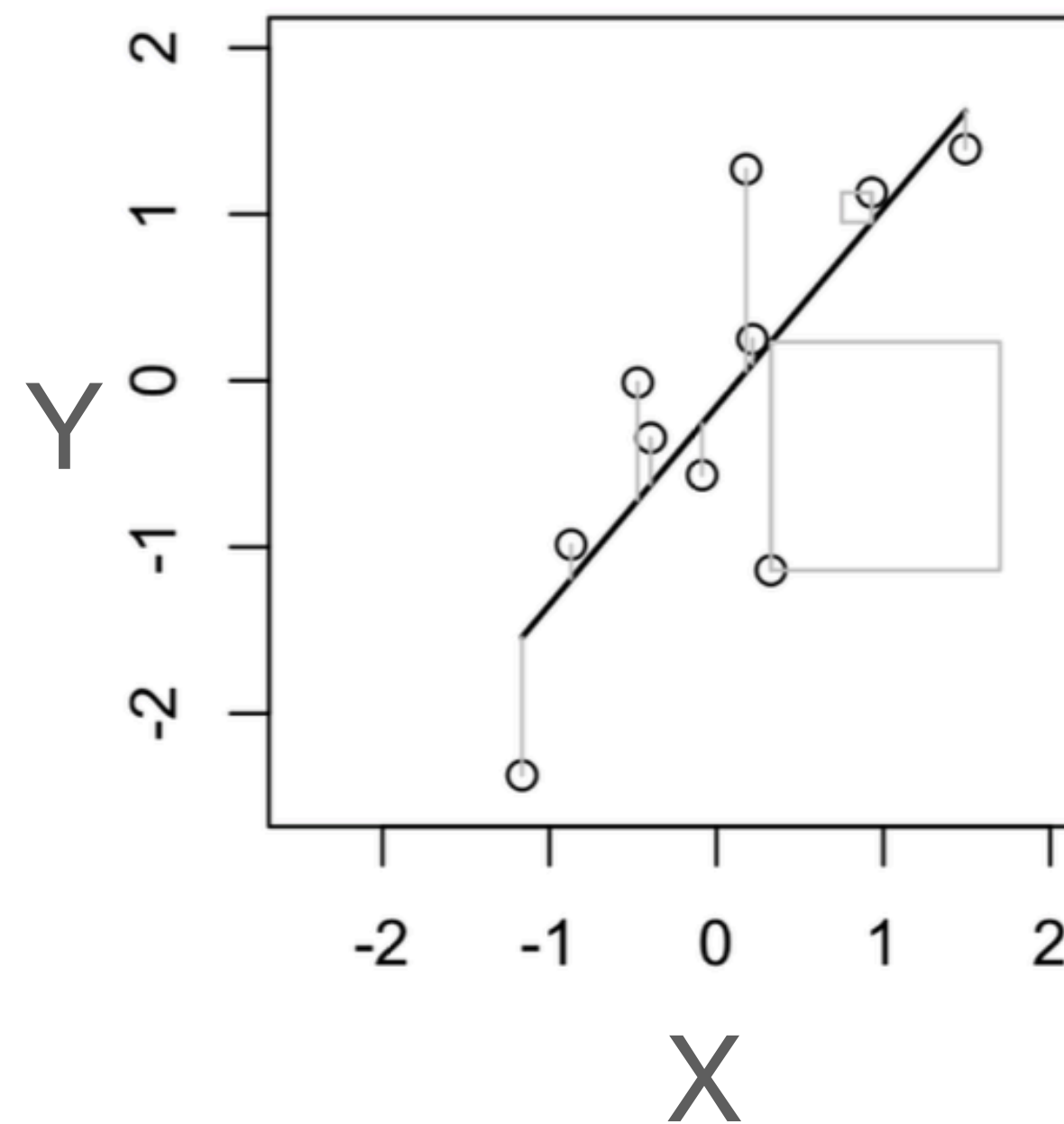
$$\text{res} = y_i - \hat{y}$$

Ordinary least squares: what does this mean?

Residuals



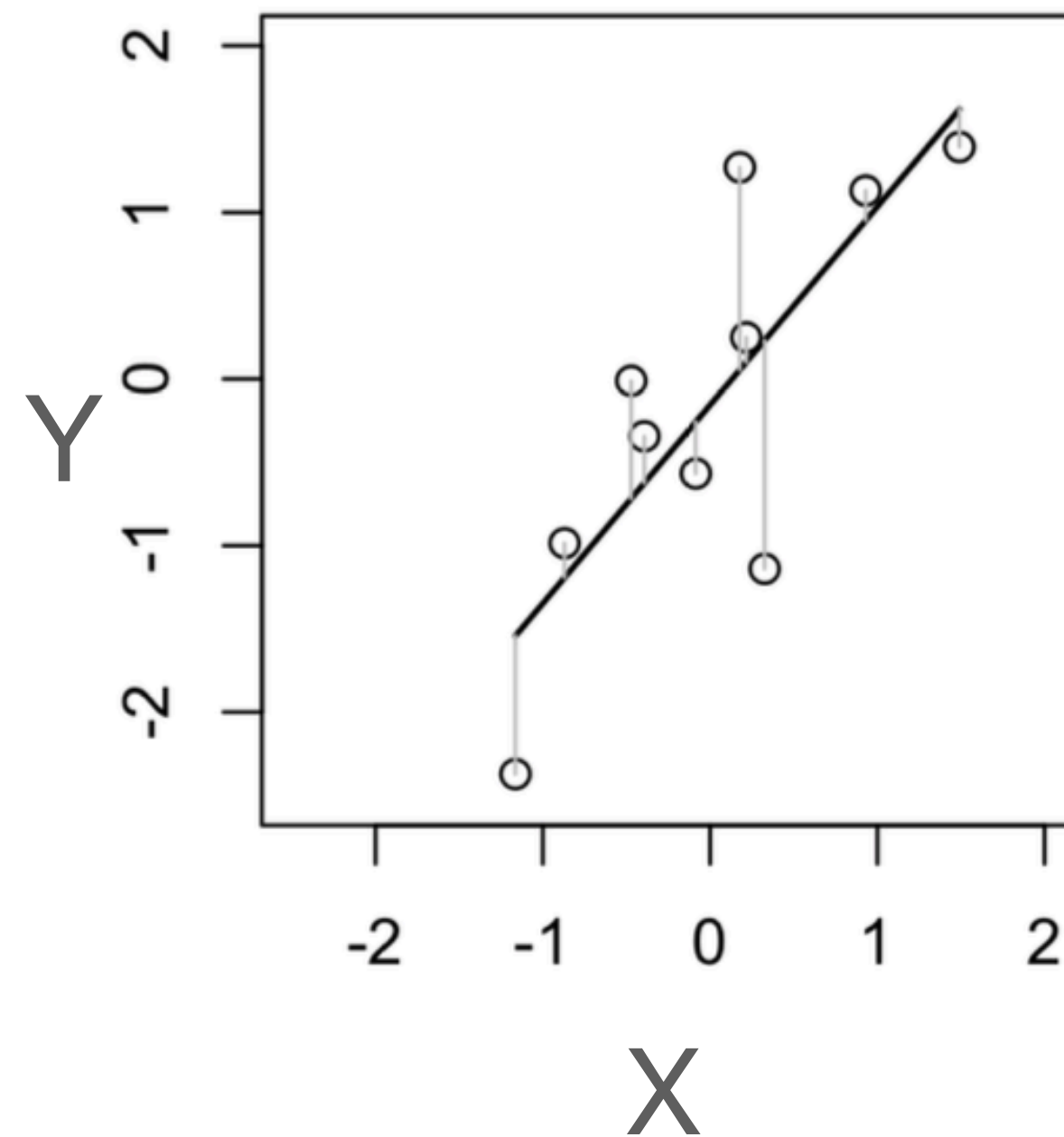
$$\text{res} = y_i - \hat{y}$$



$$(y_i - \hat{y})^2$$

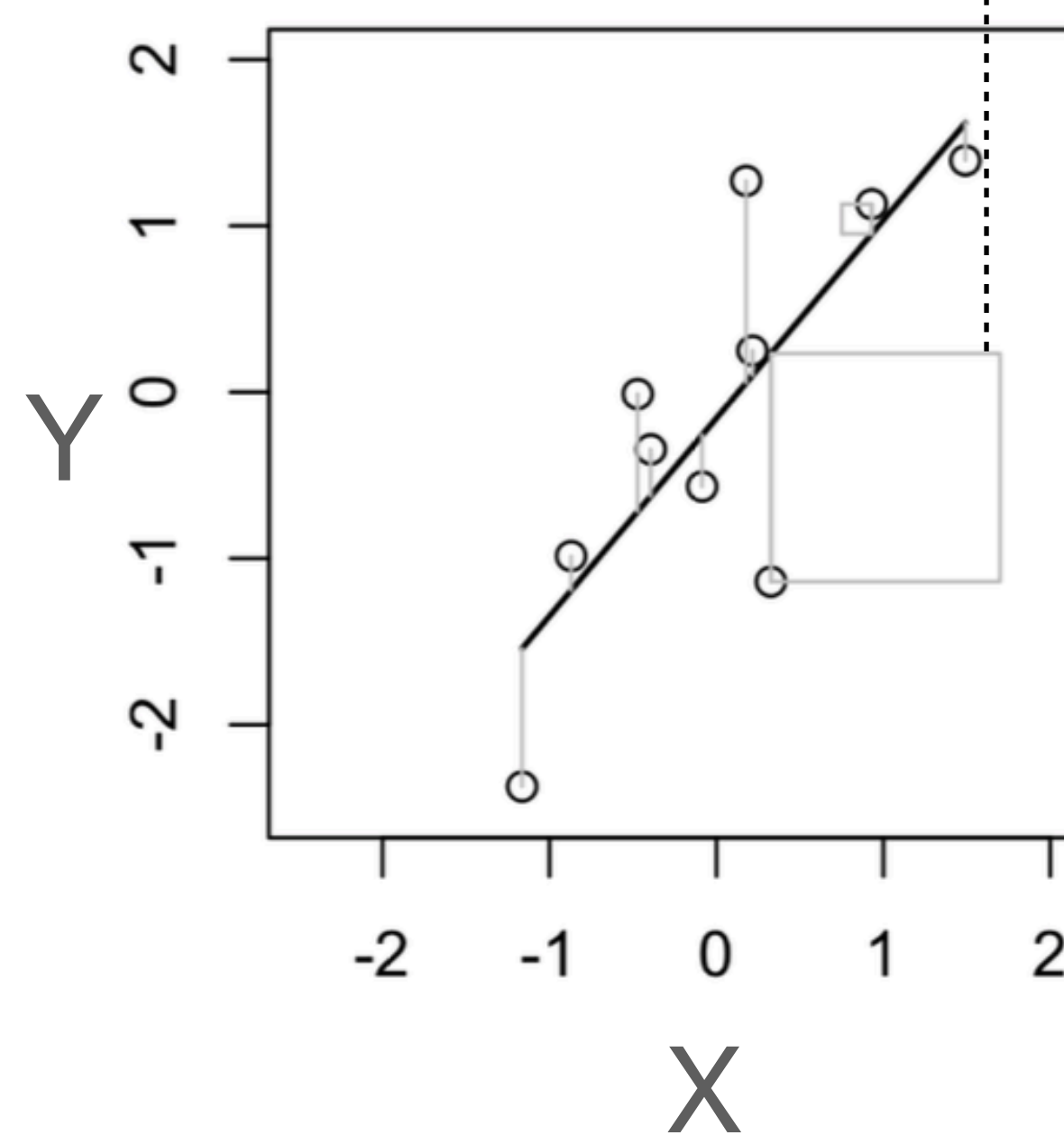
Ordinary least squares: what does this mean?

Residuals



$$\text{res} = y_i - \hat{y}$$

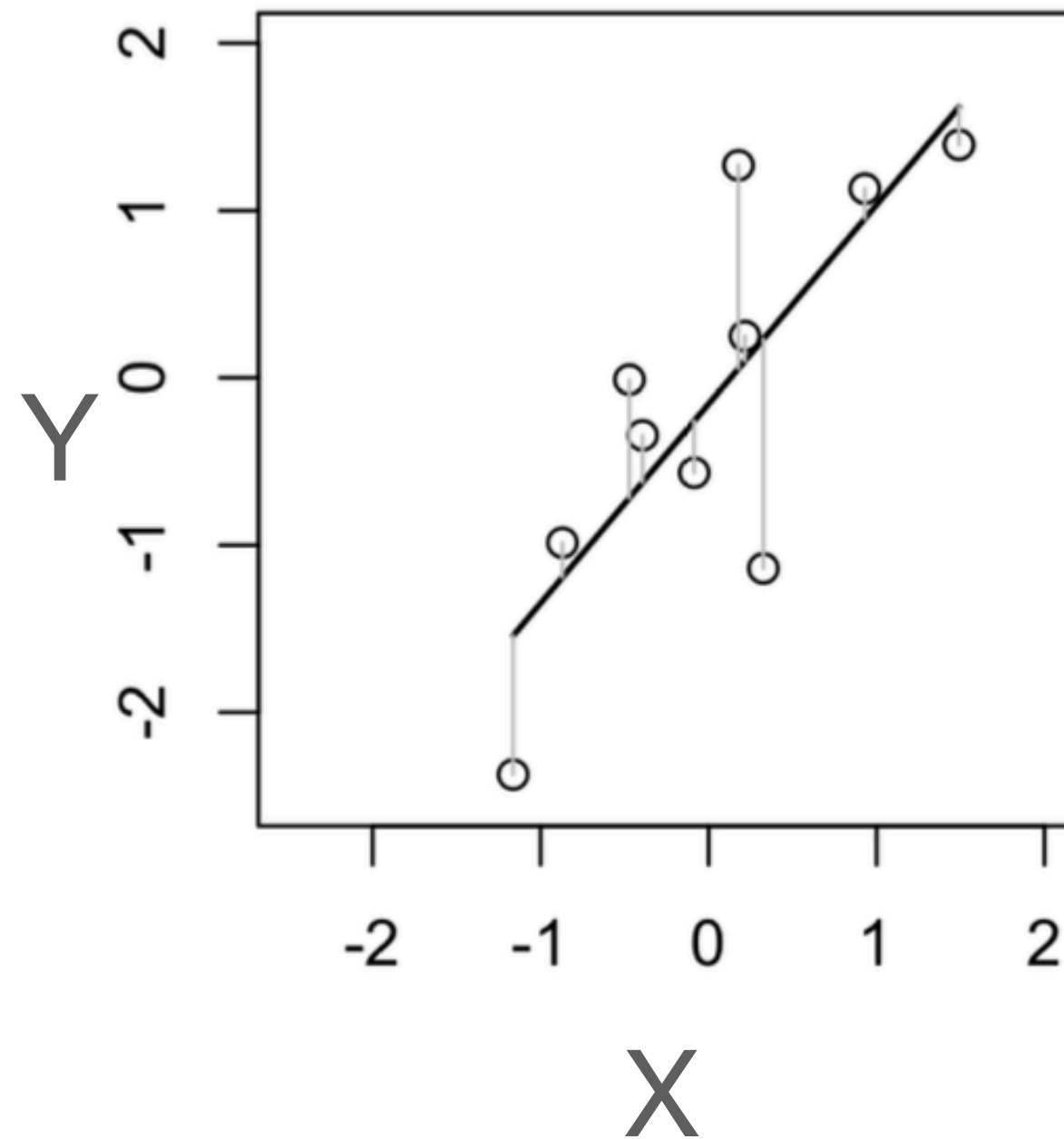
A squared residual



$$(y_i - \hat{y})^2$$

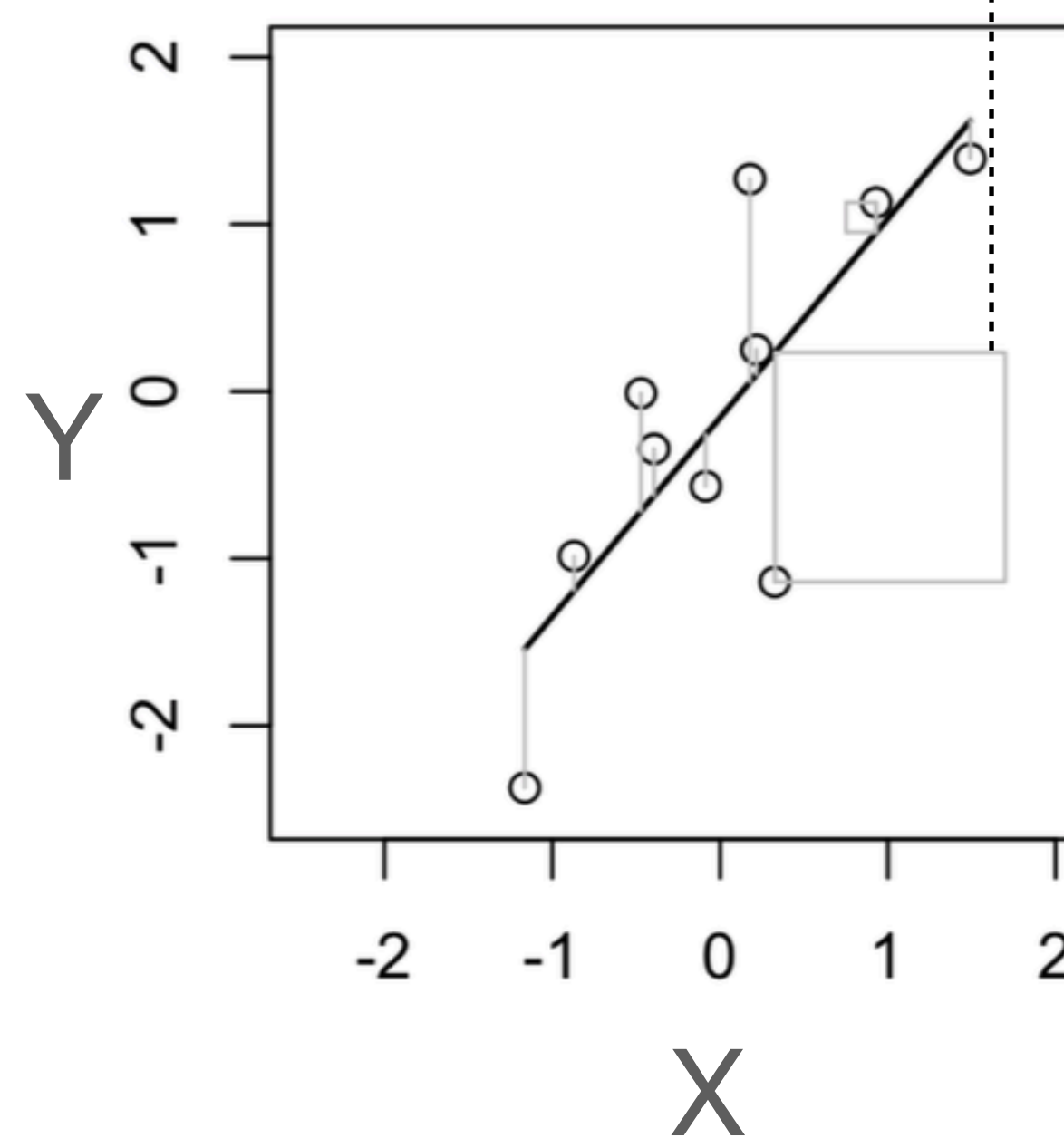
Ordinary least squares: what does this mean?

Residuals



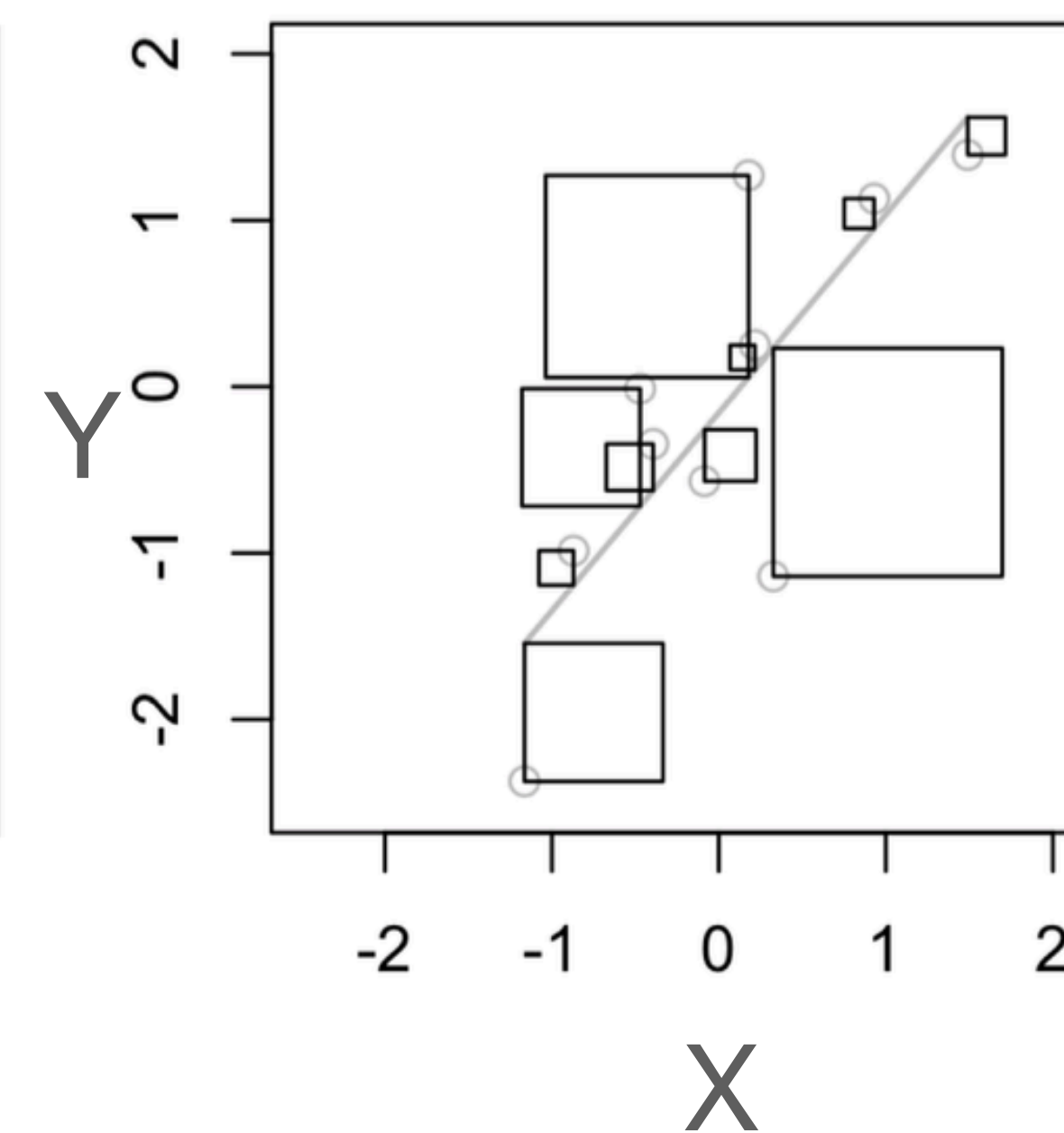
$$\text{res} = y_i - \hat{y}$$

A squared residual



$$(y_i - \hat{y})^2$$

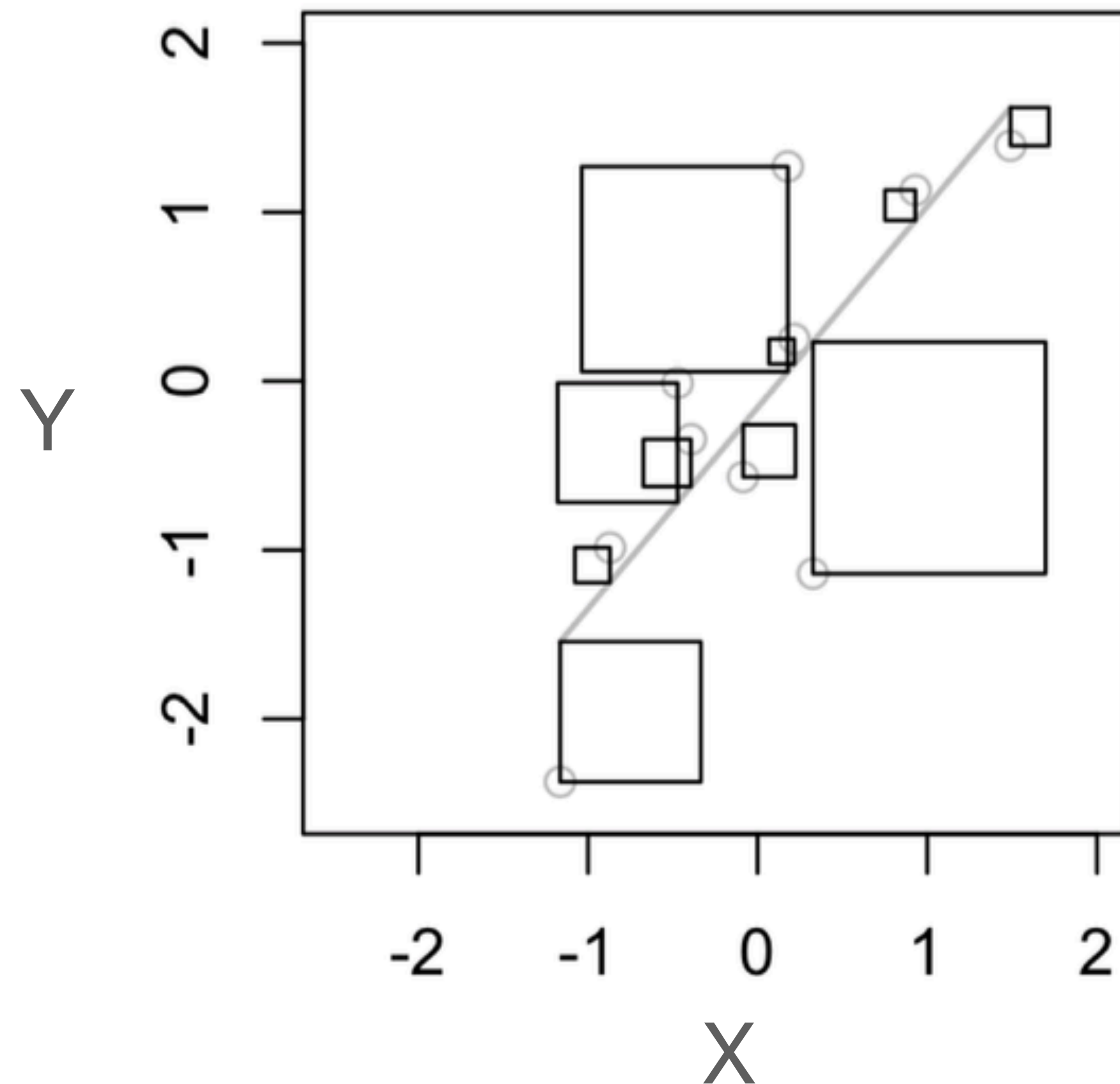
Sum of Squared Residuals



$$SS_{res} = \sum_i^n (y_i - \hat{y})^2$$

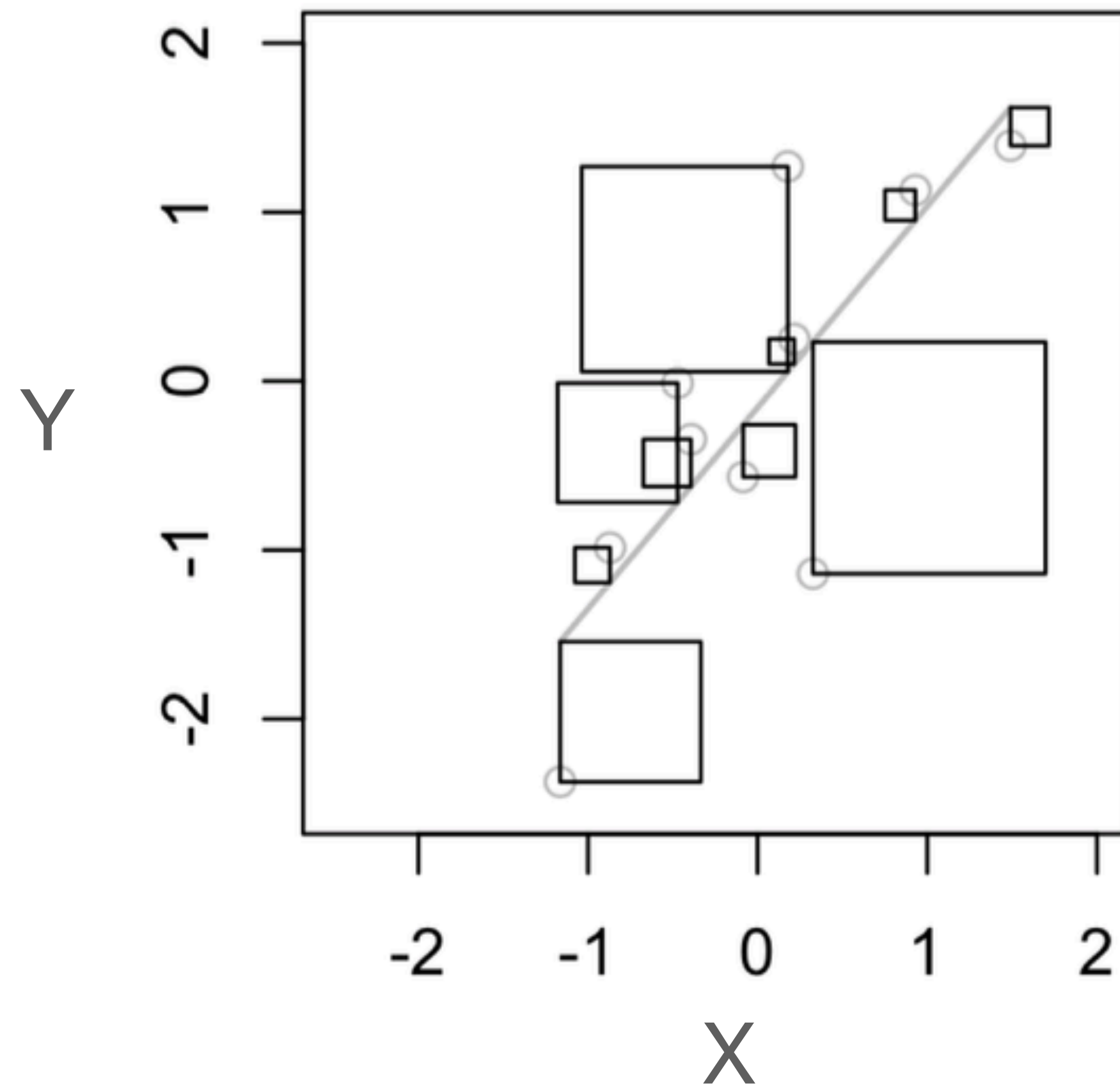
Ordinary least squares: what does this mean?

$$SS_{res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



Ordinary least squares: what does this mean?

Sum of Squared
Residuals



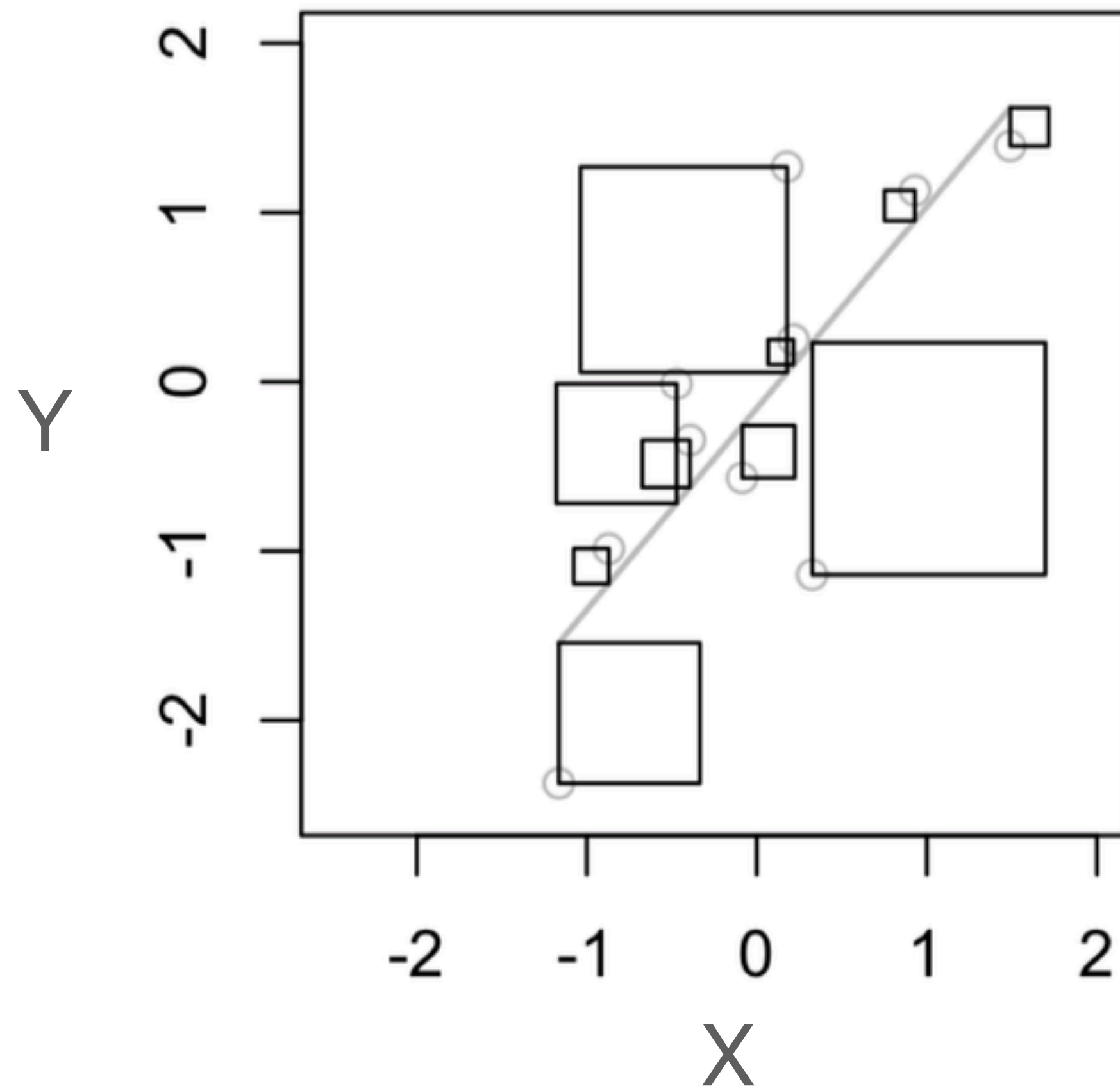
$$SS_{res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

but our estimate $\hat{y}_i$ is simply
a linear function of x_i :

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Ordinary least squares: what does this mean?

Sum of Squared
Residuals



$$SS_{res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

but our estimate $\hat{y}_i$ is simply
a linear function of x_i :

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

$$SS_{res} = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Ordinary least squares: what does this mean?

$$SS_{res} = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

The values of β_0 and β_1 that **minimize SSres** are:

Ordinary least squares: what does this mean?

$$SS_{res} = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

The values of β_0 and β_1 that **minimize SSres** are:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Ordinary least squares: what does this mean?

$$SS_{res} = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

The values of β_0 and β_1 that **minimize SSres** are:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Average of y

Average of x

Ordinary least squares: what does this mean?

$$SS_{res} = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

The values of β_0 and β_1 that **minimize SSres** are:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \qquad \hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n [(x_i - \bar{x})^2]}$$

Average of y Average of x

Ordinary least squares: what does this mean?

$$SS_{res} = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

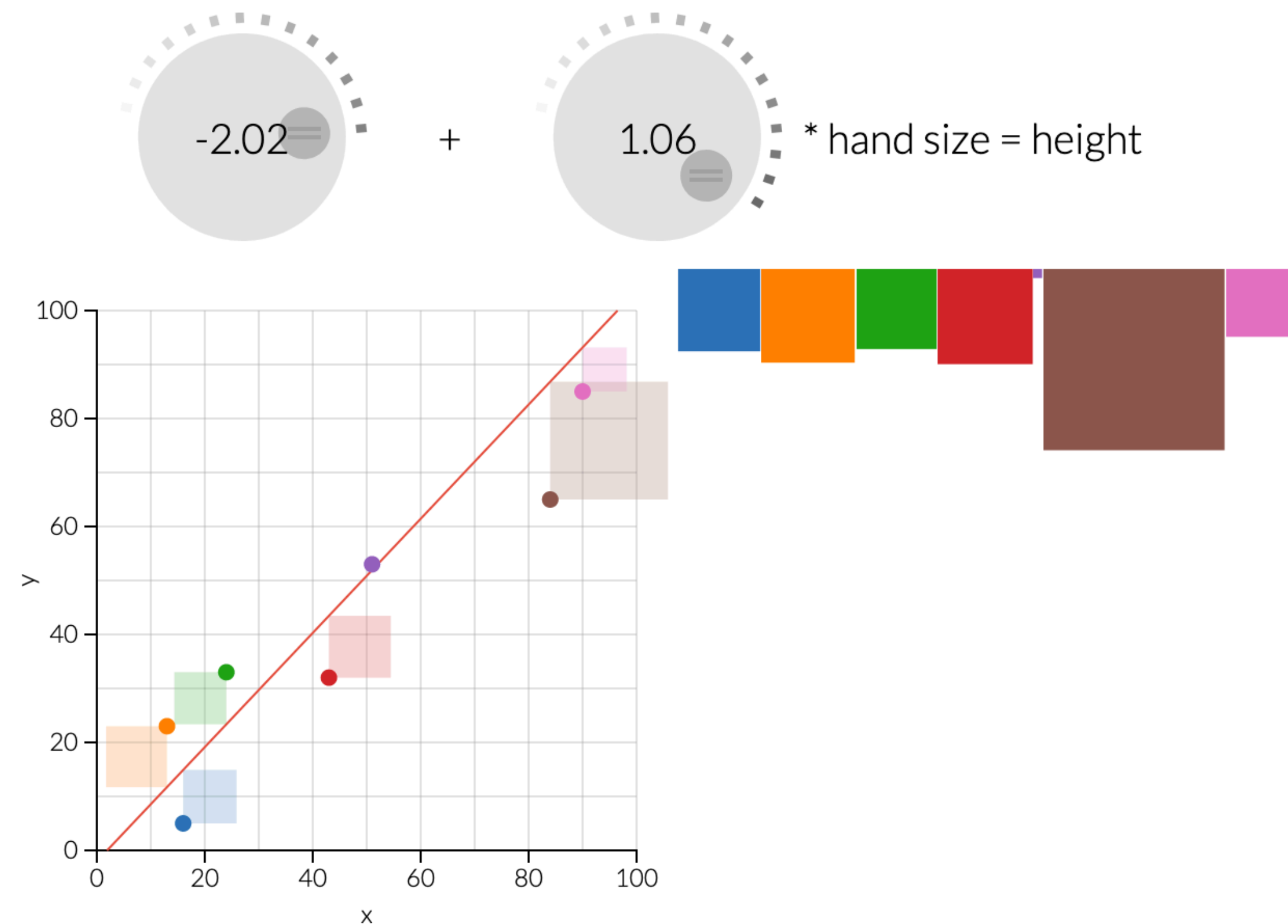
The values of β_0 and β_1 that **minimize SSres** are:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \qquad \hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n [(x_i - \bar{x})^2]}$$

Average of y Average of x

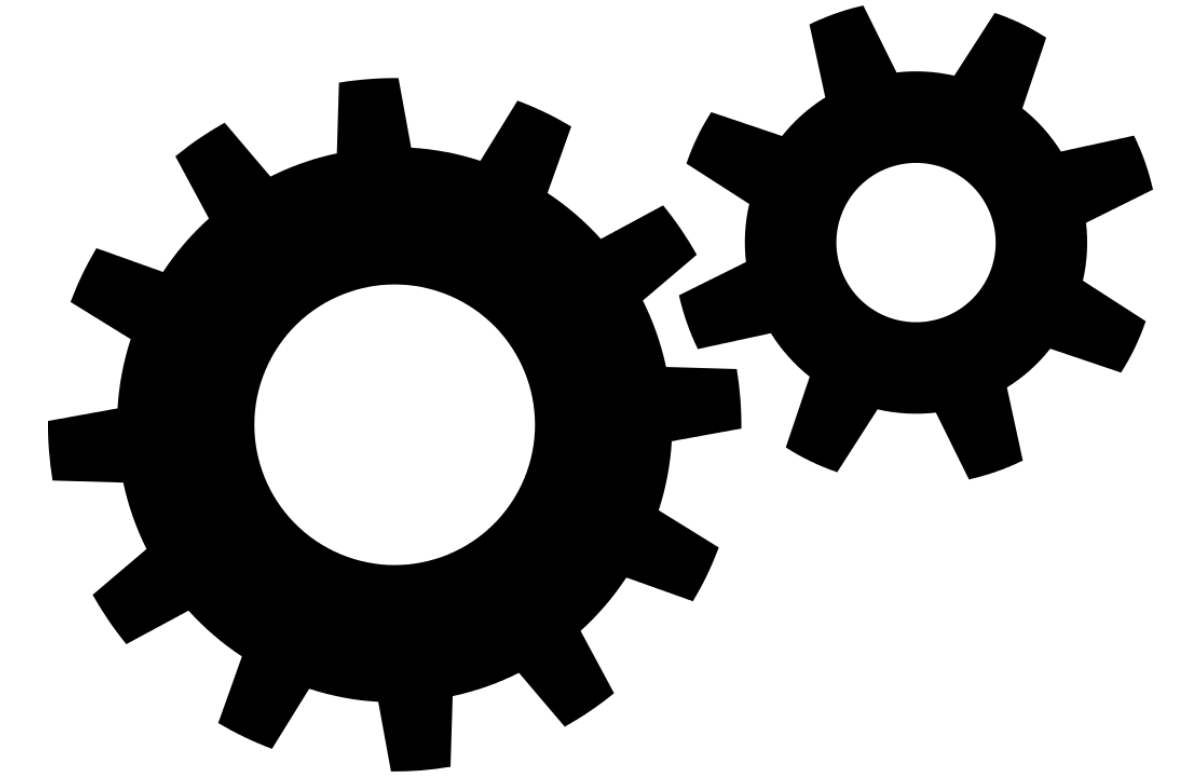
You'll have to trust me here... or not:
for proof, try taking the derivative of SSres and equalizing it to 0.

Ordinary least squares: interactive session



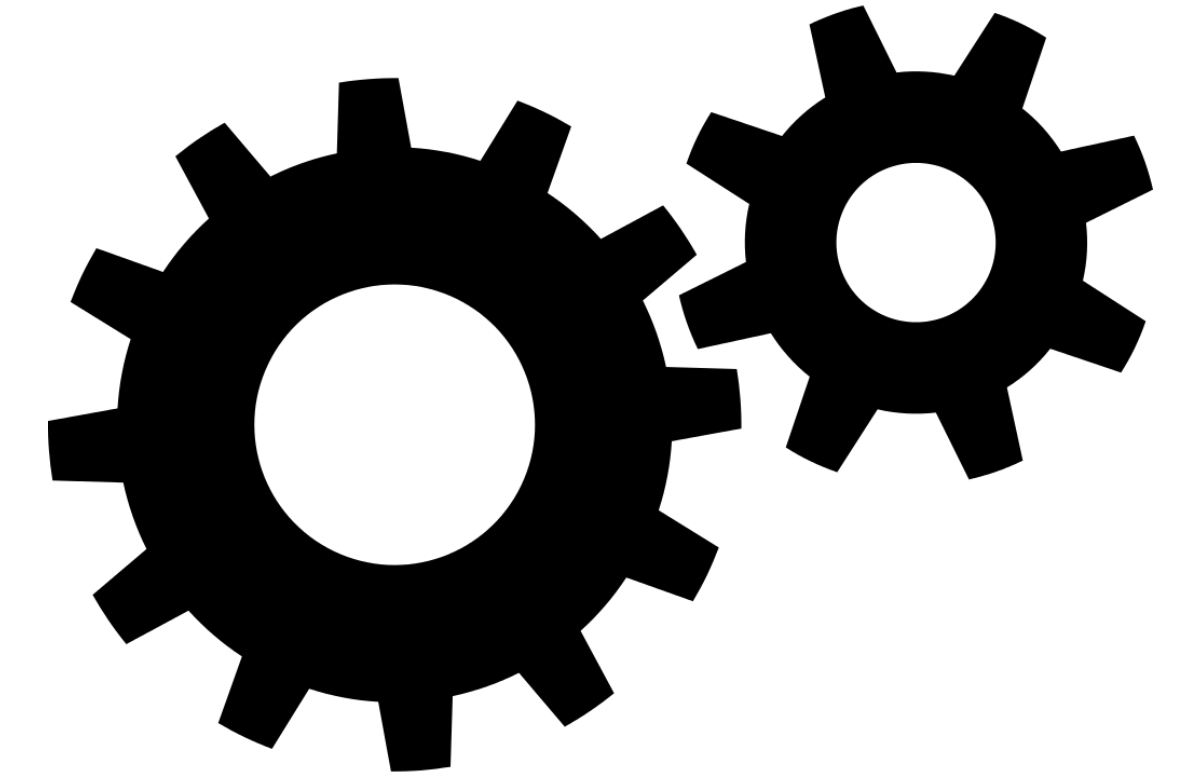
<https://setosa.io/ev/ordinary-least-squares-regression/>

Ordinary least squares



“Find estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ of the parameters β_0 and β_1 by minimizing the **Sum of Squared Residuals**”

Ordinary least squares



“Find estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ of the parameters β_0 and β_1 by minimizing the **Sum of Squared Residuals**”

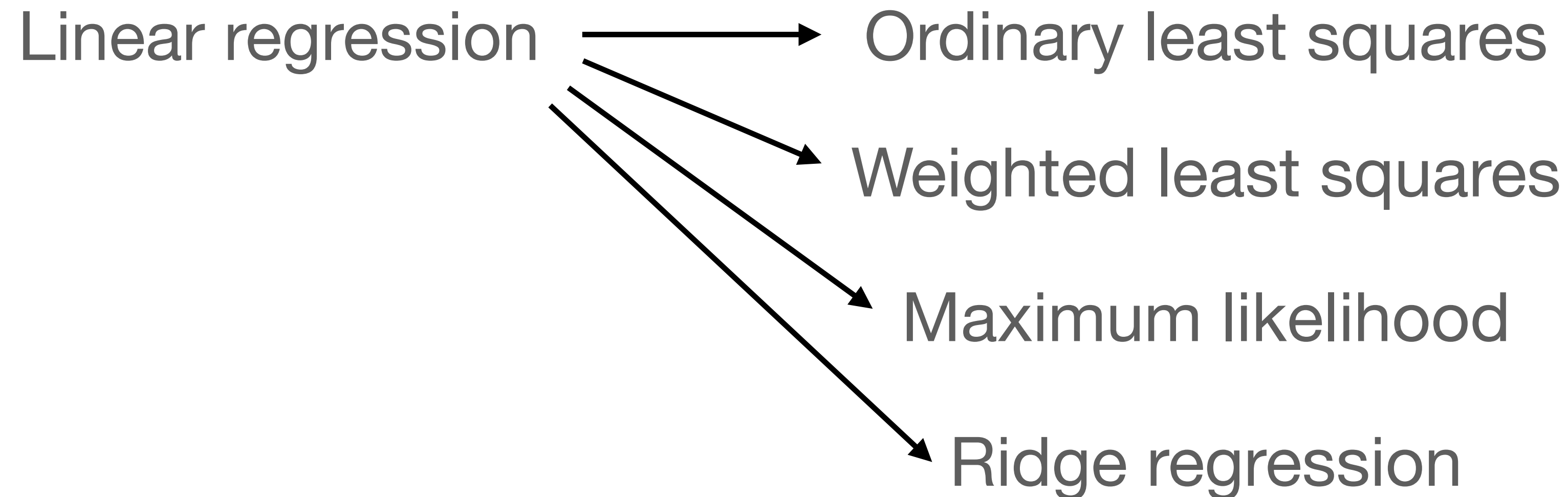


Loss function

Model vs. Inference Method

Model

Inference method



...

Each of these methods has a different loss function!

What does the slope mean?

$$\hat{\beta}_1 = ?$$

What does the slope mean?

$$\hat{\beta}_1 = ?$$

The **slope** measures the **covariance between X and Y**, as a proportion of the **variance of X**

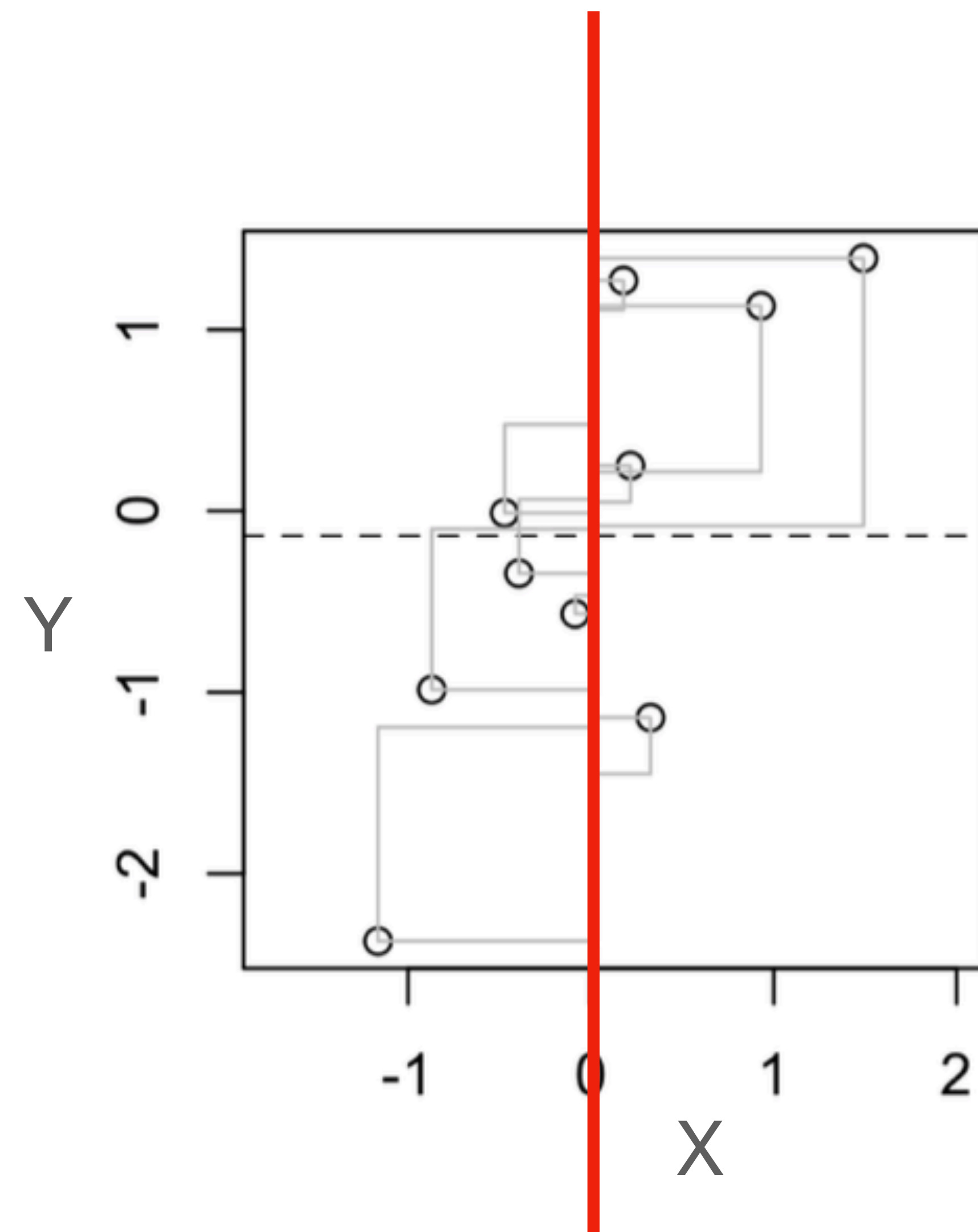
What does the slope mean?

$$\hat{\beta}_1 = ?$$

The **slope** measures the **covariance between X and Y**, as a proportion of the **variance of X**

Let's unpack this a bit...

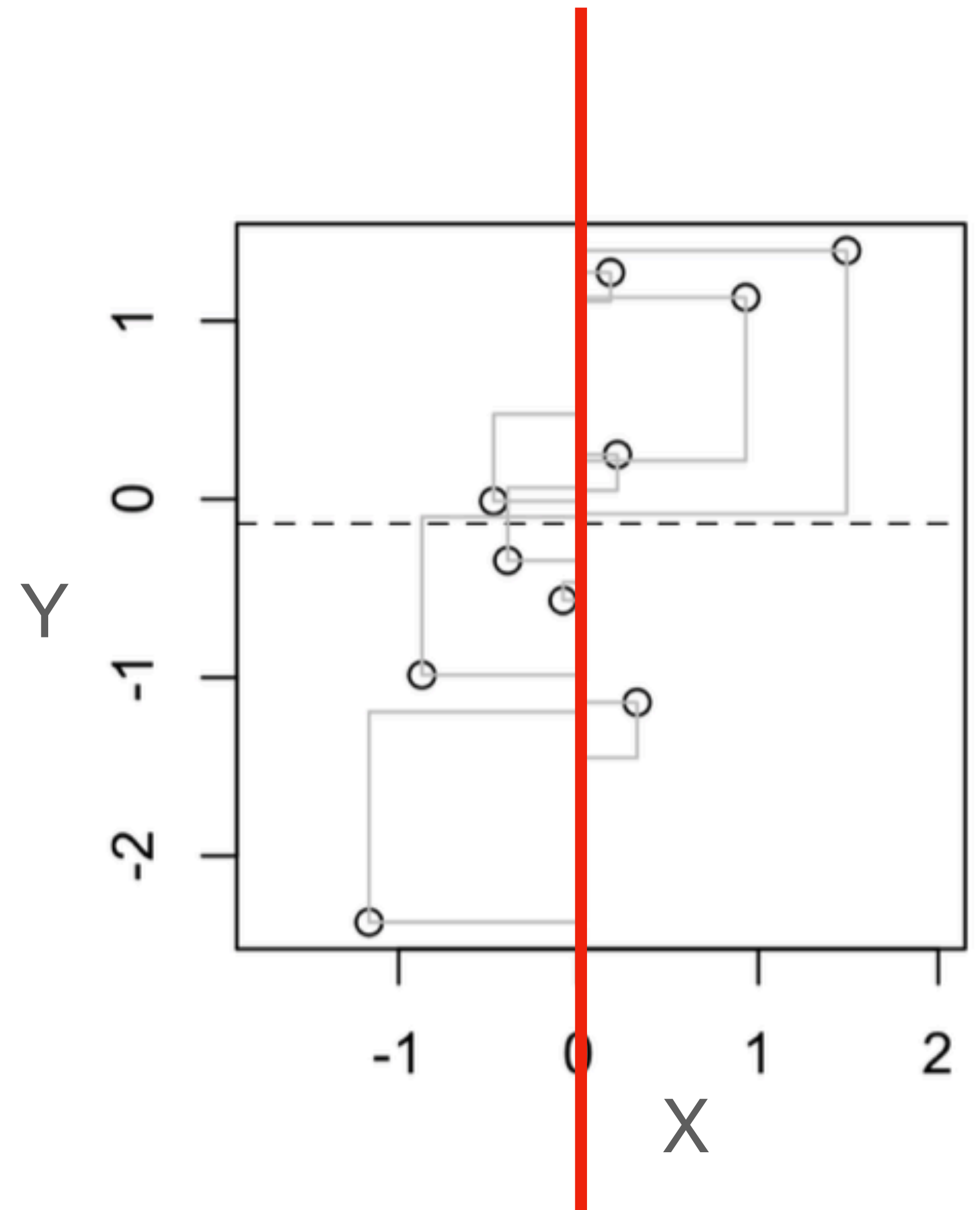
Measuring variation



Measuring variation

Sum of squares of X

$$SS_X = \sum_{i=1}^n (x_i - \bar{x})^2$$



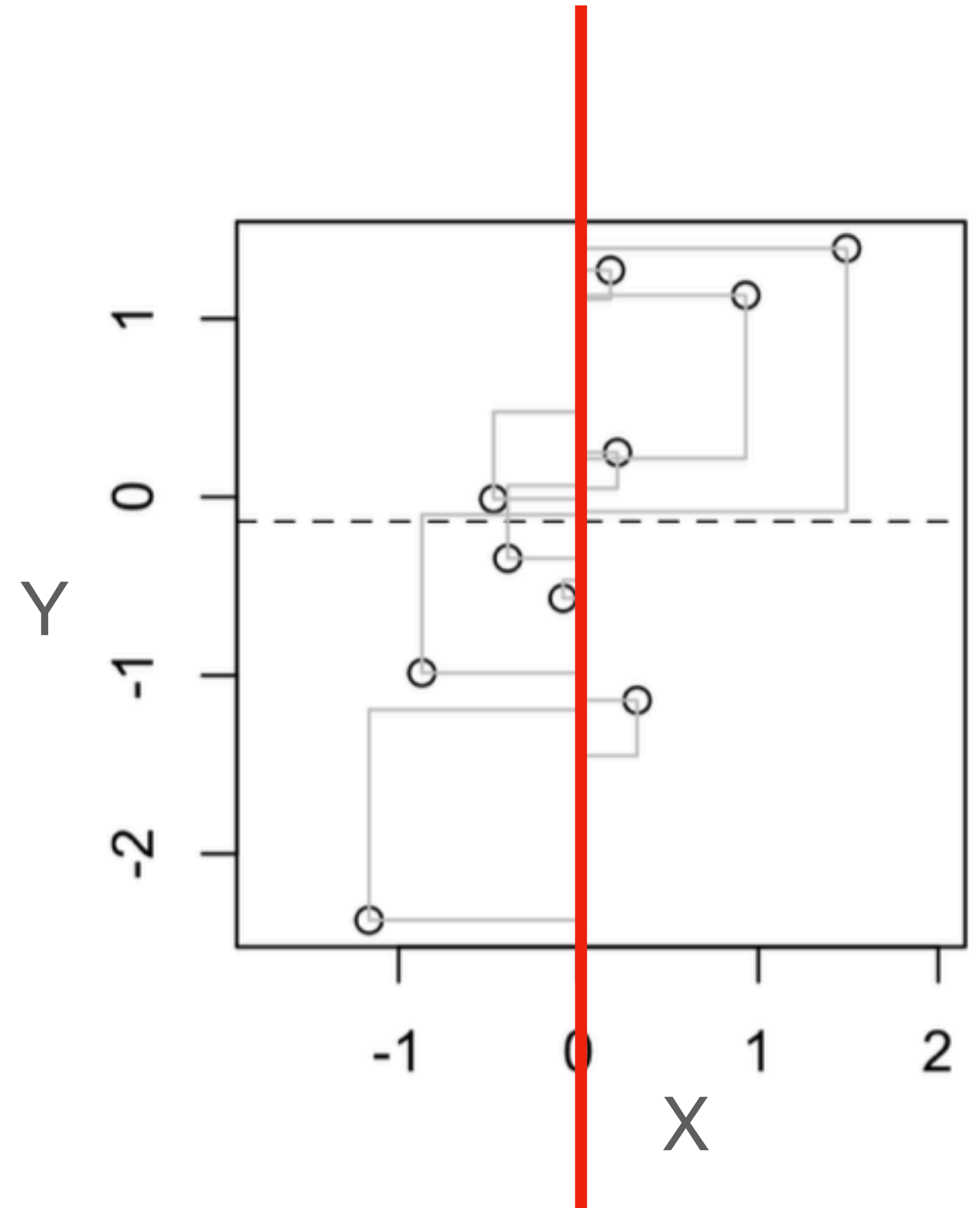
Measuring variation

Sum of squares of X

$$SS_X = \sum_{i=1}^n (x_i - \bar{x})^2$$

Sample variance of X:

$$s_X = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$



Measuring variation

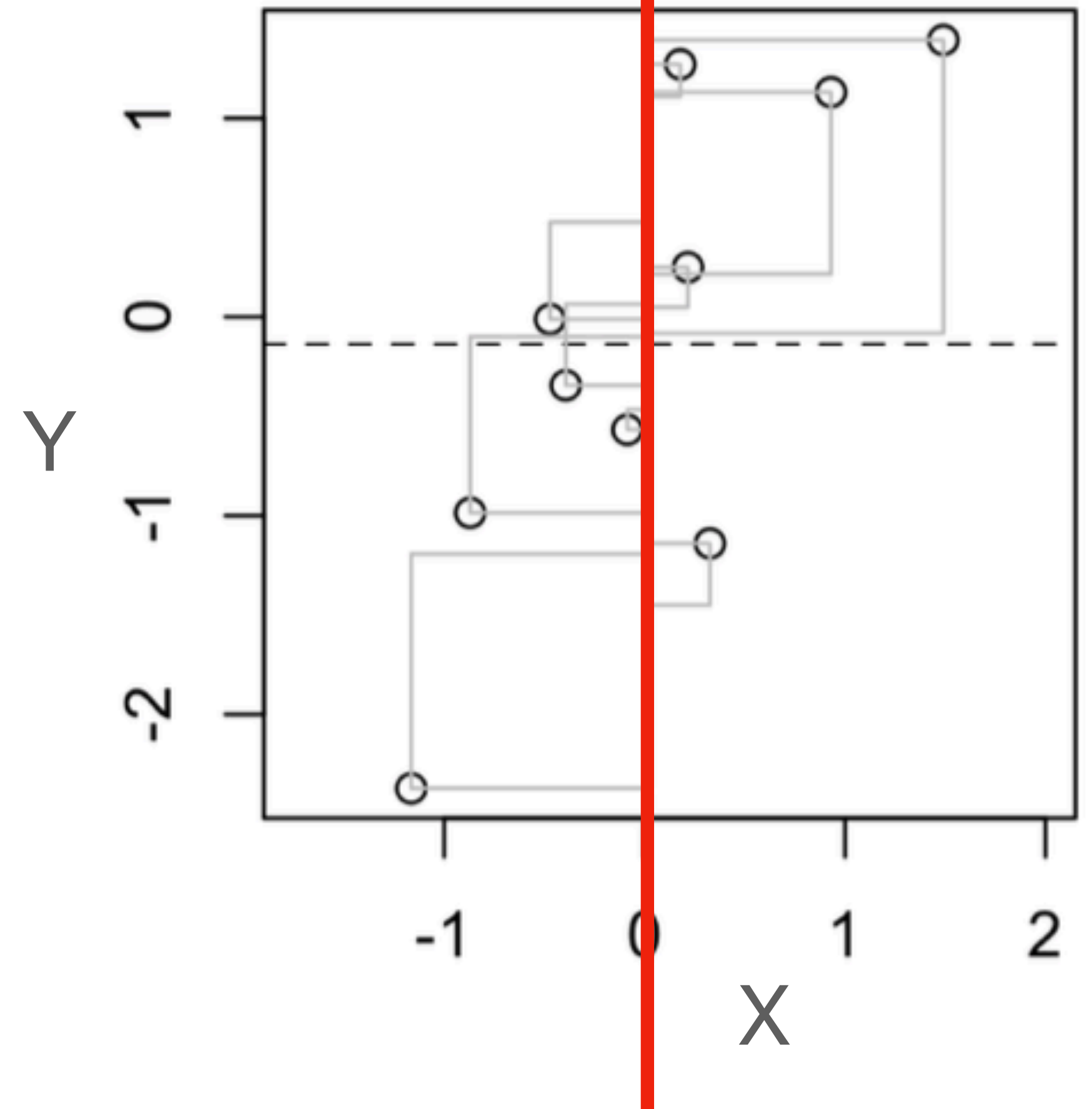
Sum of squares of X

$$SS_X = \sum_{i=1}^n (x_i - \bar{x})^2$$

Sample variance of X:

$$s_X = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

This constant is applied to the sum of squares to obtain an **unbiased estimate of the true variance** when we only have **finite samples**

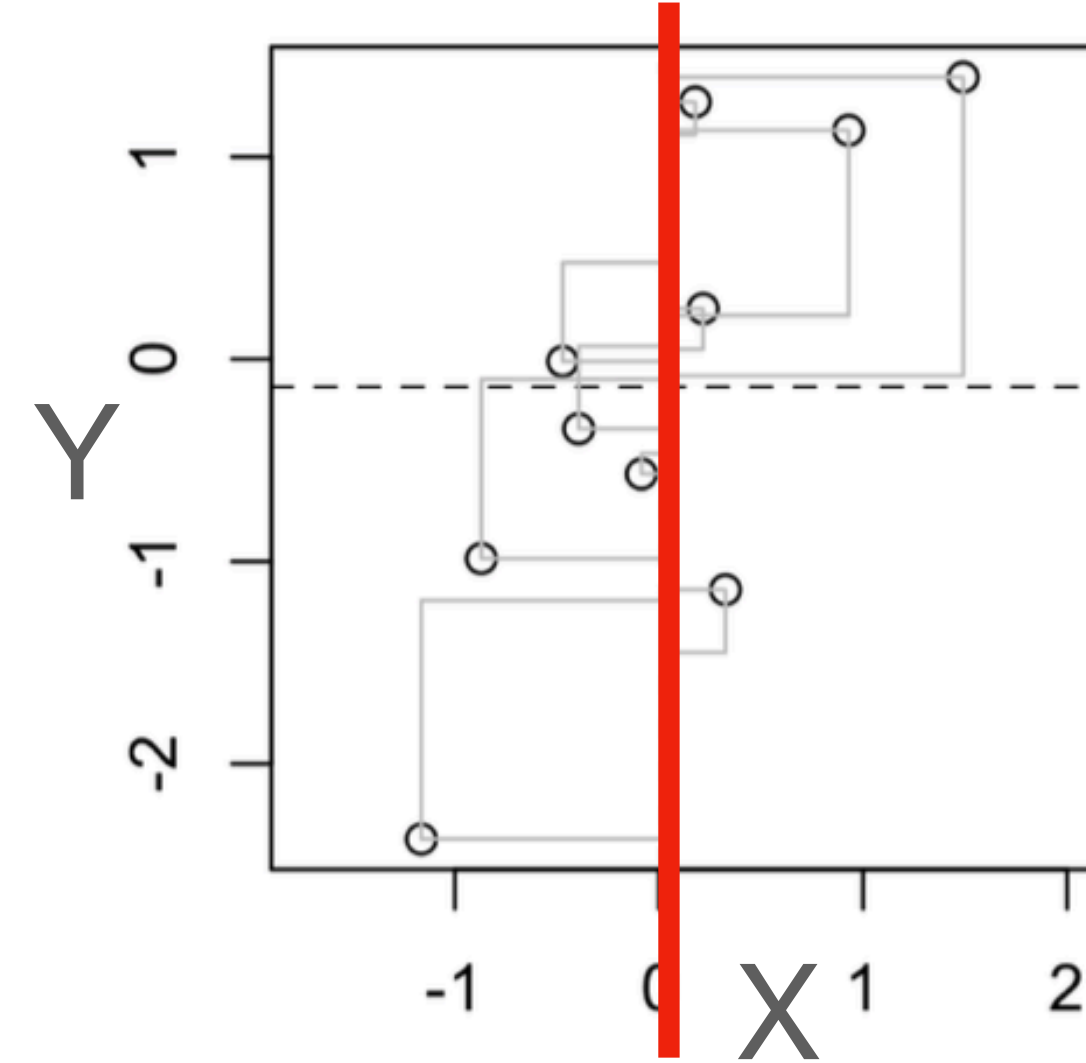


Measuring variation

Sample variance of X:

$$s_X = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

“Sum of squares of X”

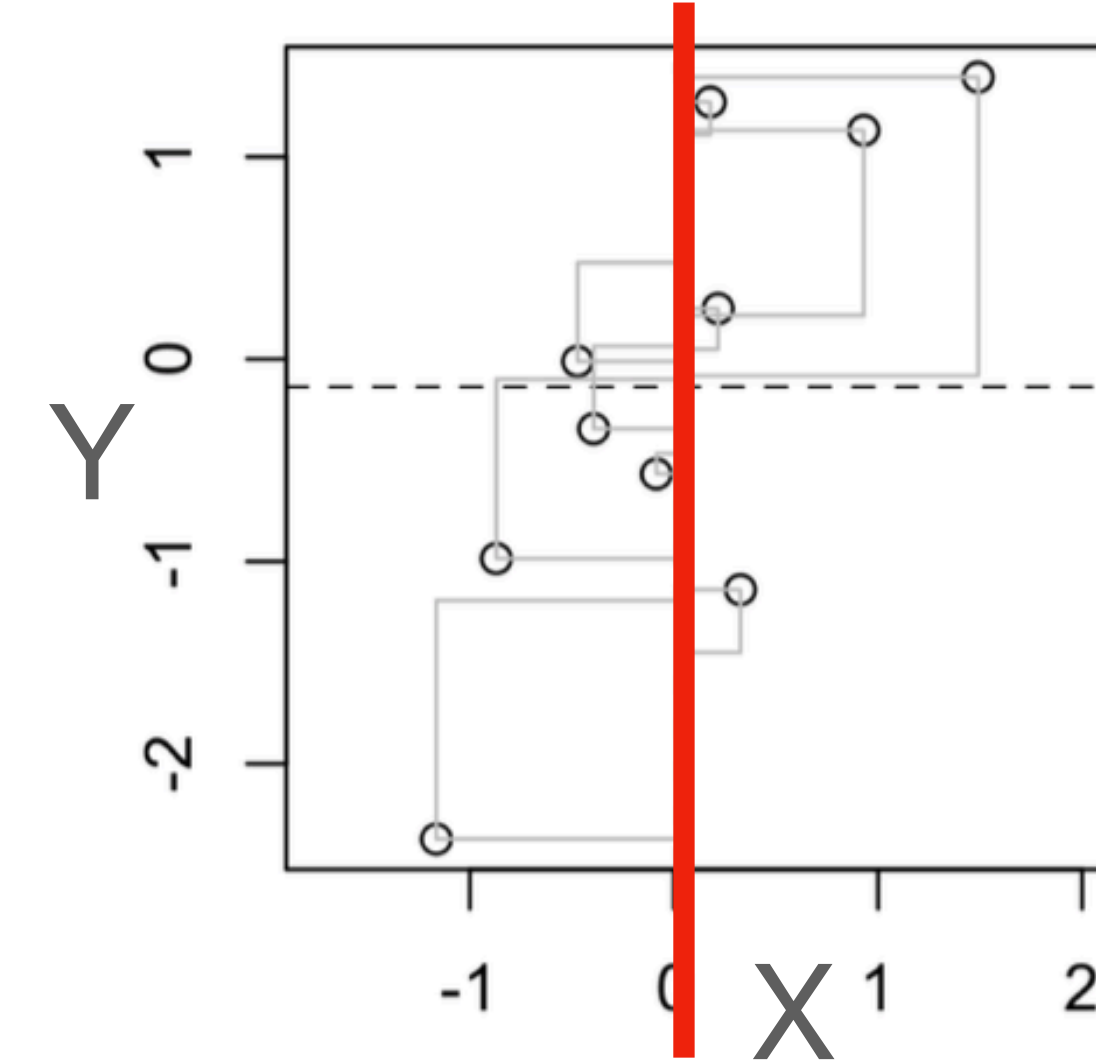


Measuring variation

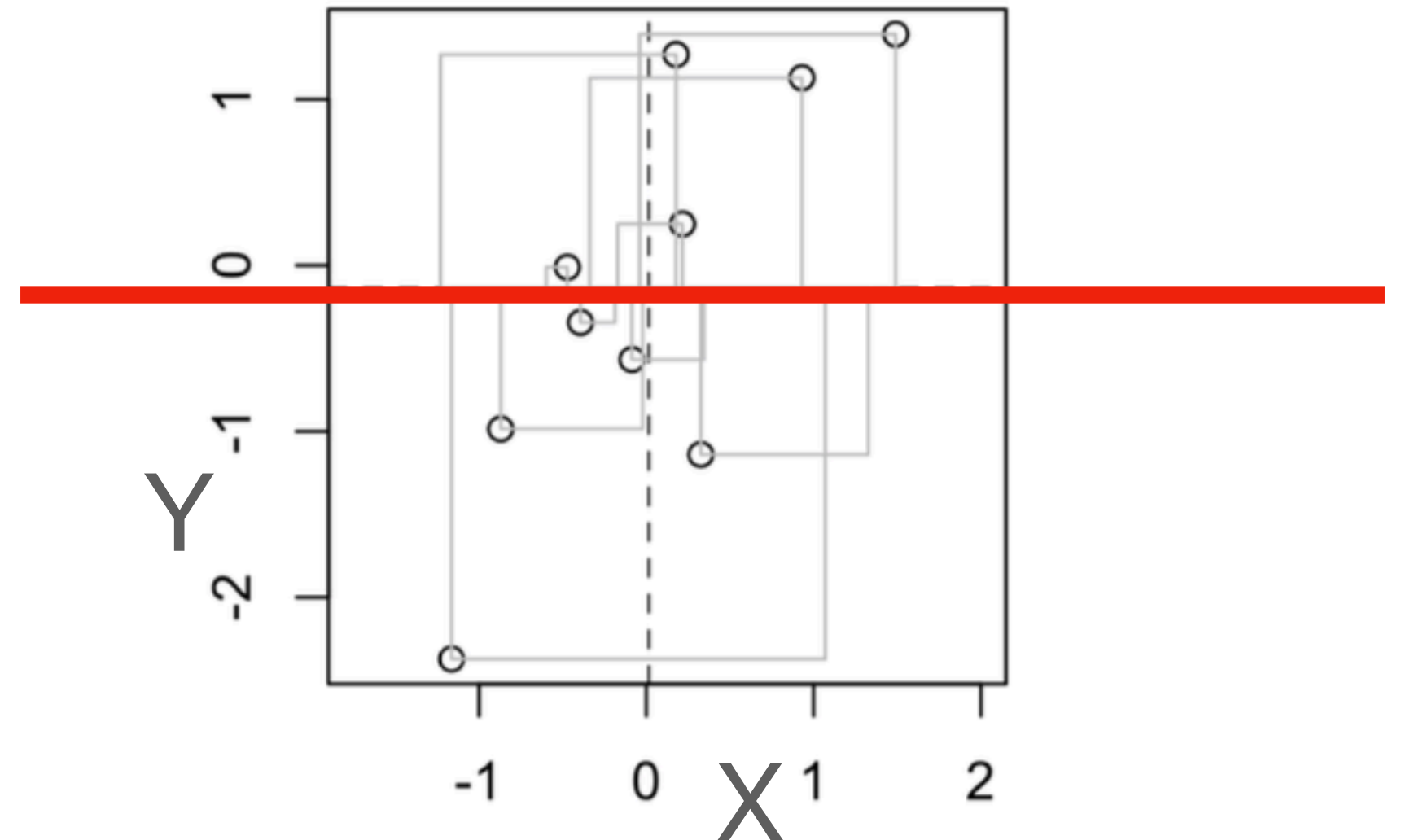
Sample variance of X:

$$s_X = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

“Sum of squares of X”



“Sum of squares of Y”



Measuring variation

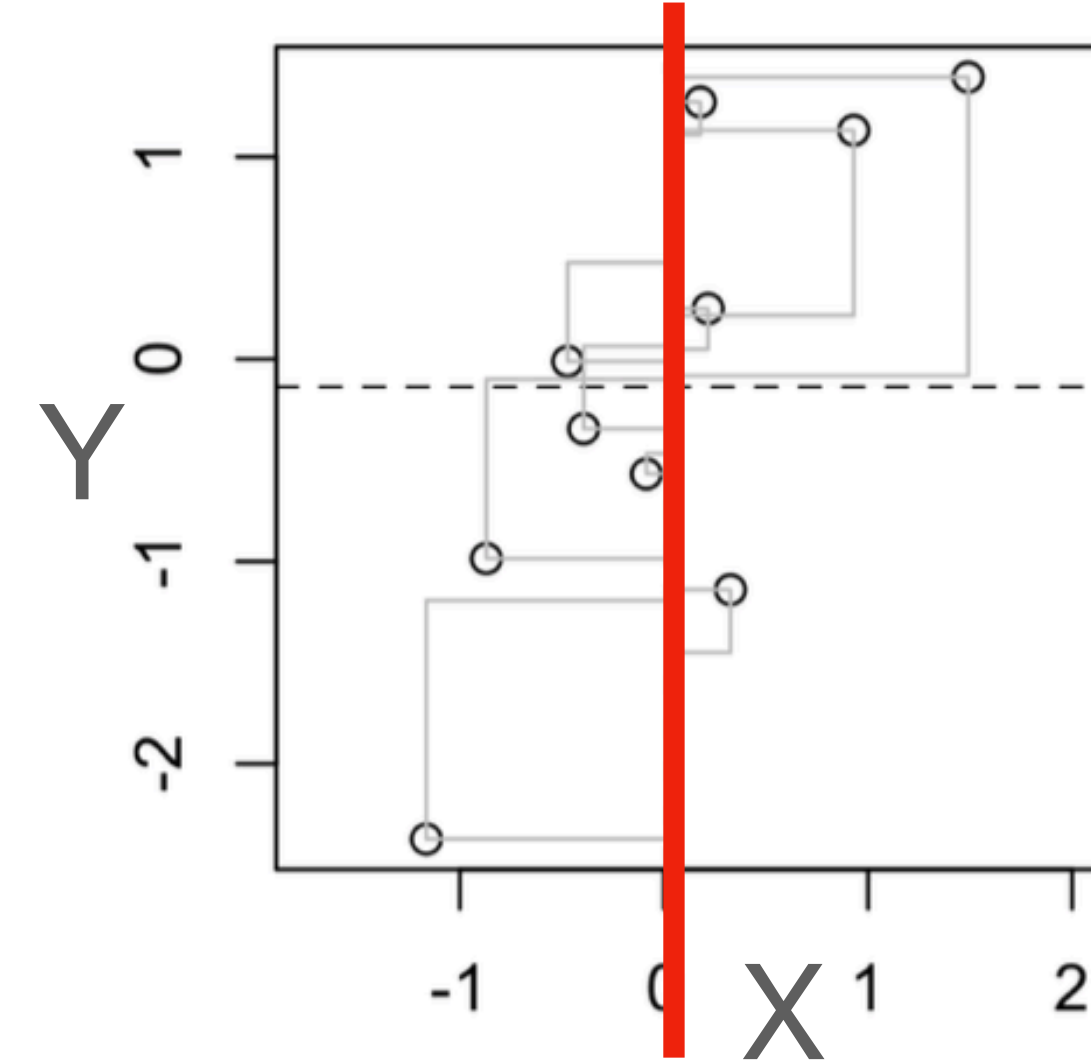
Sample variance of X:

$$s_X = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

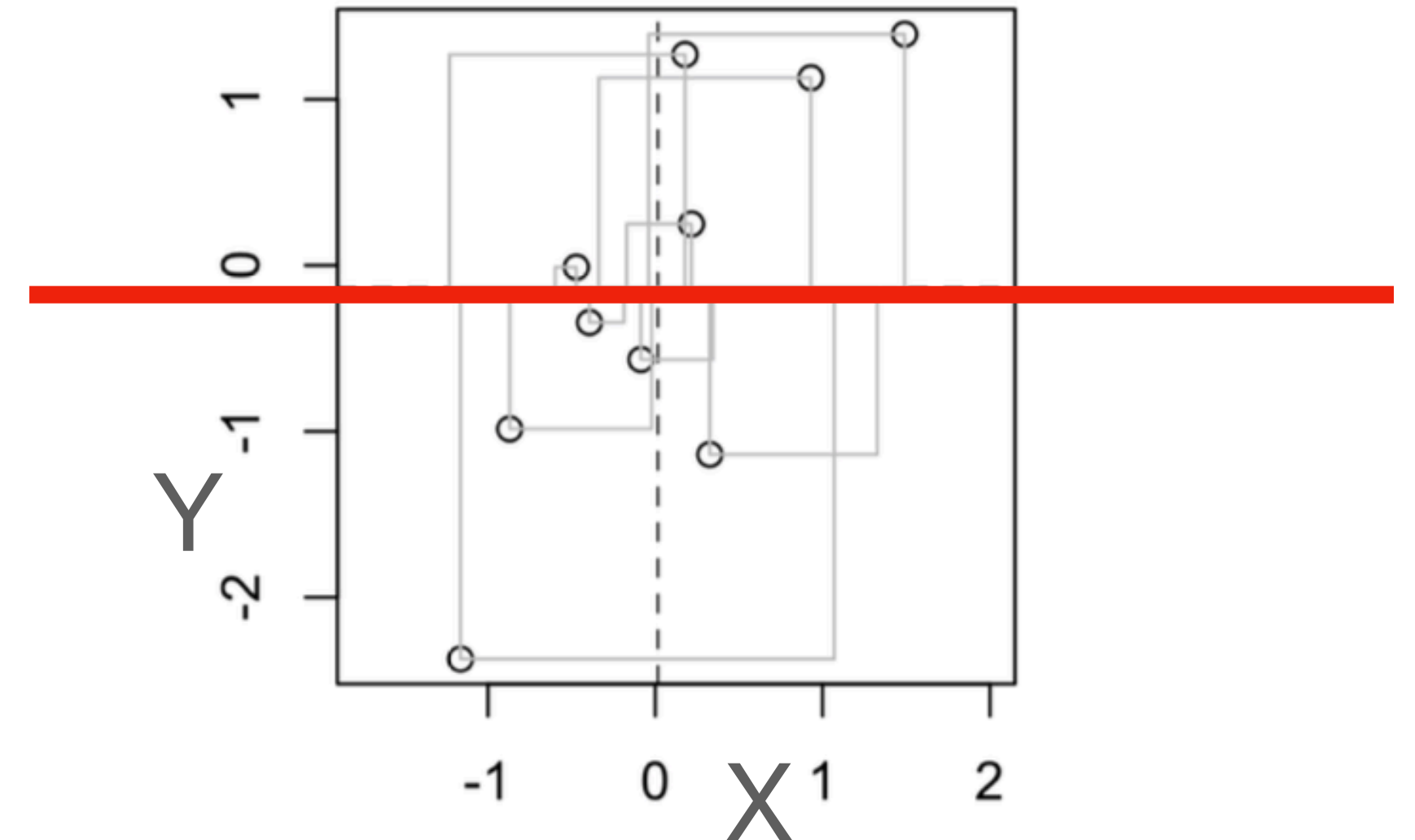
Sample variance of Y:

$$s_Y = \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n - 1}$$

“Sum of squares of X”



“Sum of squares of Y”

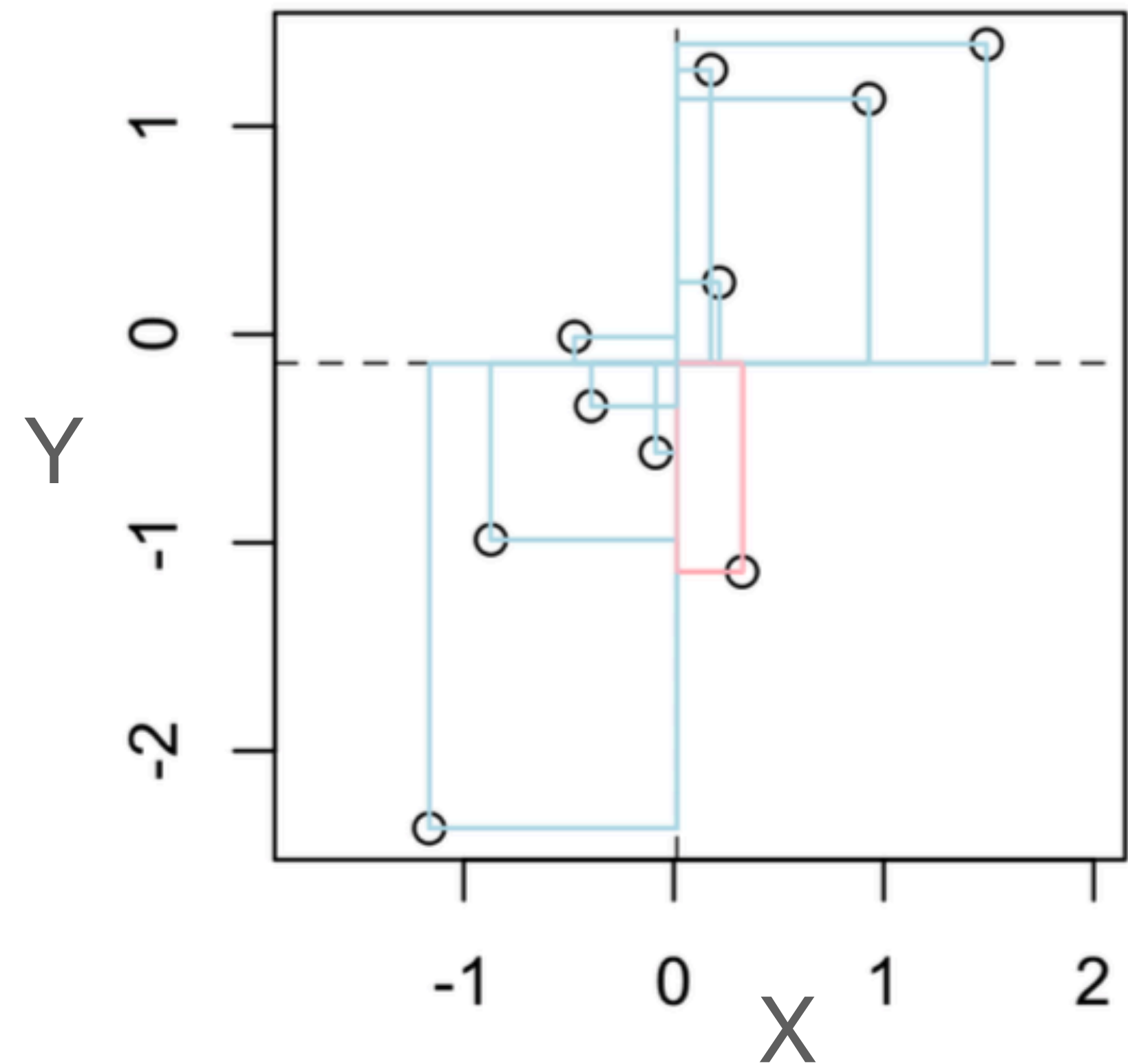


Measuring co-variation

Sample covariance of X and Y:

$$s_{X,Y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

“Sum of XY rectangles”



The Slope

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n [(x_i - \bar{x})^2]} \longrightarrow \begin{array}{l} \text{Sample variance} \\ \text{of } X \end{array}$$

The Slope

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n [(x_i - \bar{x})^2]}$$

→ Sample covariance between X and Y

→ Sample variance of X

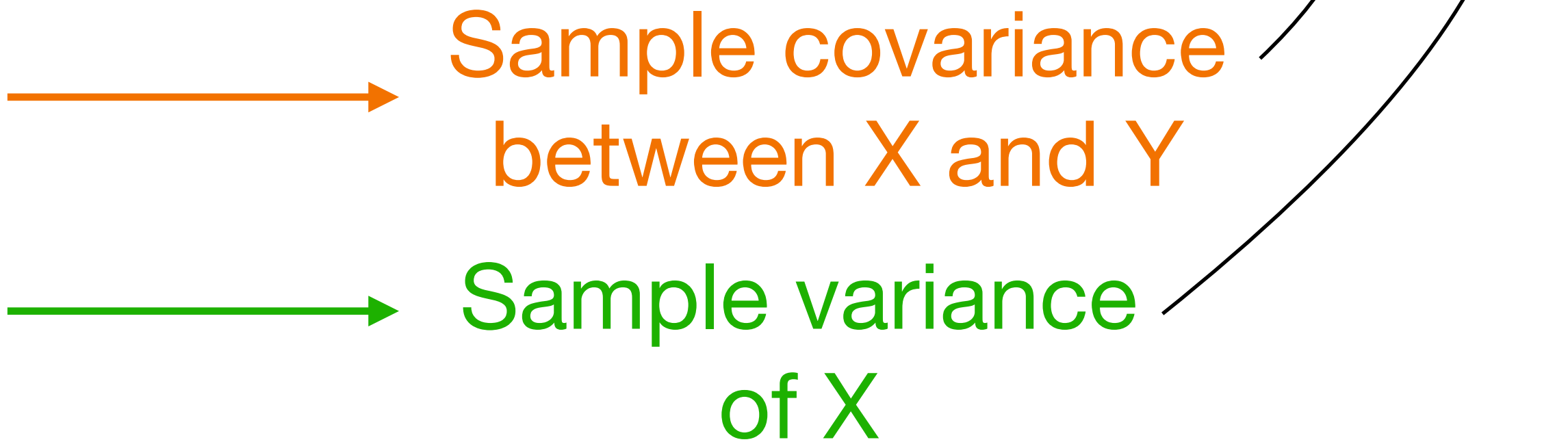
The Slope

The constant $n-1$ is in both the numerator and denominator,
so it cancels out

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n [(x_i - \bar{x})^2]}$$

Sample covariance
between X and Y

Sample variance
of X



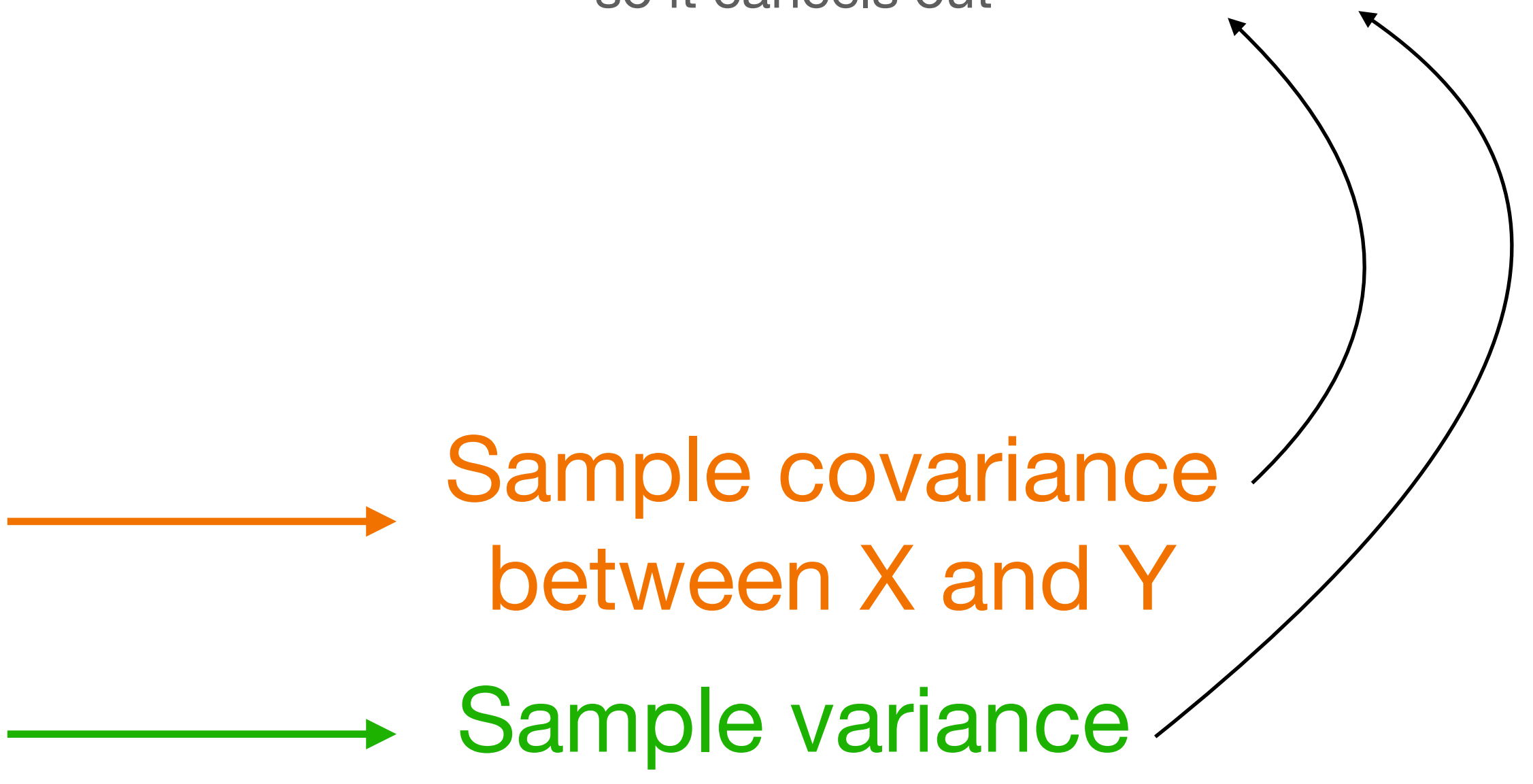
The Slope

The constant $n-1$ is in both the numerator and denominator,
so it cancels out

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n [(x_i - \bar{x})^2]}$$

Sample covariance
between X and Y

Sample variance
of X

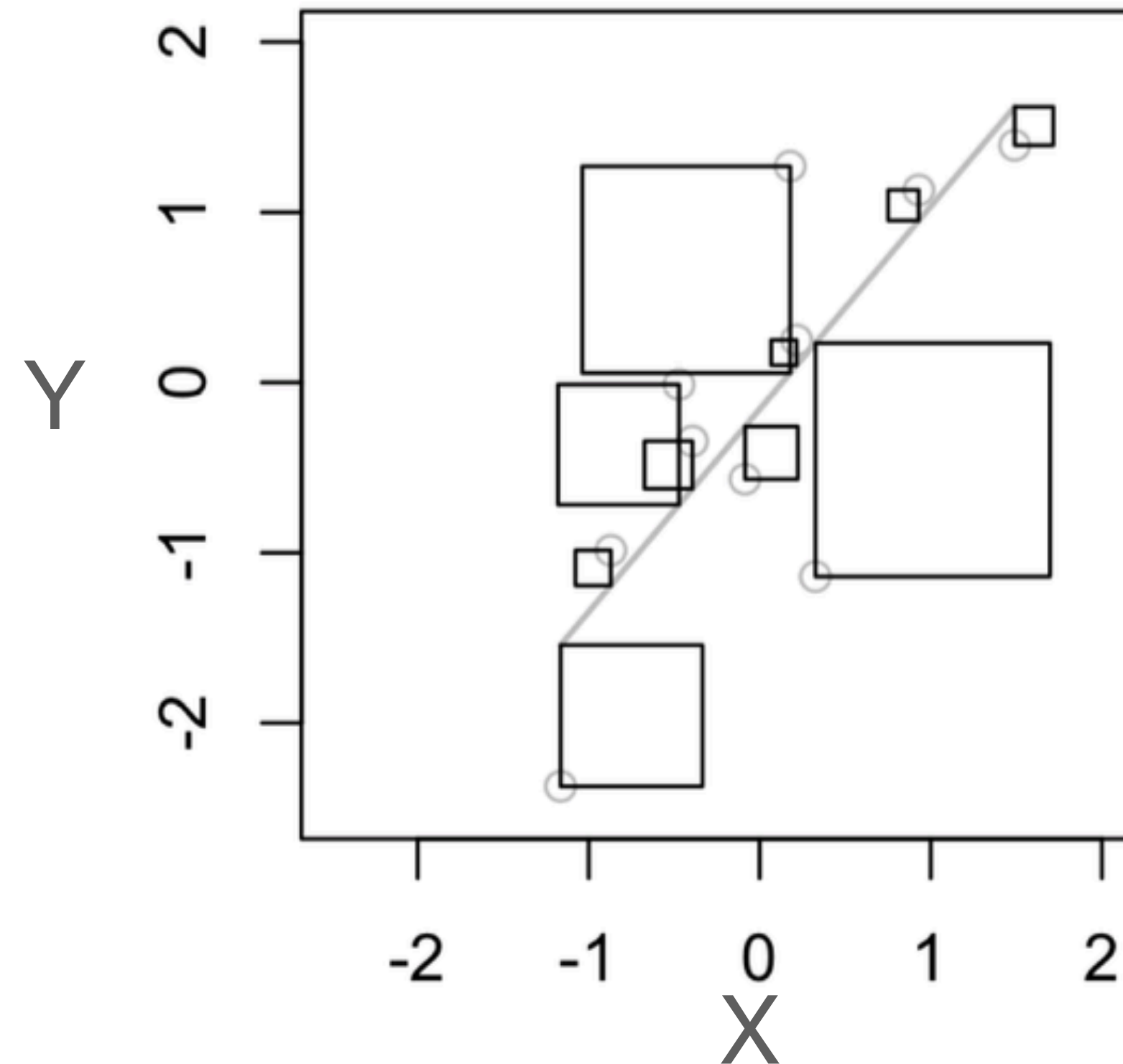


The **slope** measures the **covariance between X and Y**, as a proportion of the **variance of X**

How good is our model?

- No model is perfect, but some are more useful than others
- After fitting our model, we still have **unexplained variation in the dependent variable**: the sums of squares of the residuals (SS_{res}).

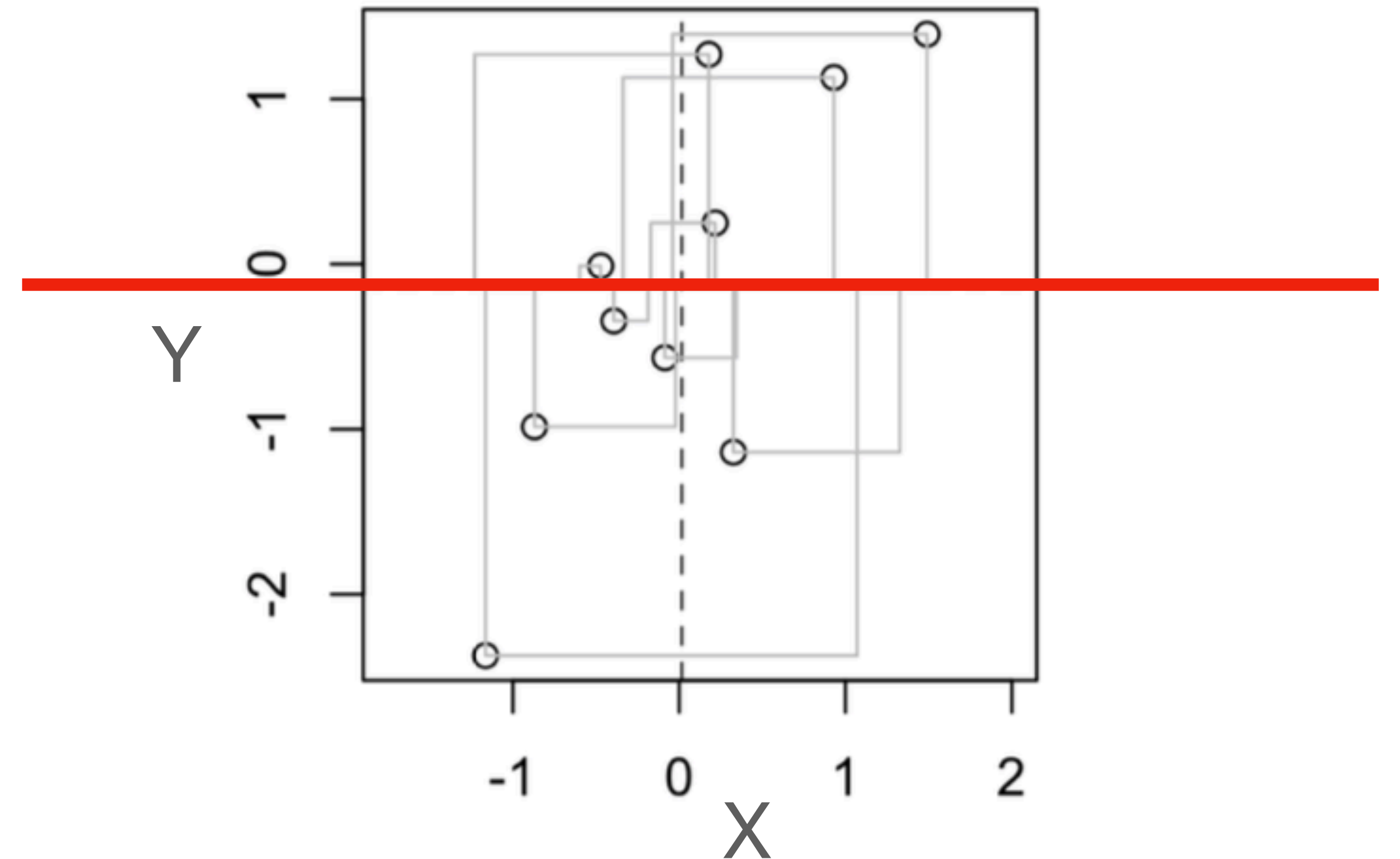
$$SS_{res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



How good is our model?

- We also have a measure of the **total variation in the dependent variable**: the sum of the squares of Y (SS_Y)

$$SS_Y = \sum_{i=1}^n (y_i - \bar{y})^2$$



How good is our model?

- A natural measure of fit: the **coefficient of determination (R^2)**

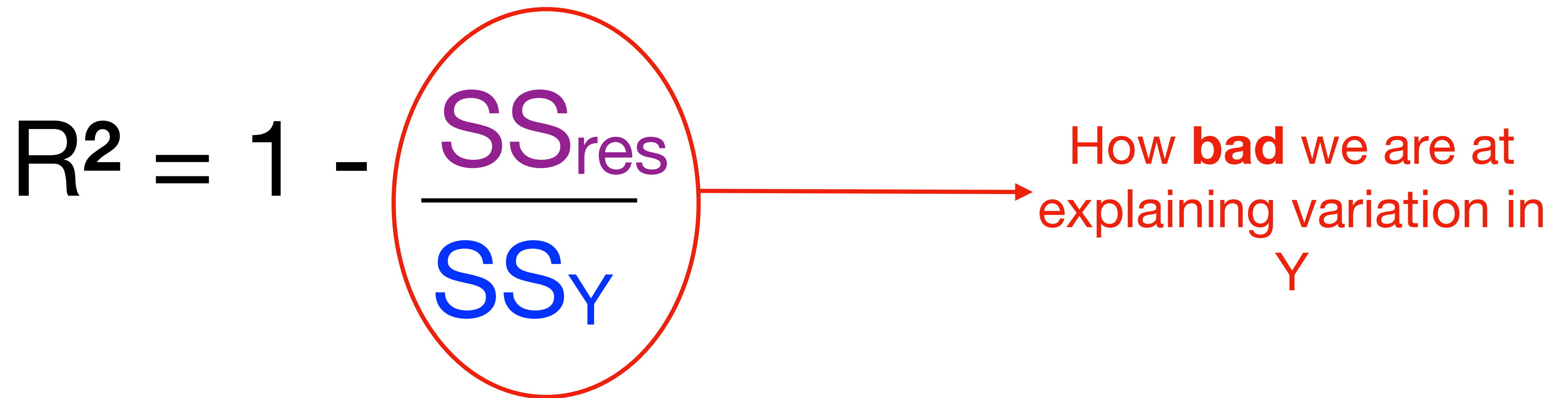
$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_Y}$$

How good is our model?

- A natural measure of fit: the **coefficient of determination** (R^2)

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_Y}$$

How **bad** we are at explaining variation in Y

The diagram shows the formula for the coefficient of determination, R-squared, as 1 minus the ratio of residual sum of squares (SS_res) to total sum of squares (SS_Y). The fraction is enclosed in a red circle. A red arrow points from the circle to the text 'How bad we are at explaining variation in Y', where 'bad' is in bold and 'Y' is a subscript.

How good is our model?

- A natural measure of fit: the **coefficient of determination (R^2)**

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_Y}$$

How **good** we are at
explaining variation in
Y

Hypothesis testing via a linear model

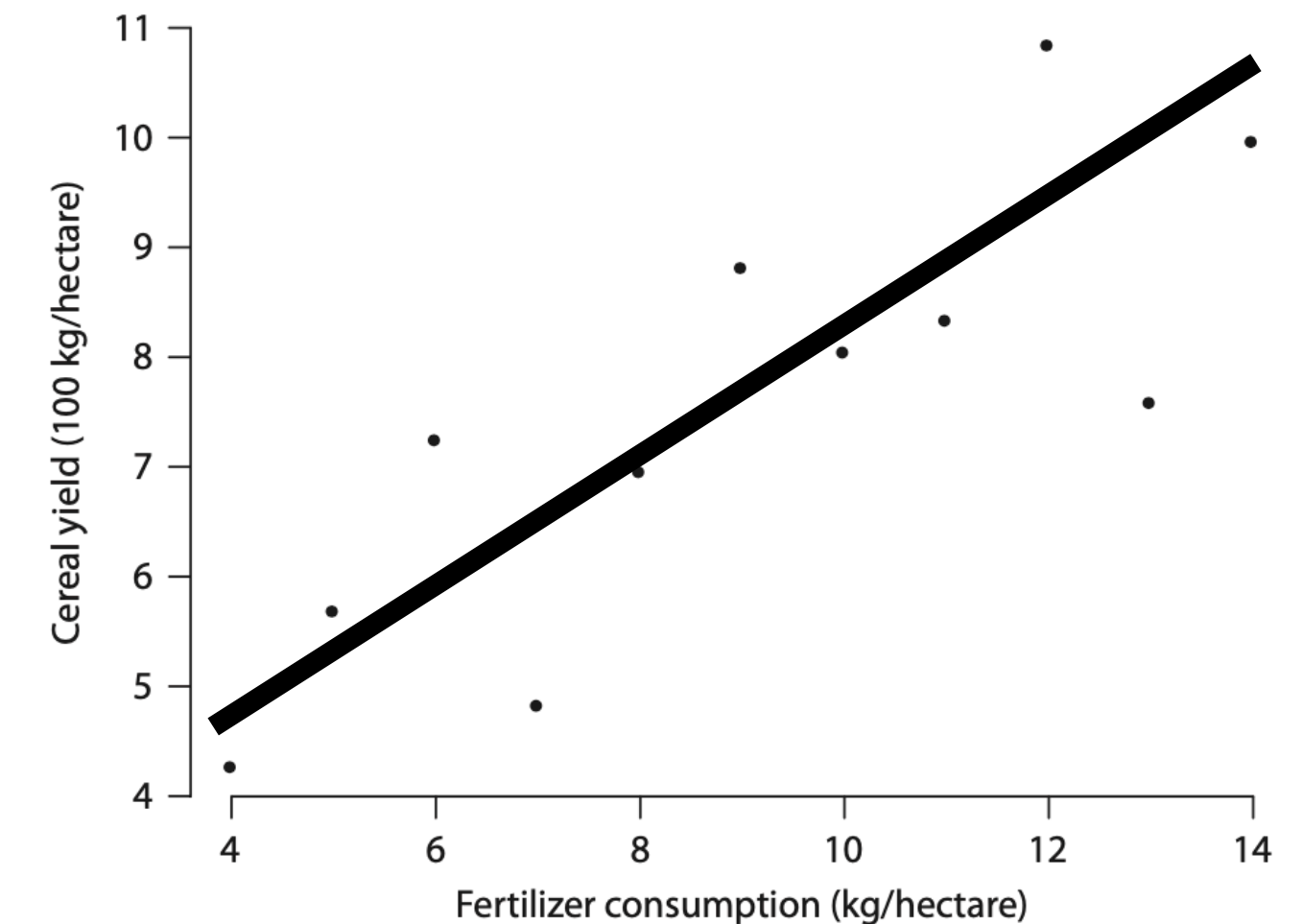
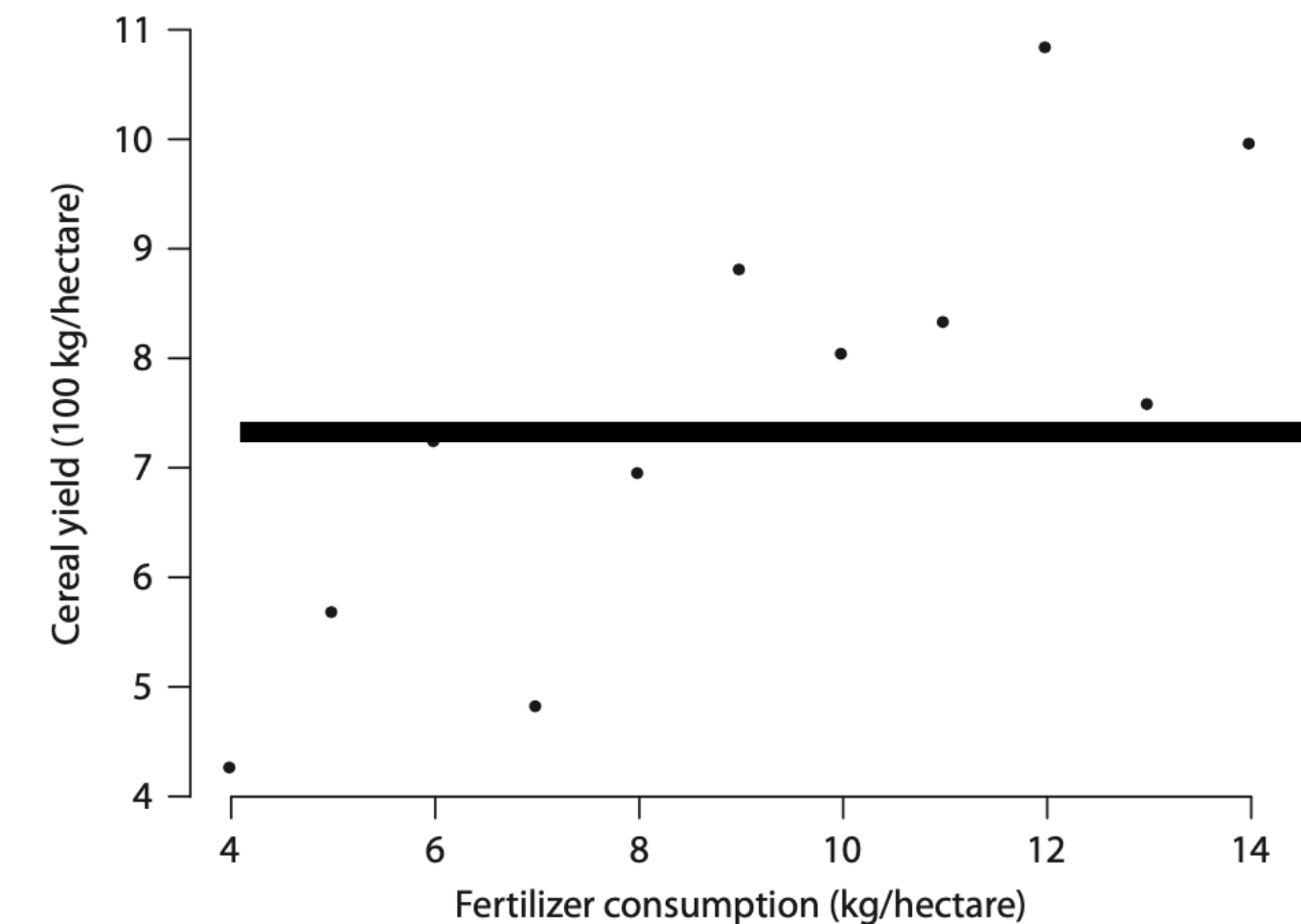
Hypothesis testing

H_0 : there is **no** relationship between fertilizer consumption and cereal yield

Null (simpler) hypothesis $\longrightarrow$ Can we reject it?

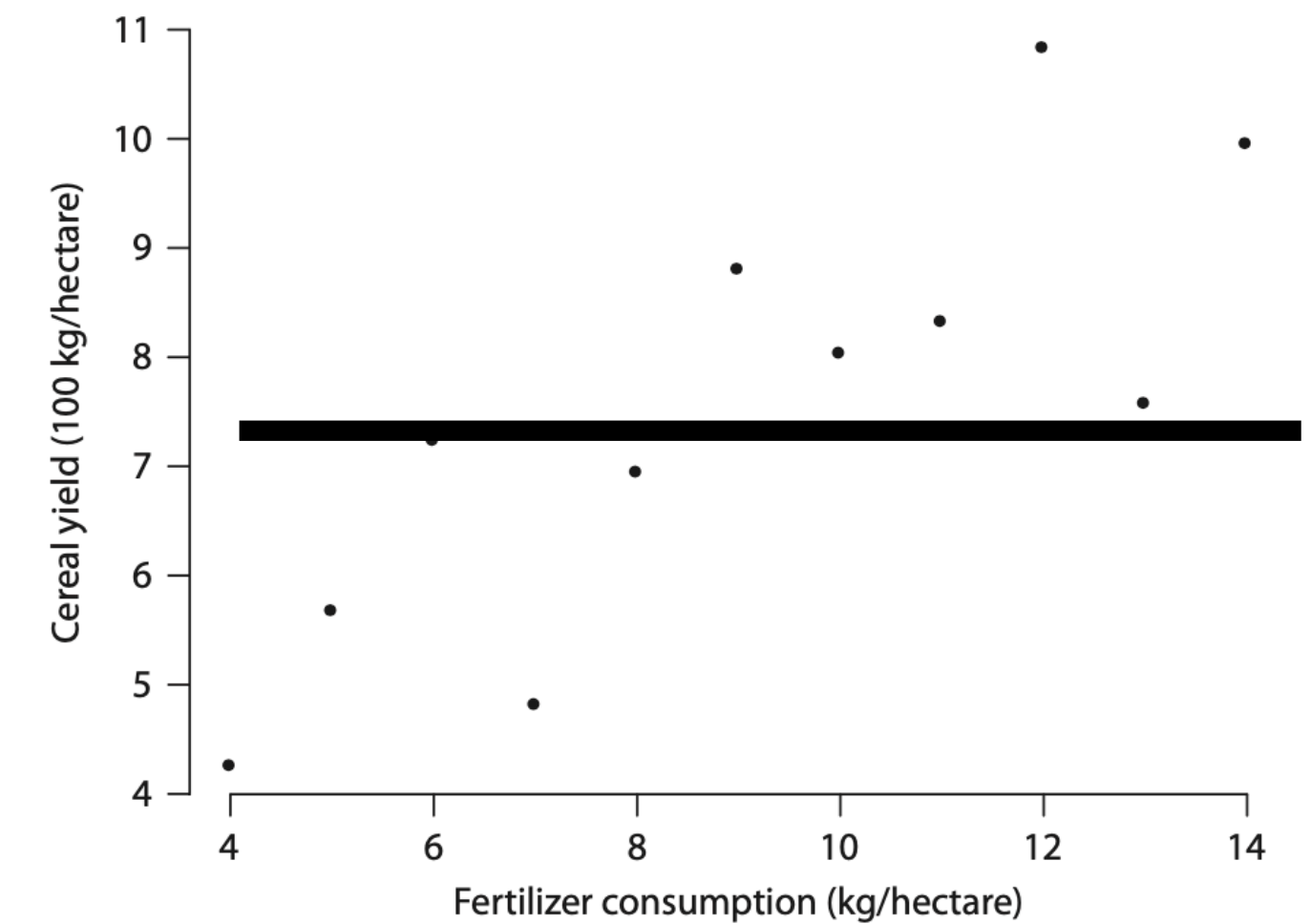
H_1 : there is **some** relationship between fertilizer consumption and cereal yield

Alternative (more complicated) hypothesis

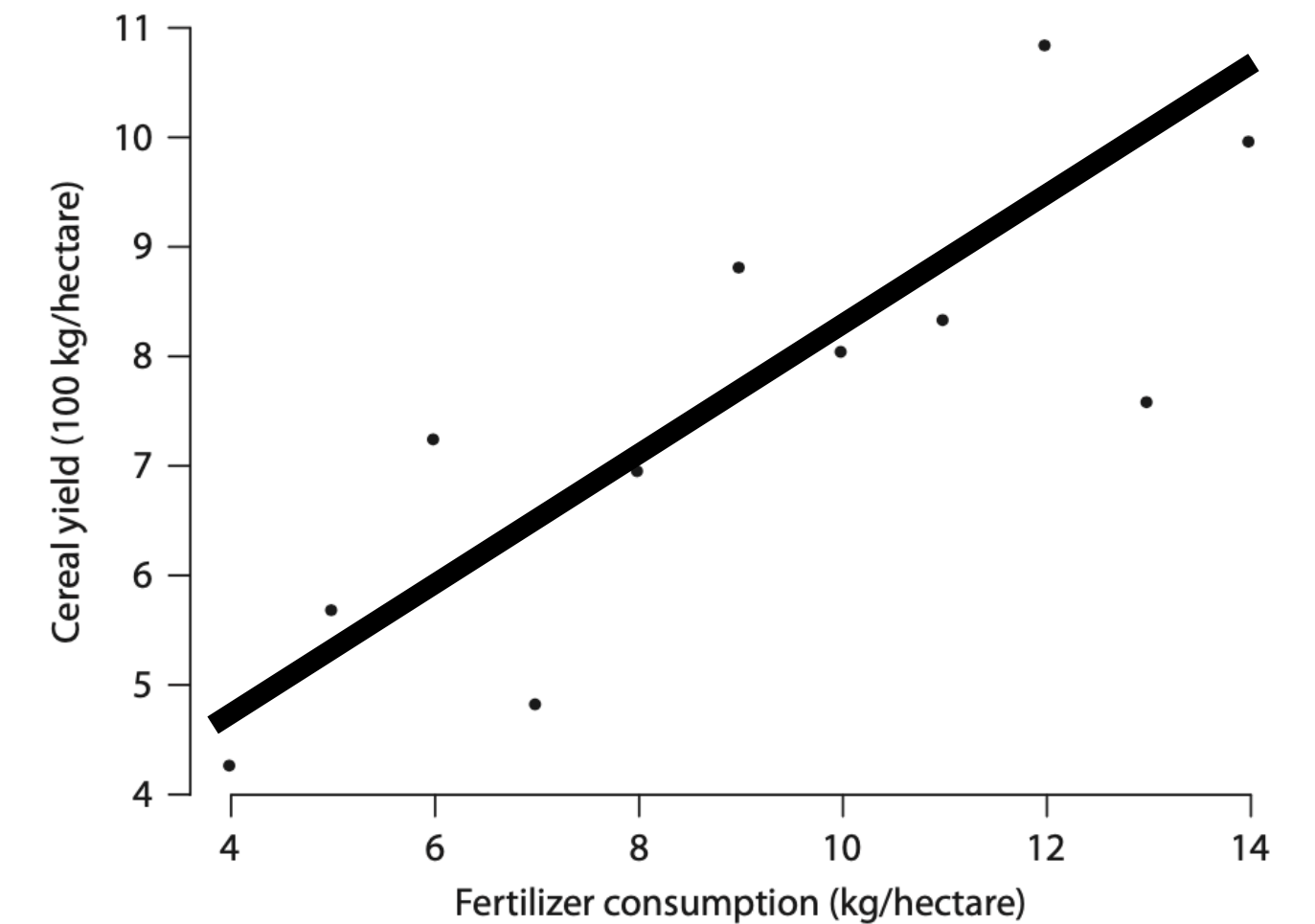


Hypothesis testing

$$y = \beta_0 + \epsilon$$



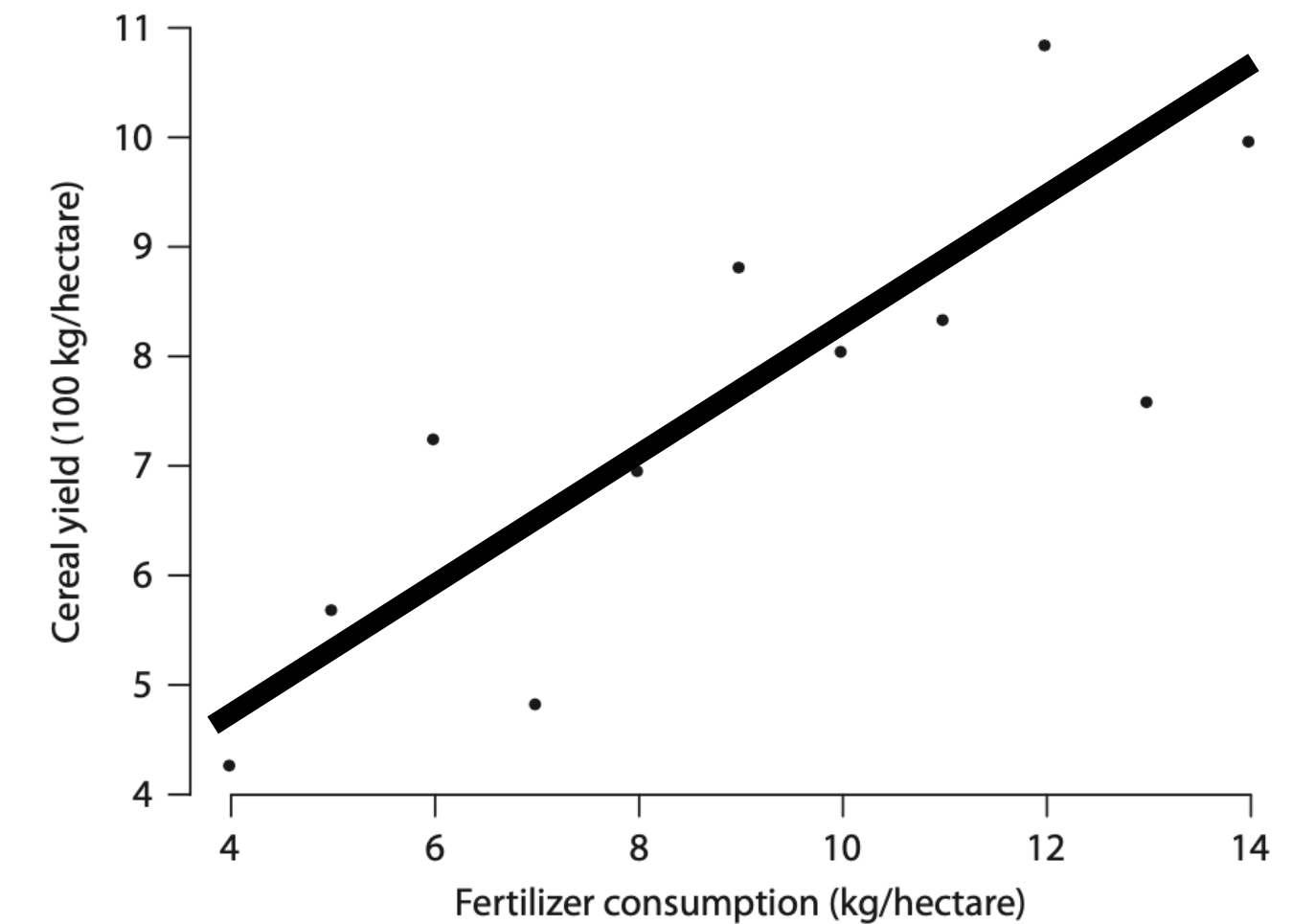
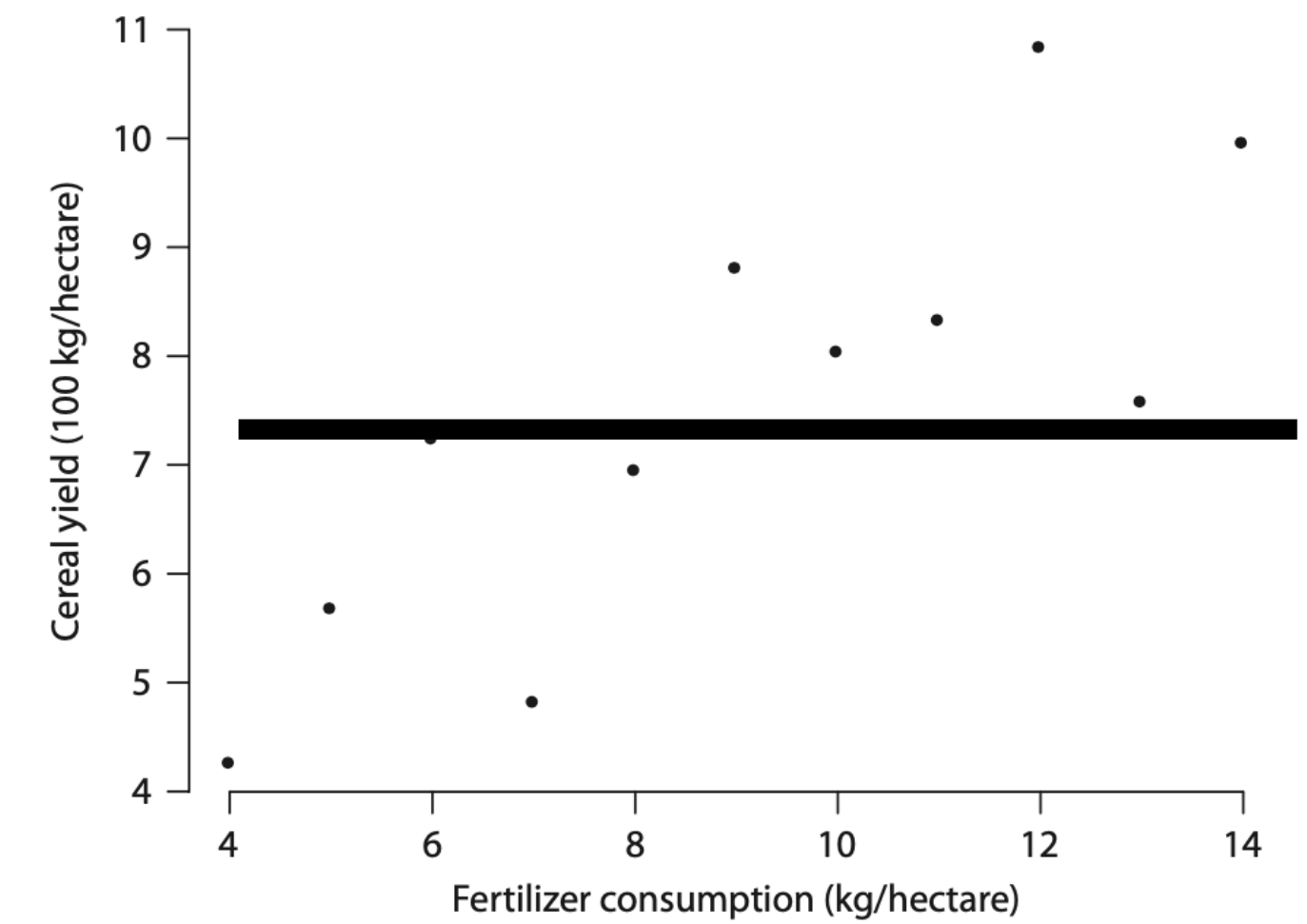
$$y = \beta_0 + \beta_1 x + \epsilon$$



Hypothesis testing

$$\beta_1 = 0$$

$$\beta_1 \neq 0$$



Test statistic: t

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$$

Test statistic: t

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \longrightarrow \text{Difference between our estimate } \hat{\beta}_1 \text{ and } 0$$

Test statistic: t

The bigger the difference, the bigger is t

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \longrightarrow \text{Difference between our estimate } \hat{\beta}_1 \text{ and } 0$$

Test statistic: t

The bigger the difference, the bigger is t

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$$

→ Difference between our estimate $\hat{\beta}_1$ and 0

→ Standard Error: a measure of how inaccurate our estimate $\hat{\beta}_1$ is at estimating β_1

Test statistic: t

The bigger the difference, the bigger is t

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$$

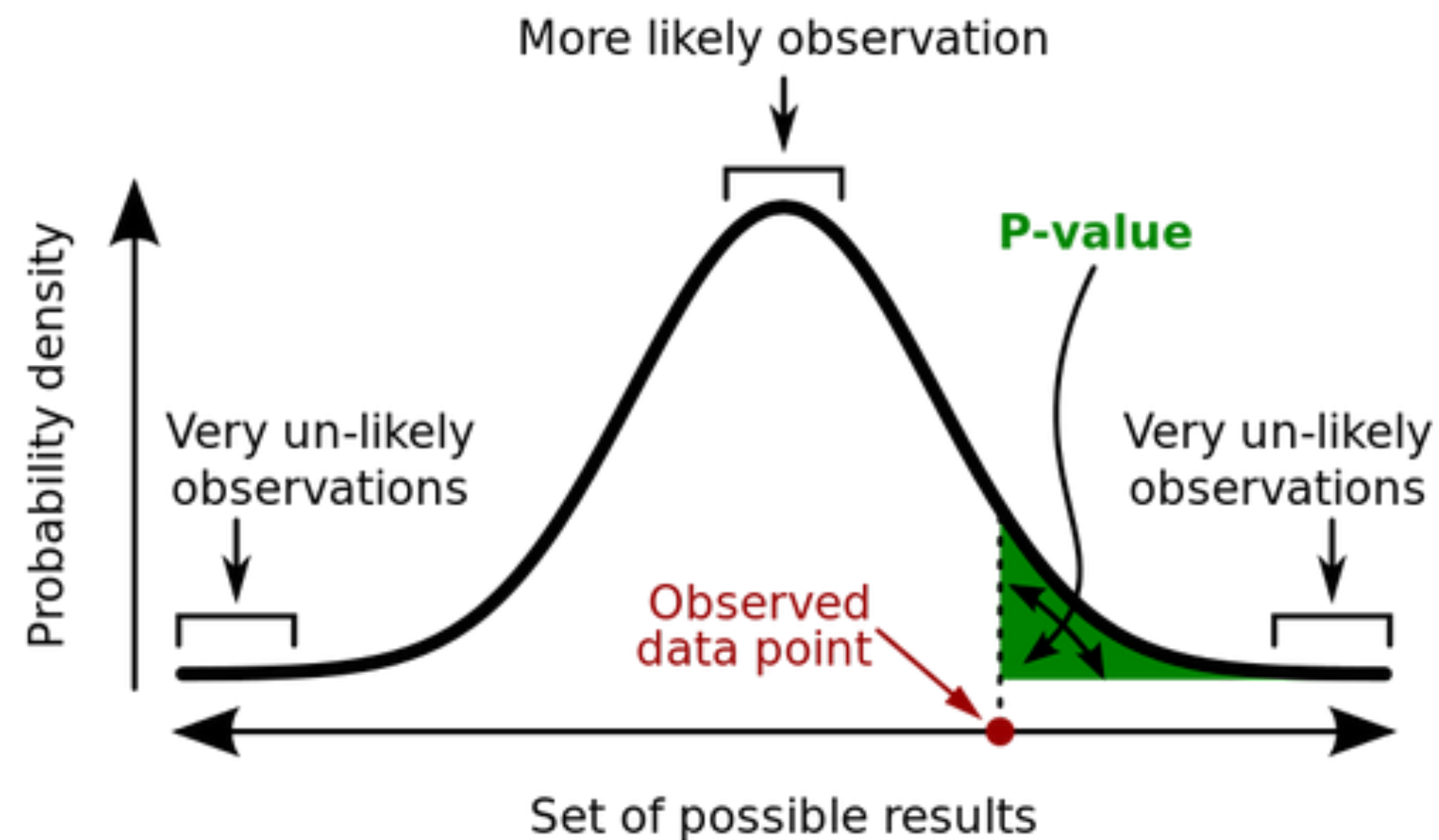
→ Difference between our estimate $\hat{\beta}_1$ and 0

→ Standard Error: a measure of how inaccurate our estimate $\hat{\beta}_1$ is at estimating β_1

The bigger the error, the smaller is t

Test statistic: t

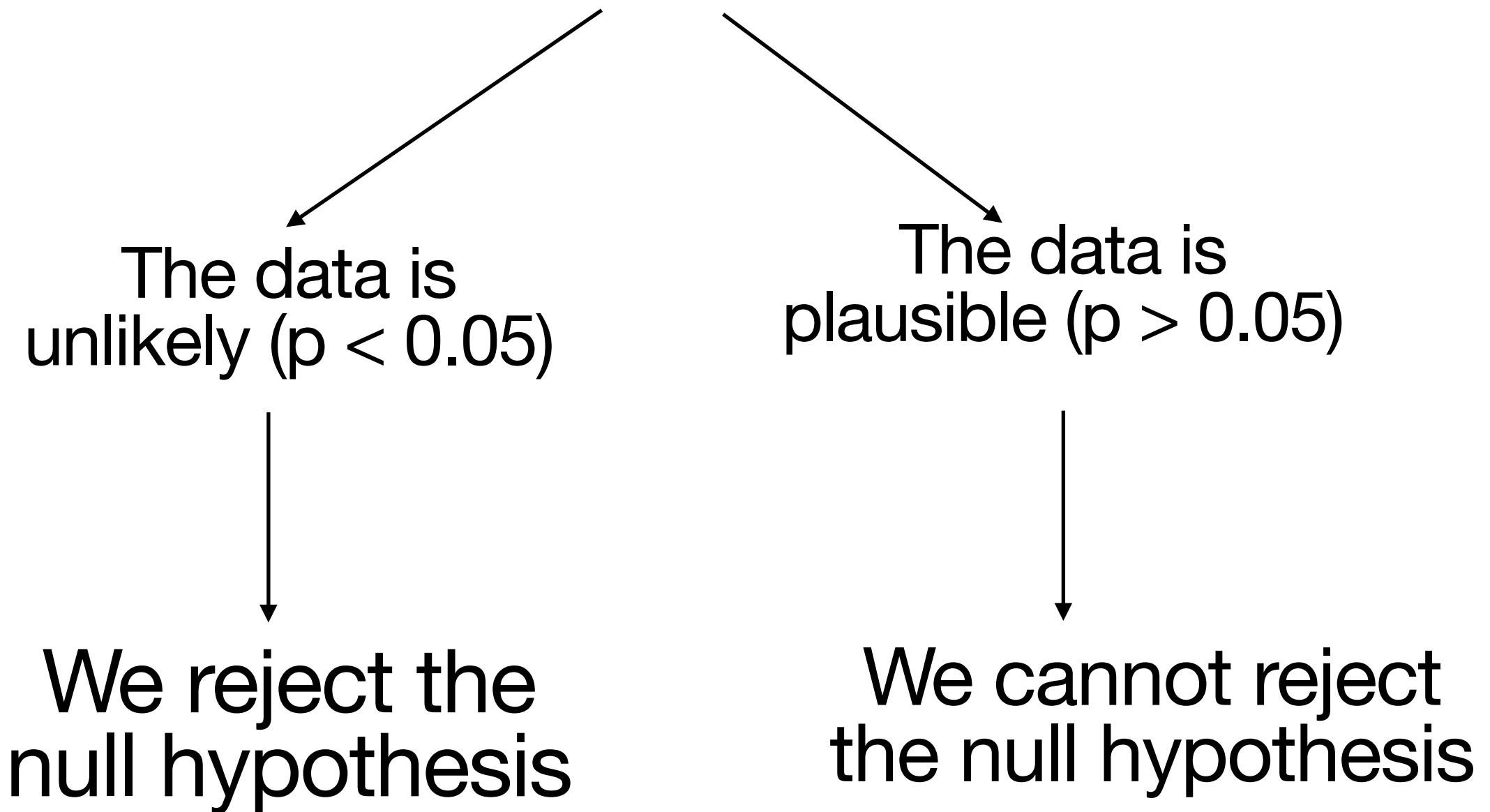
Assuming $\beta_1 = 0$, then the t-statistic should follow a well-known distribution: the t-distribution.



If our estimate is **too unlikely** under this distribution, then we can **reject the hypothesis** $\beta_1 = 0$

Frequentist hypothesis testing

Assuming the null hypothesis is true...



Problems with hypothesis testing

- Who decides what is “too unlikely”? ($p < 0.05$ is an arbitrary cutoff)
- Could a “rejected” model still be a plausible explanation of the data?
- If we “reject” a null model, are we necessarily “accepting” the alternative?
- What if there are multiple alternative models?

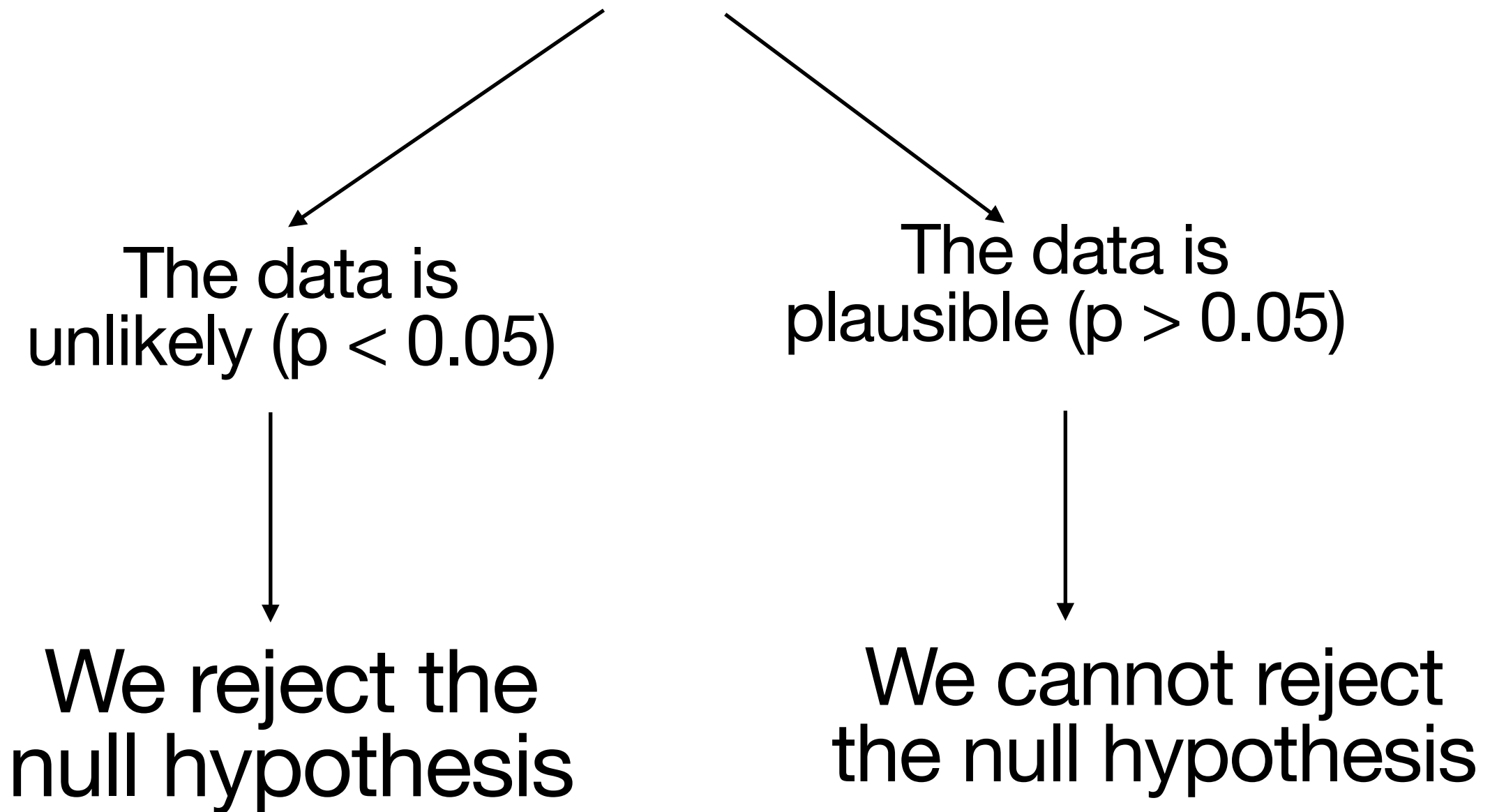
Problems with hypothesis testing

- Who decides what is “too unlikely”? ($p < 0.05$ is an arbitrary cutoff)
- Could a “rejected” model still be a plausible explanation of the data?
- If we “reject” a null model, are we necessarily “accepting” the alternative?
- What if there are multiple alternative models?



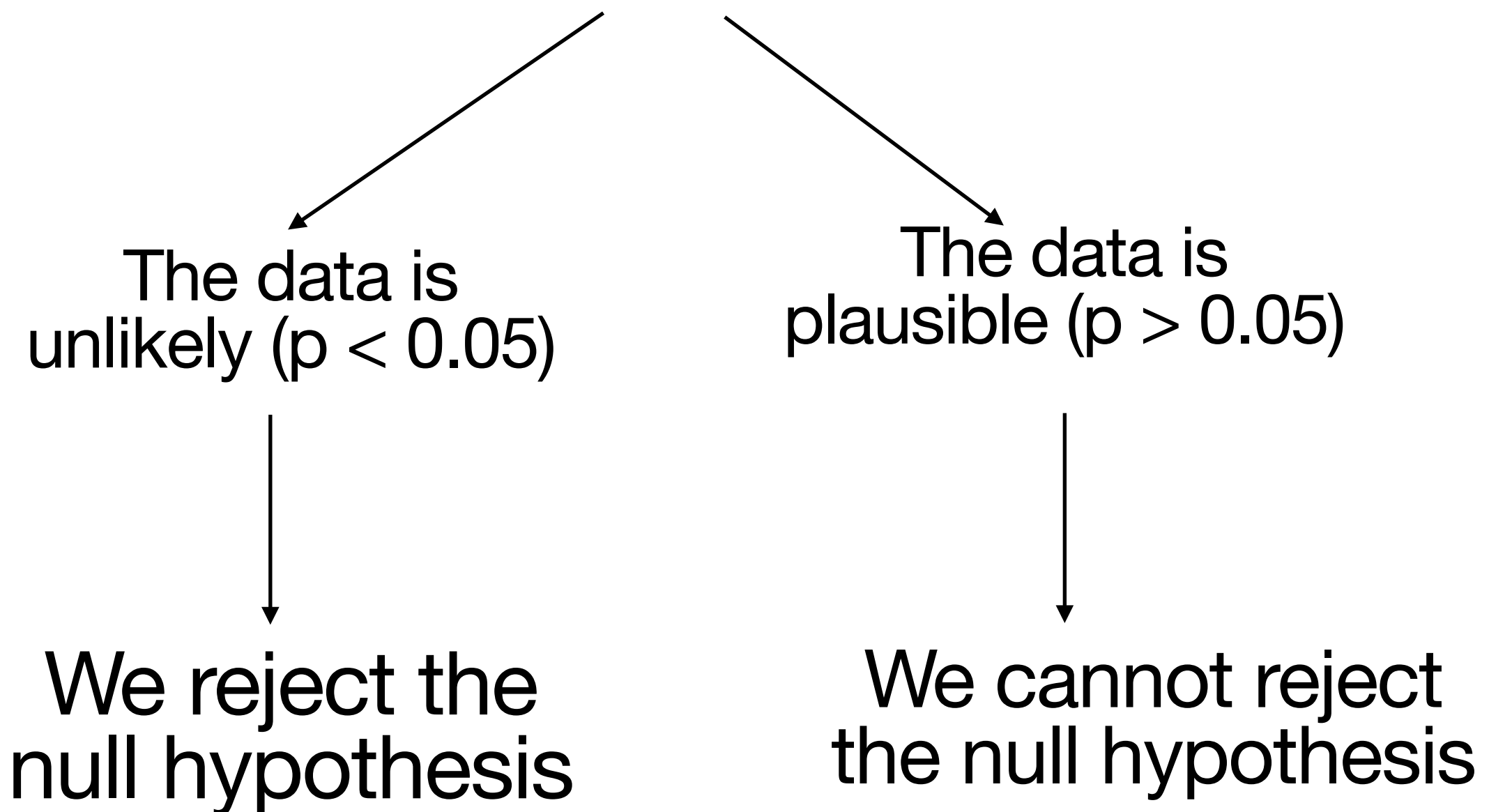
Frequentist hypothesis testing

Assuming the null hypothesis is true...



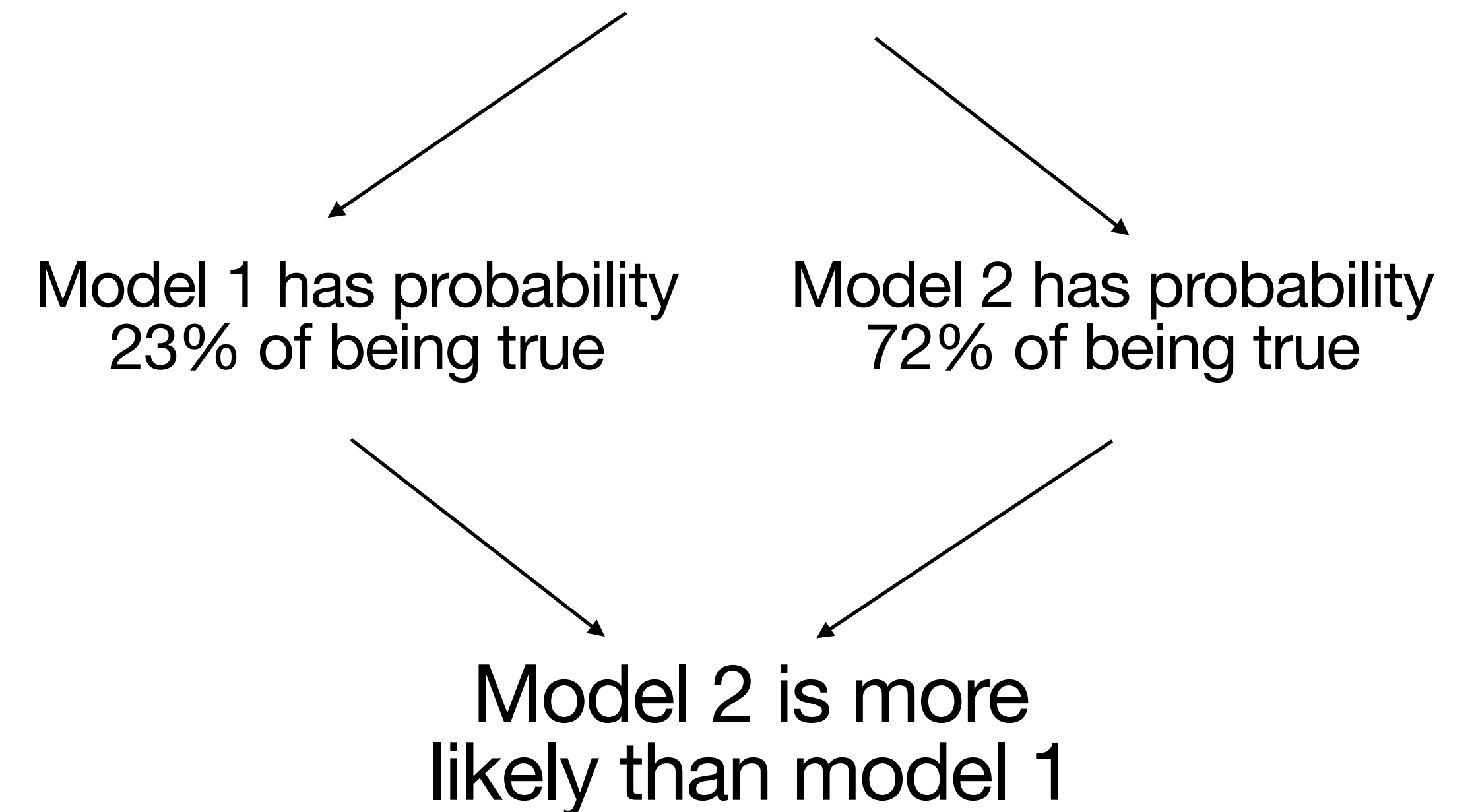
Frequentist hypothesis testing

Assuming the null hypothesis is true...



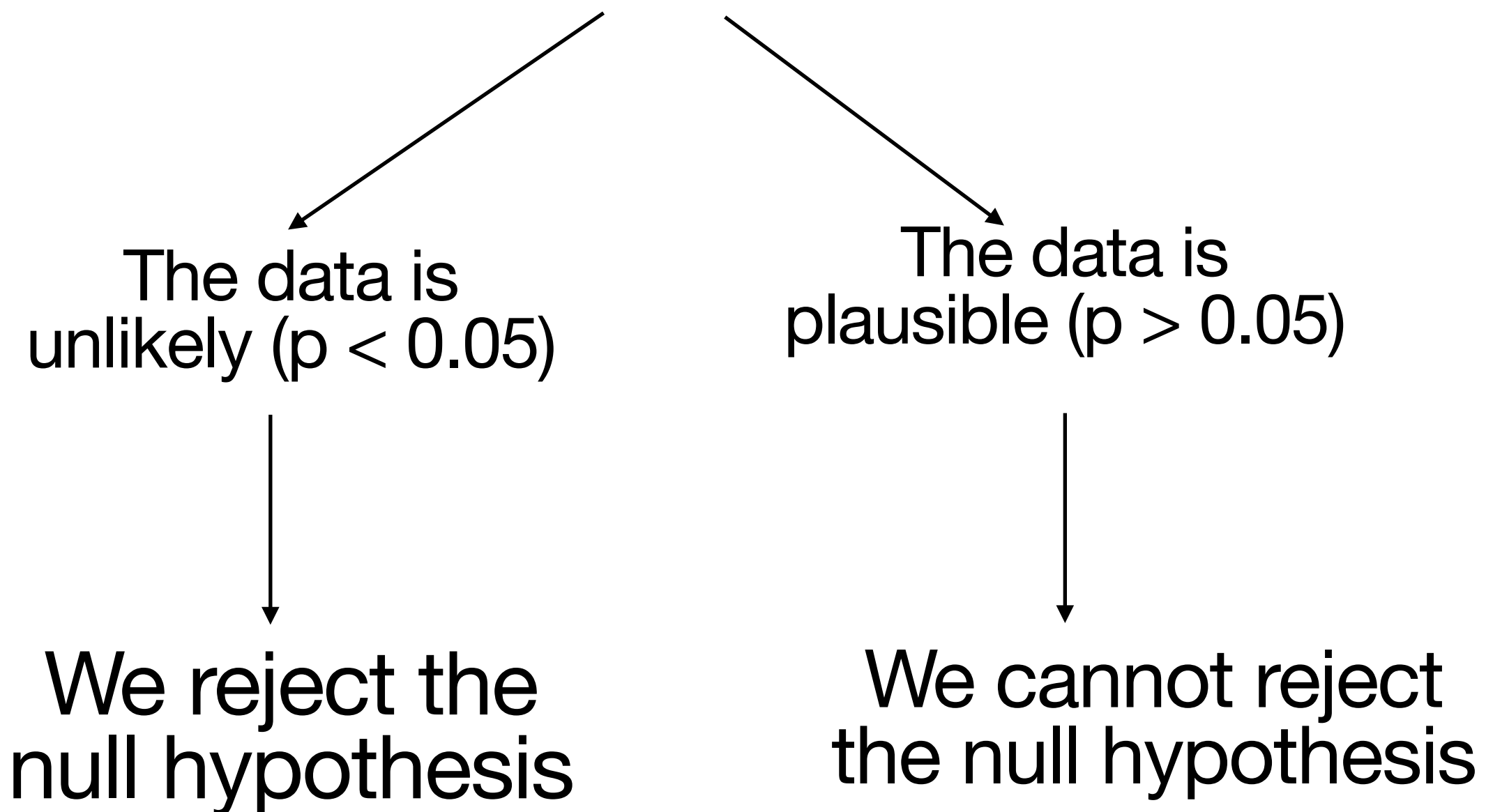
Bayesian model choice

Given the data...



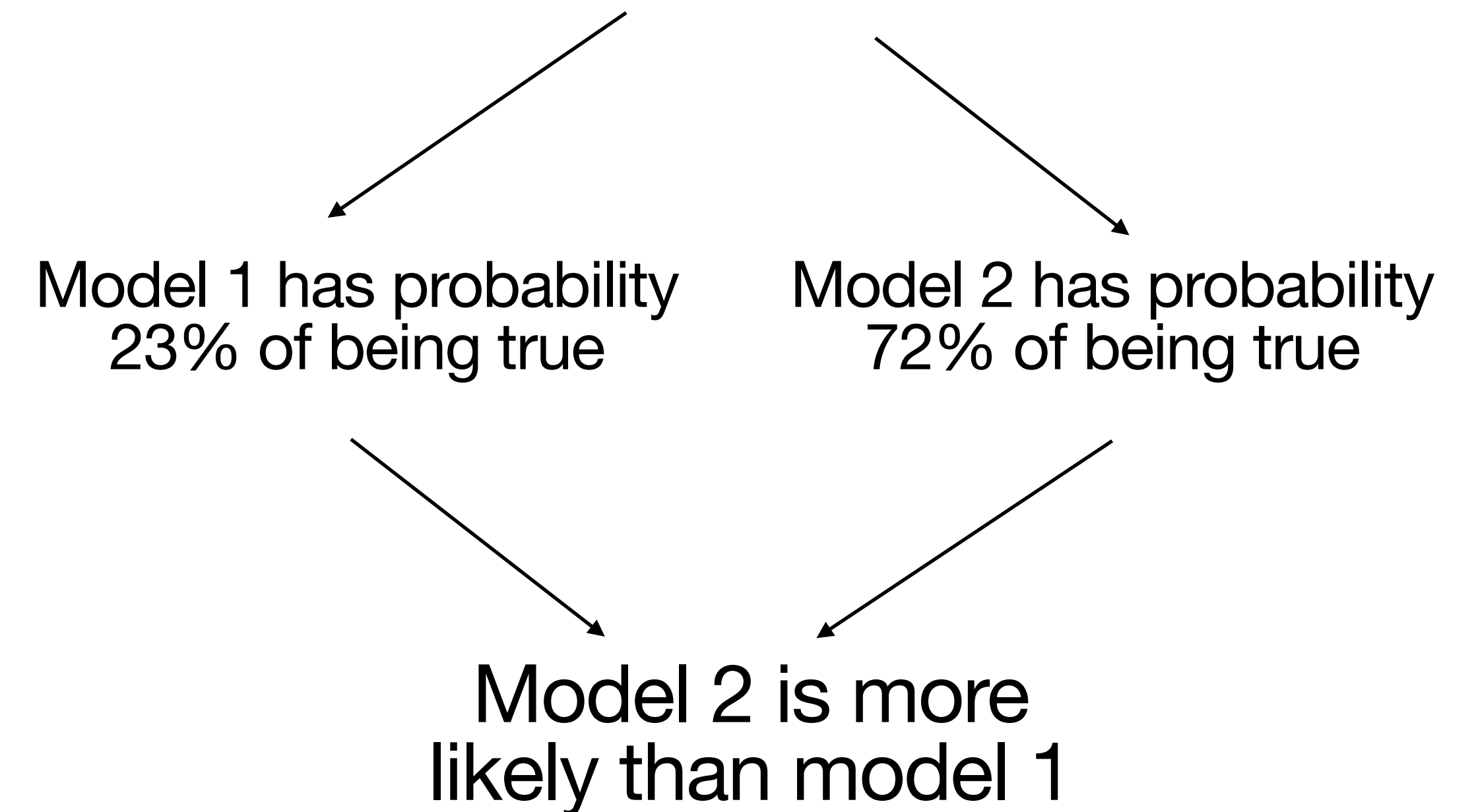
Frequentist hypothesis testing

Assuming the null hypothesis is true...



Bayesian model choice

Given the data...

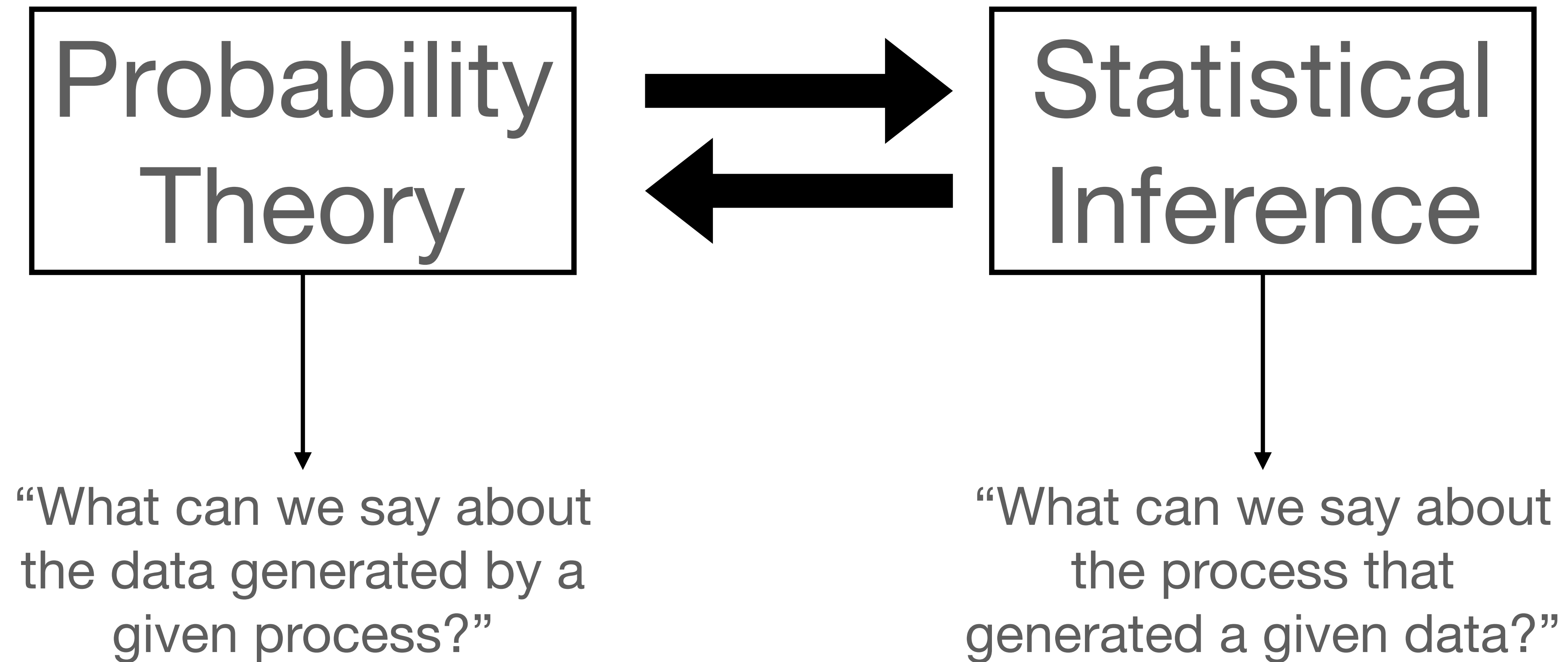


To find out more: take the course
“Advanced Topics in Data Analysis”!

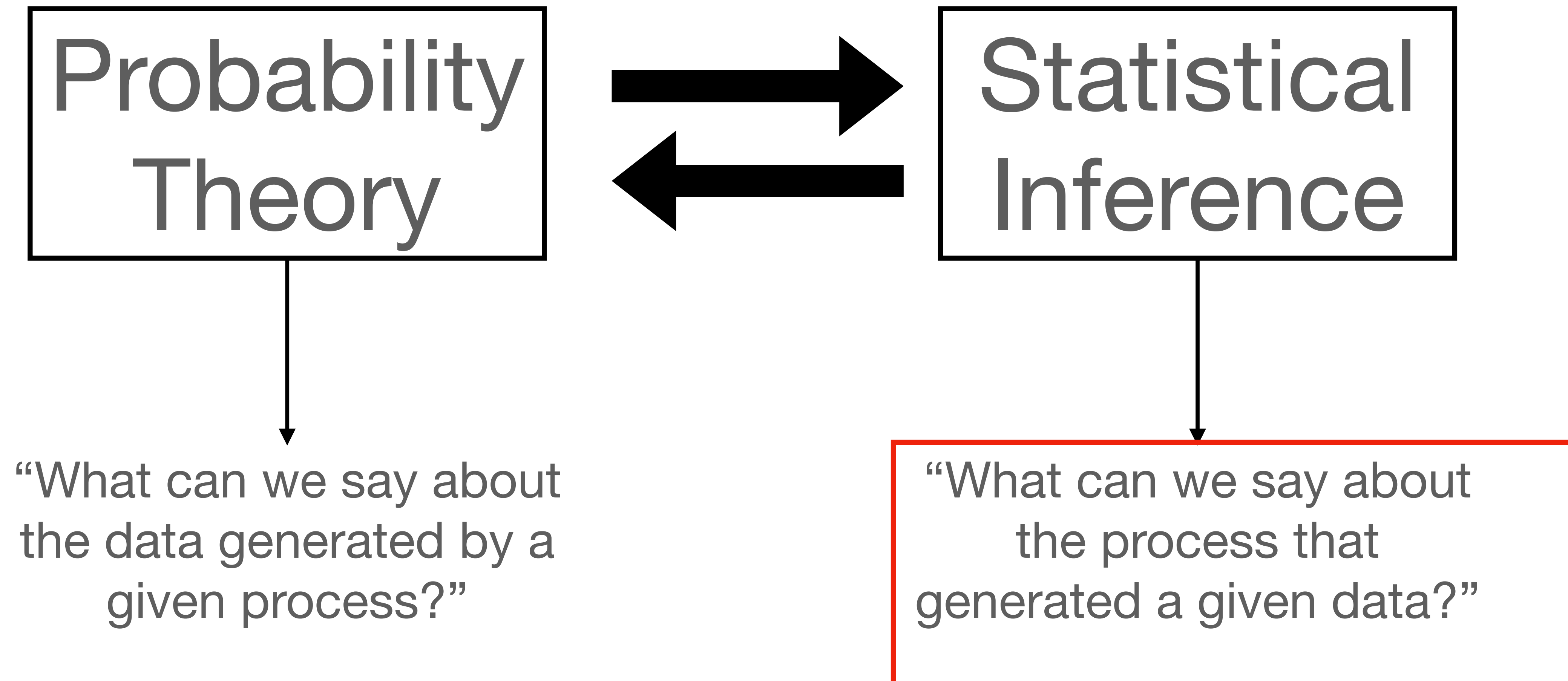
- **Fitting a simple linear model in R**
- **Interpreting a linear model in R**



The two sides of statistics



The two sides of statistics



Linear Regression: a probabilistic interpretation

Question: What can we say about Y after we've fitted our model?



Linear Regression: a probabilistic interpretation

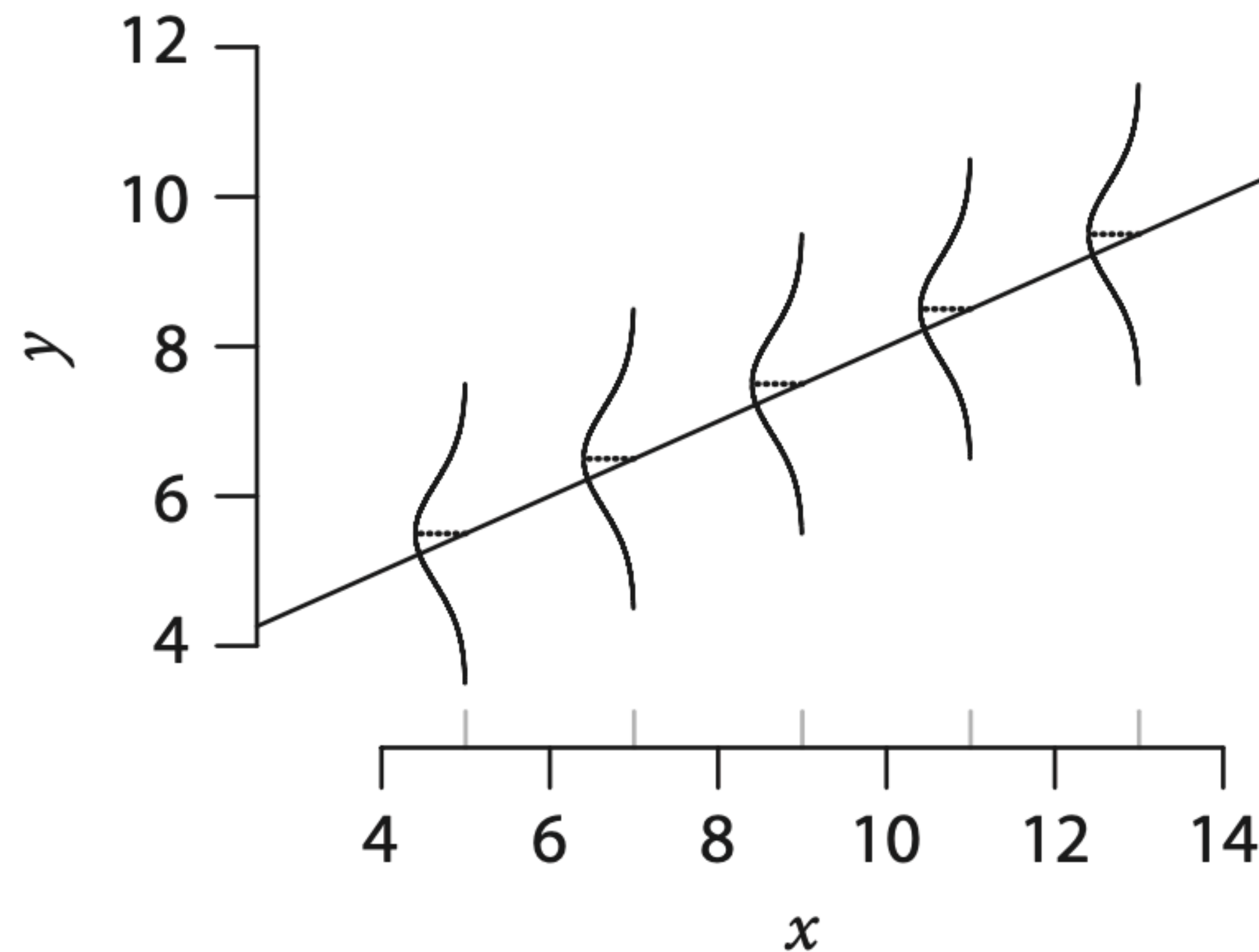
Question: What can we say about Y after we've fitted our model?



Answer: It depends on what we're willing to assume...

Linear Regression: a probabilistic interpretation

Let's treat the variable Y as a random variable with some (possibly unknown) distribution



“Linearity” assumption

$$E[\epsilon \mid X = x] = 0$$

“The expected value of the error term is zero, regardless of the value of X ”

“Linearity” assumption

$$E[\epsilon | X = x] = 0$$

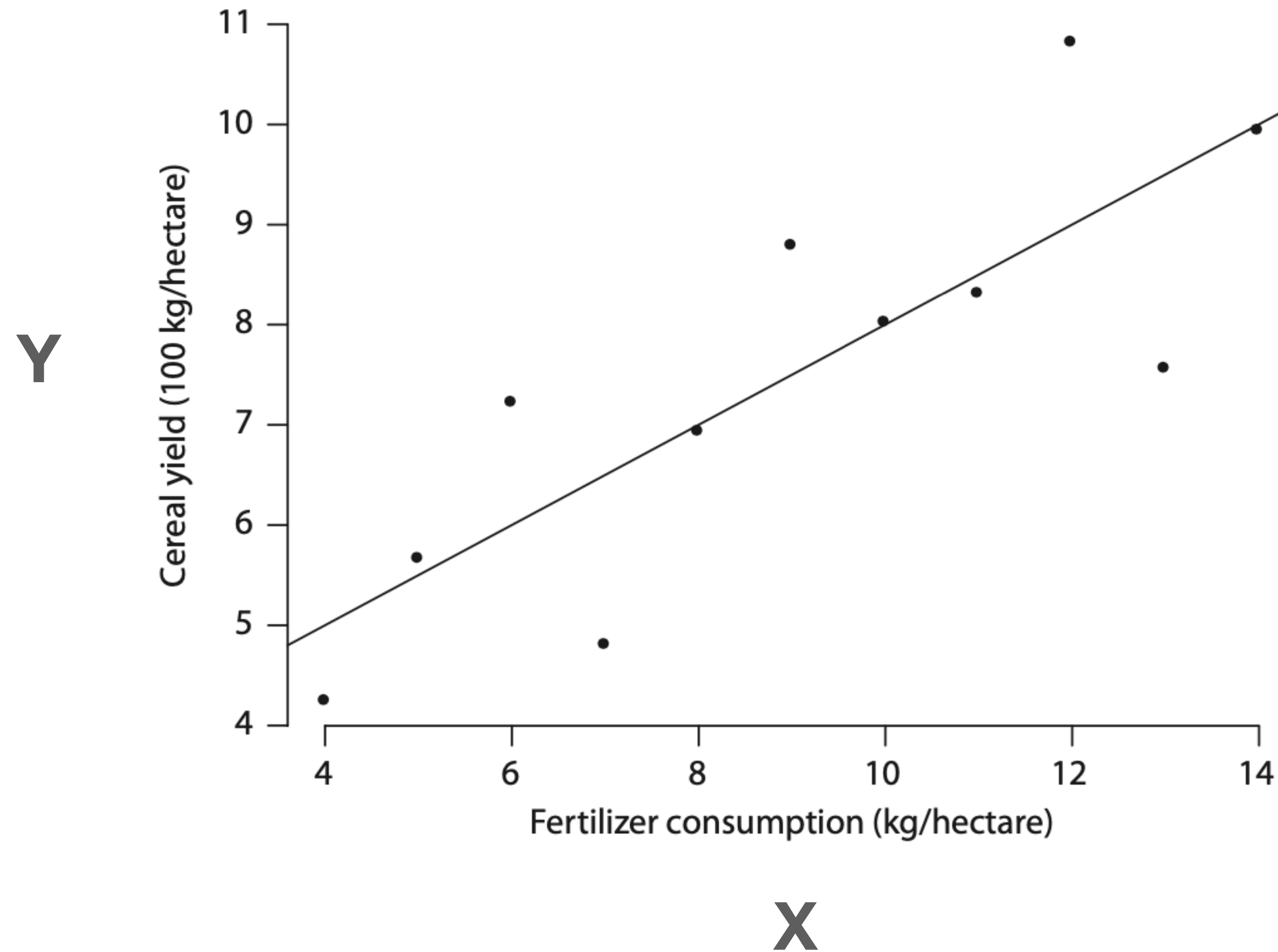
“The expected value of the error term is zero, regardless of the value of X ”

Then:

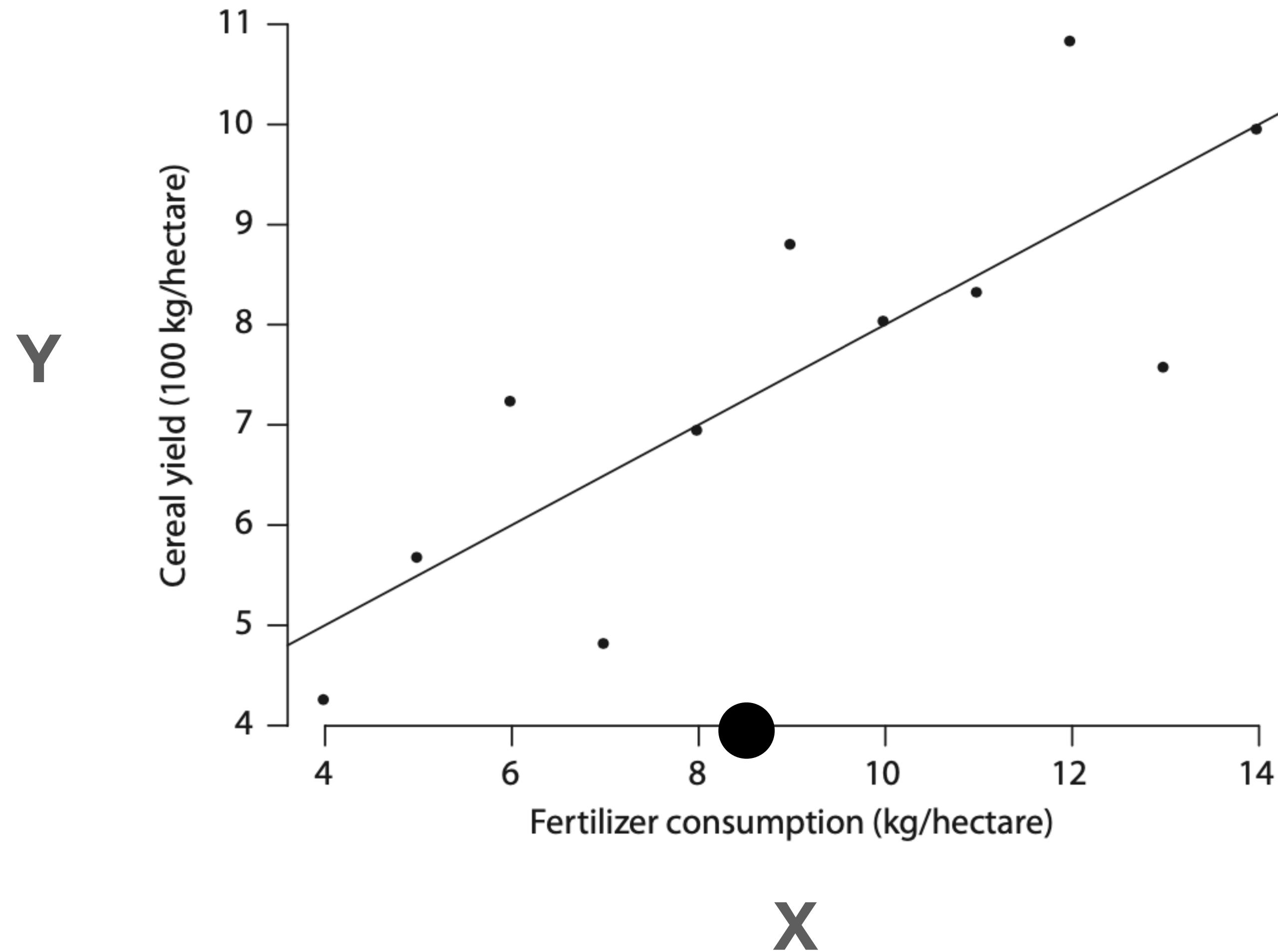
$$E[Y | X = x] = \beta_0 + \beta_1 x$$

“Given a value of X , the expected value of Y is a linear function of that value of X ”

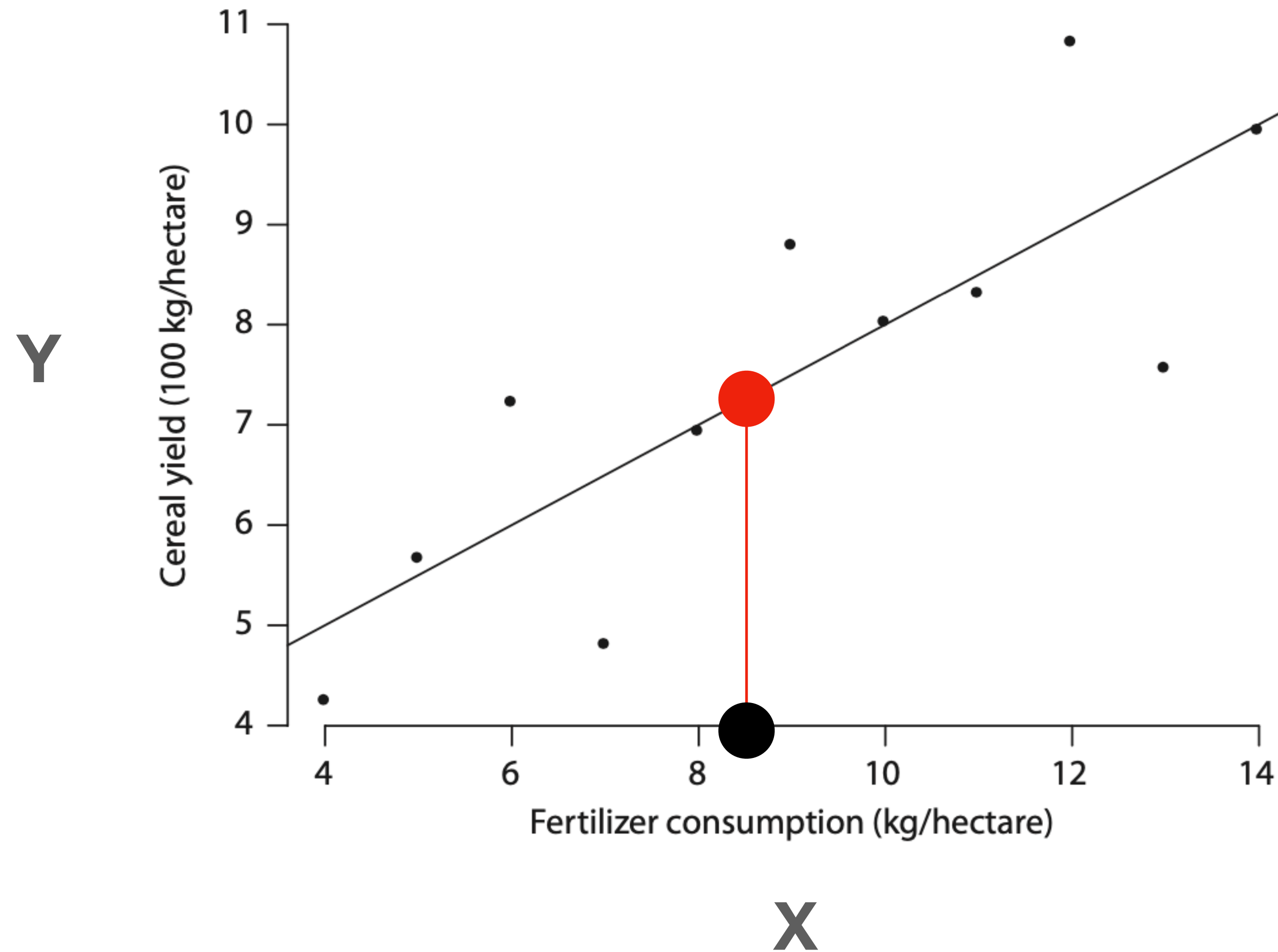
$E[Y|X=x]$ is the best linear predictor for $X=x$



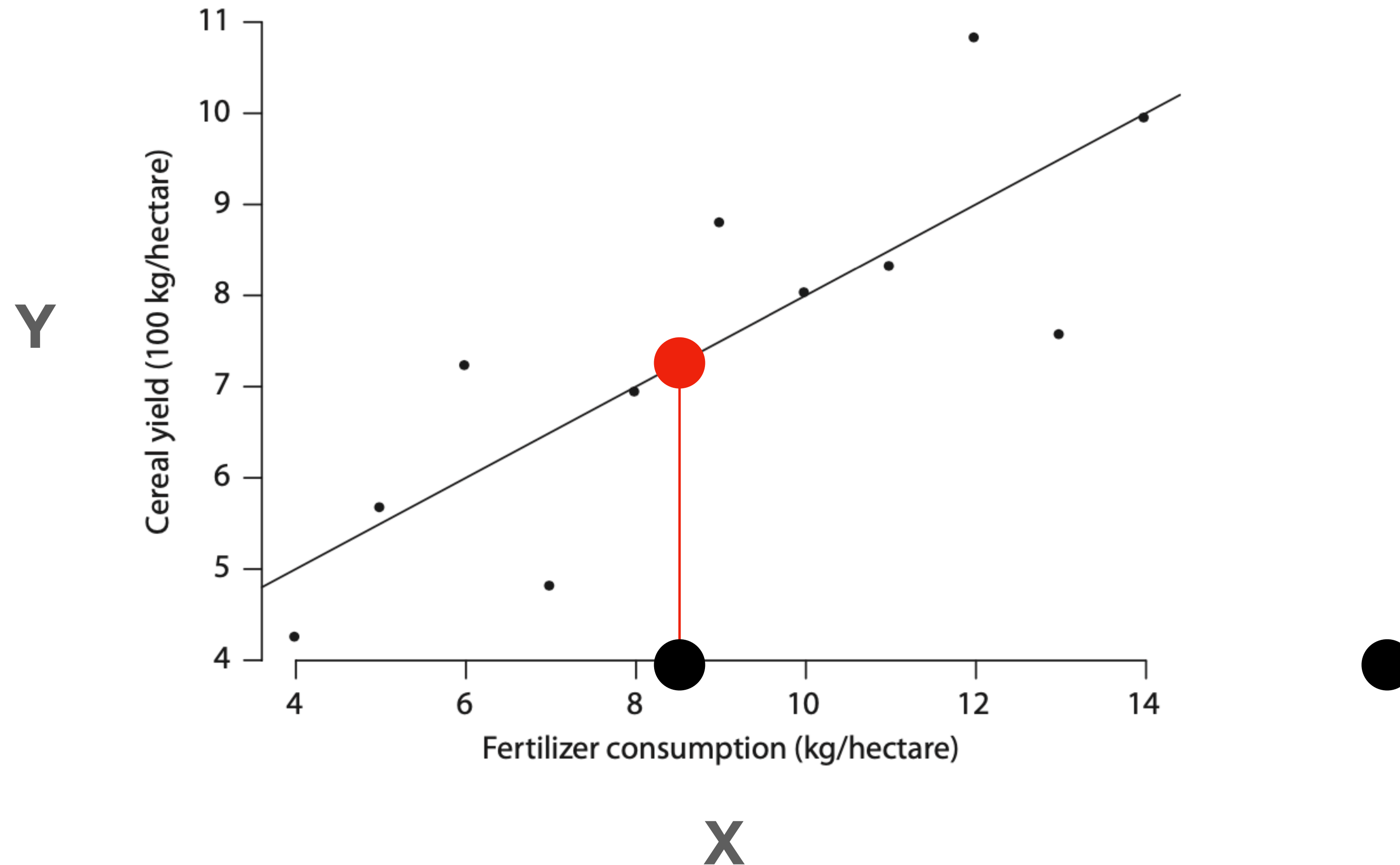
$E[Y|X=x]$ is the best linear predictor for $X=x$



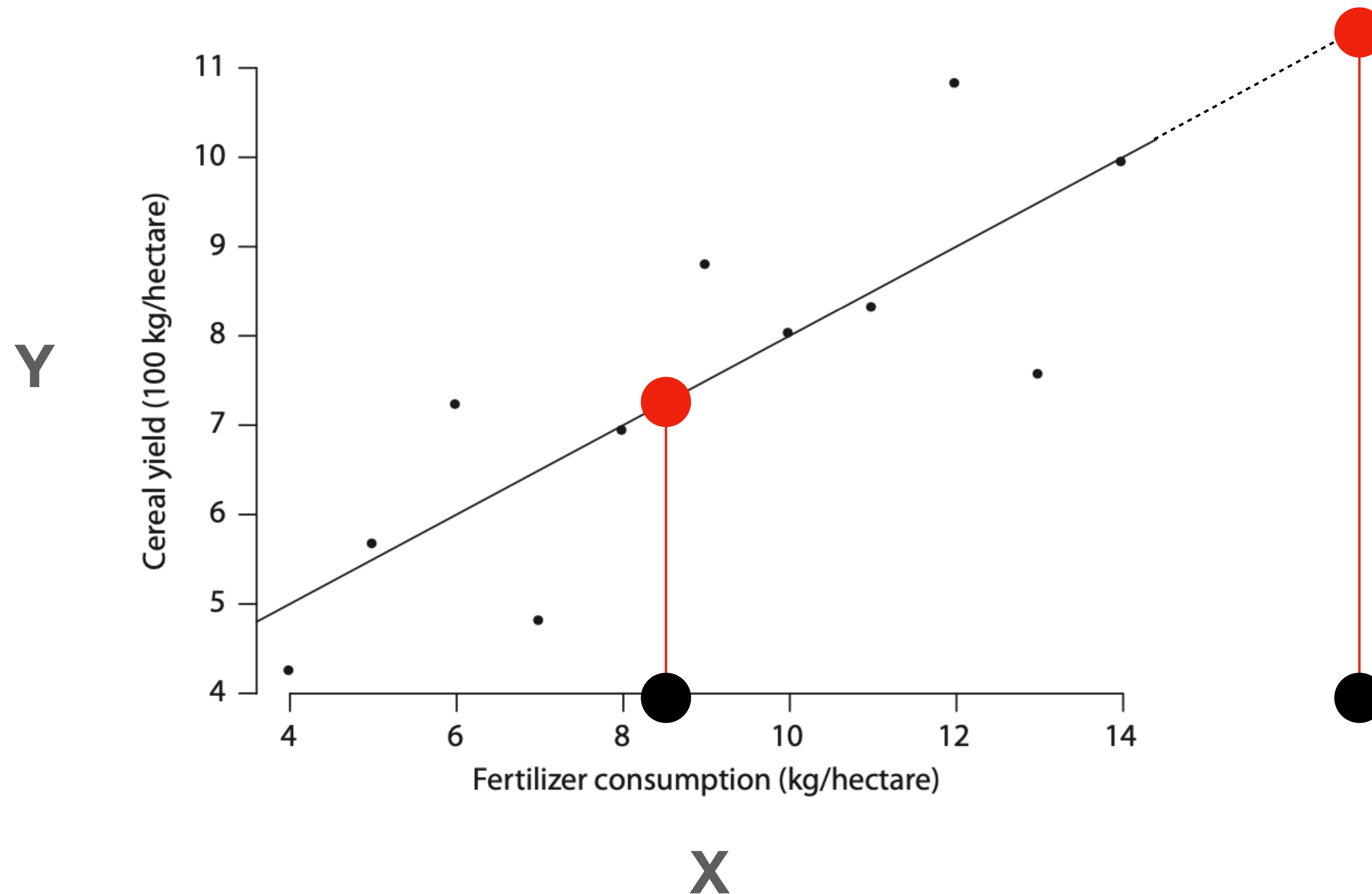
$E[Y|X=x]$ is the best linear predictor for $X=x$



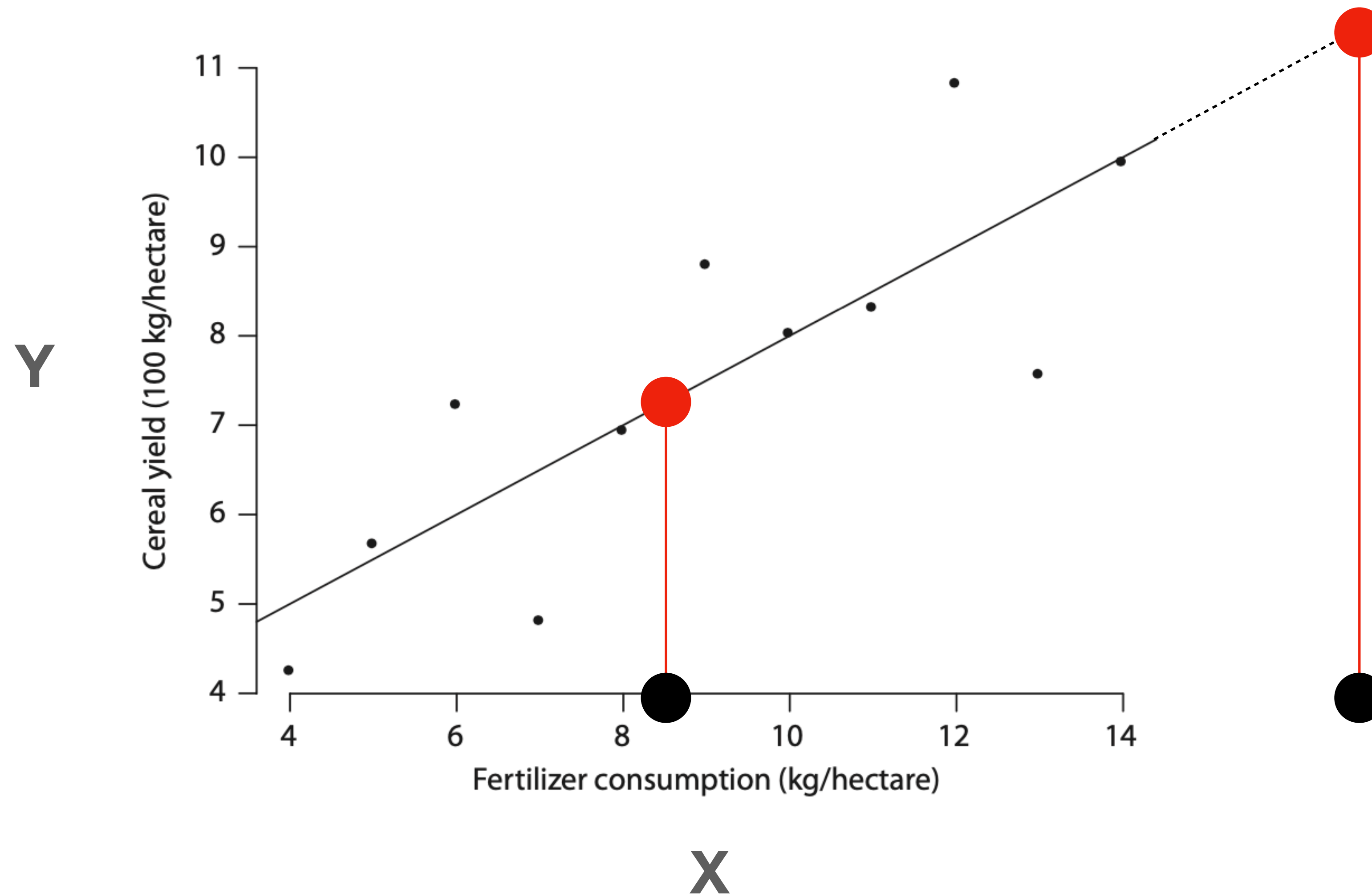
$E[Y|X=x]$ is the best linear predictor for $X=x$



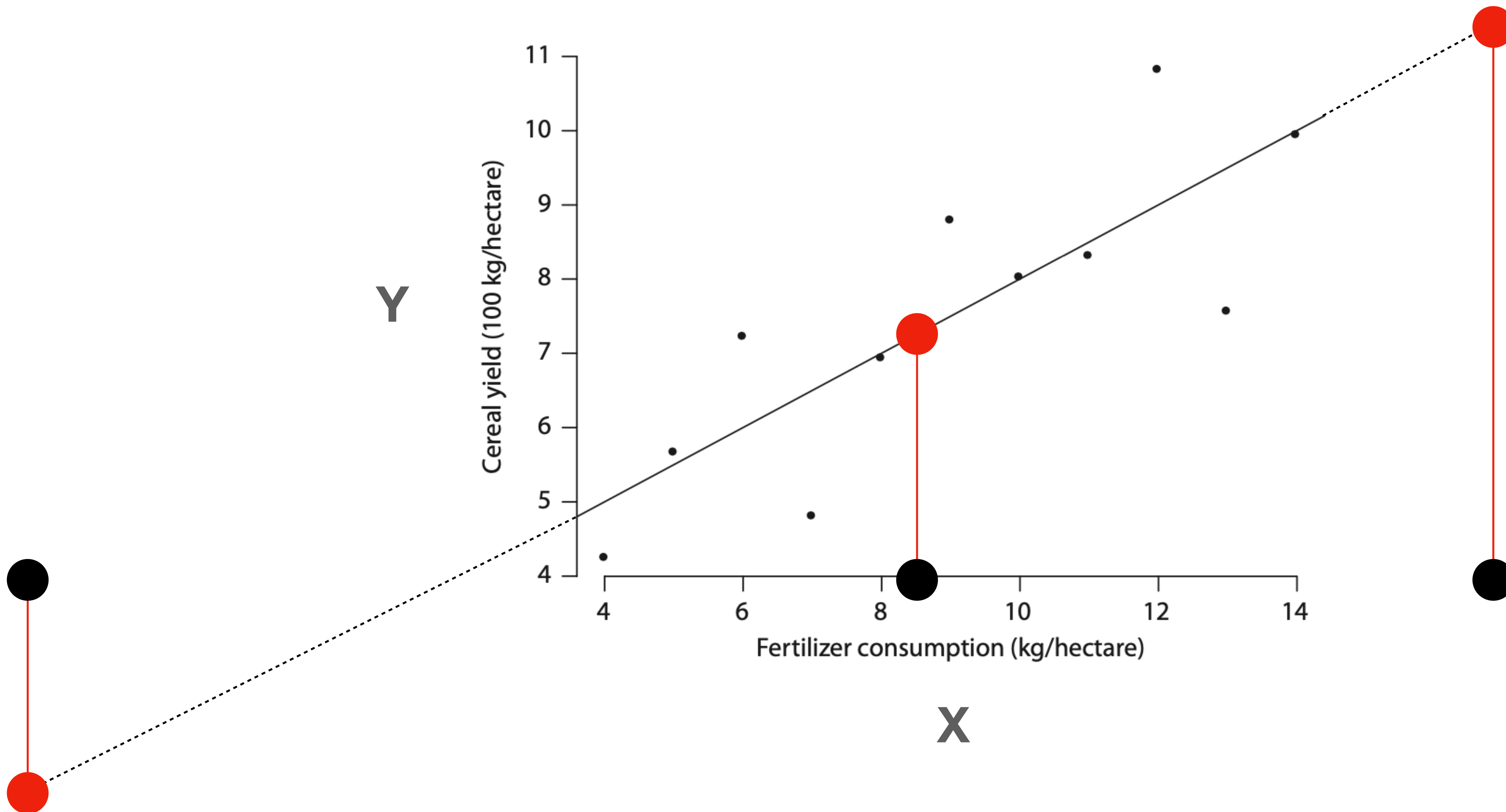
$E[Y|X=x]$ is the best linear predictor for $X=x$



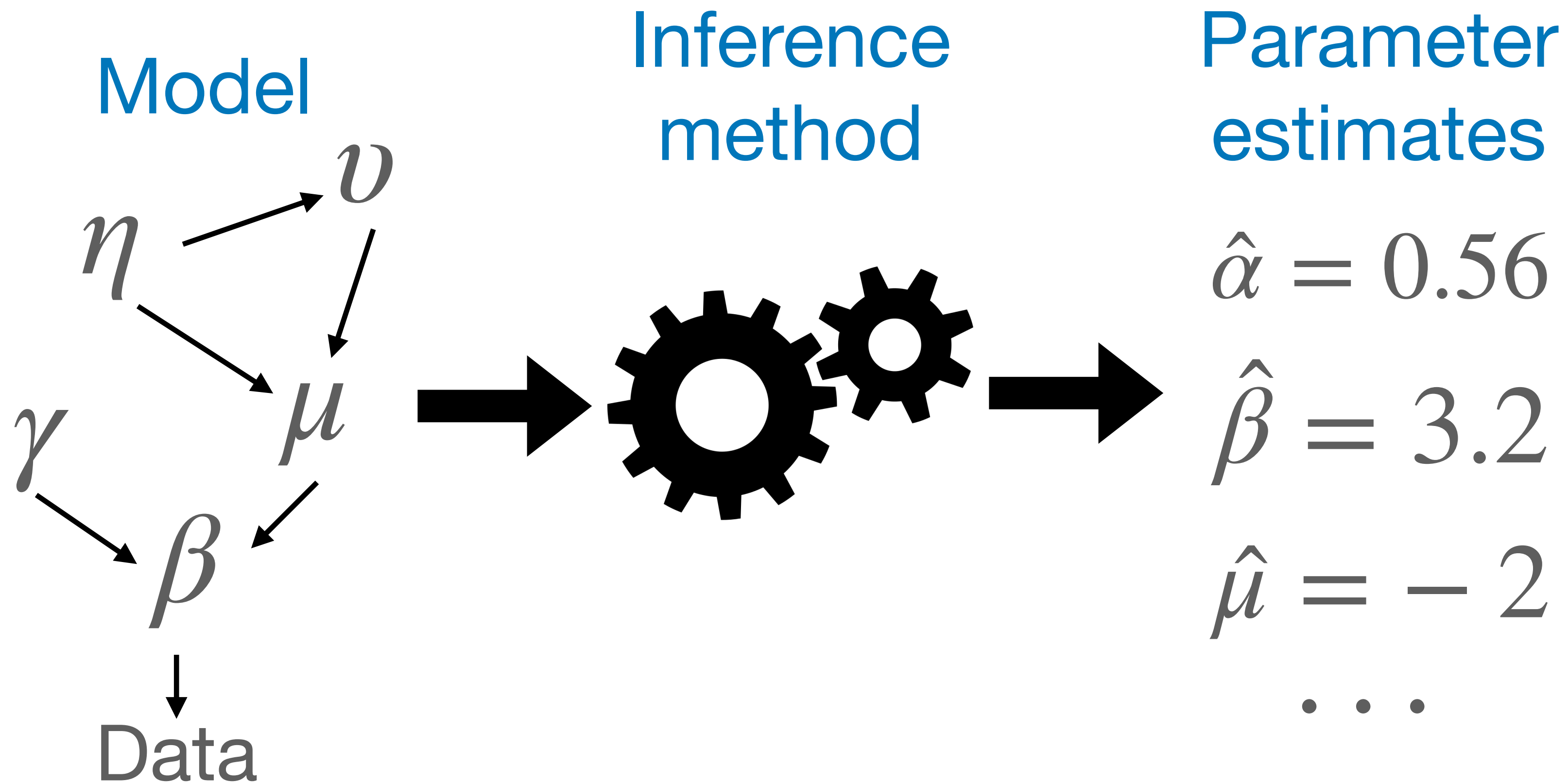
$E[Y|X=x]$ is the best linear predictor for $X=x$



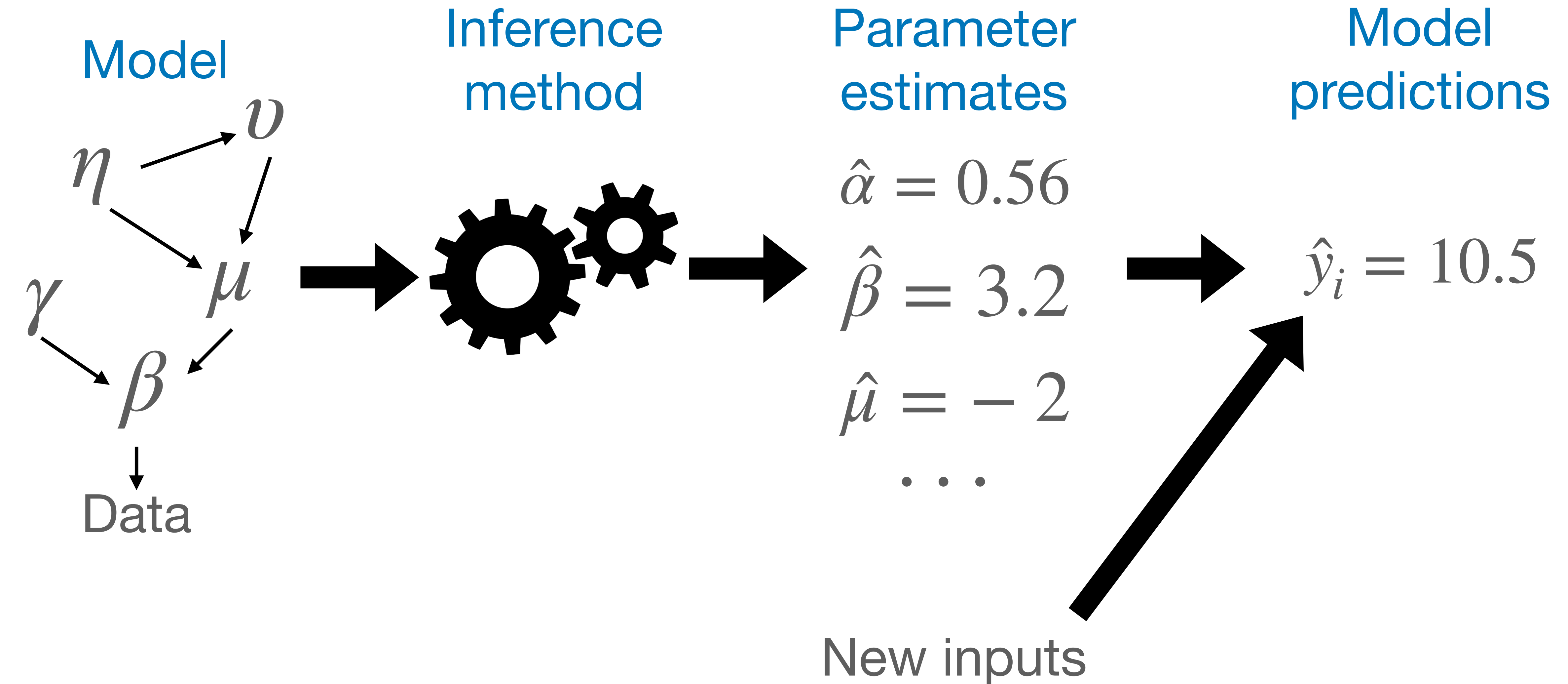
$E[Y|X=x]$ is the best linear predictor for $X=x$



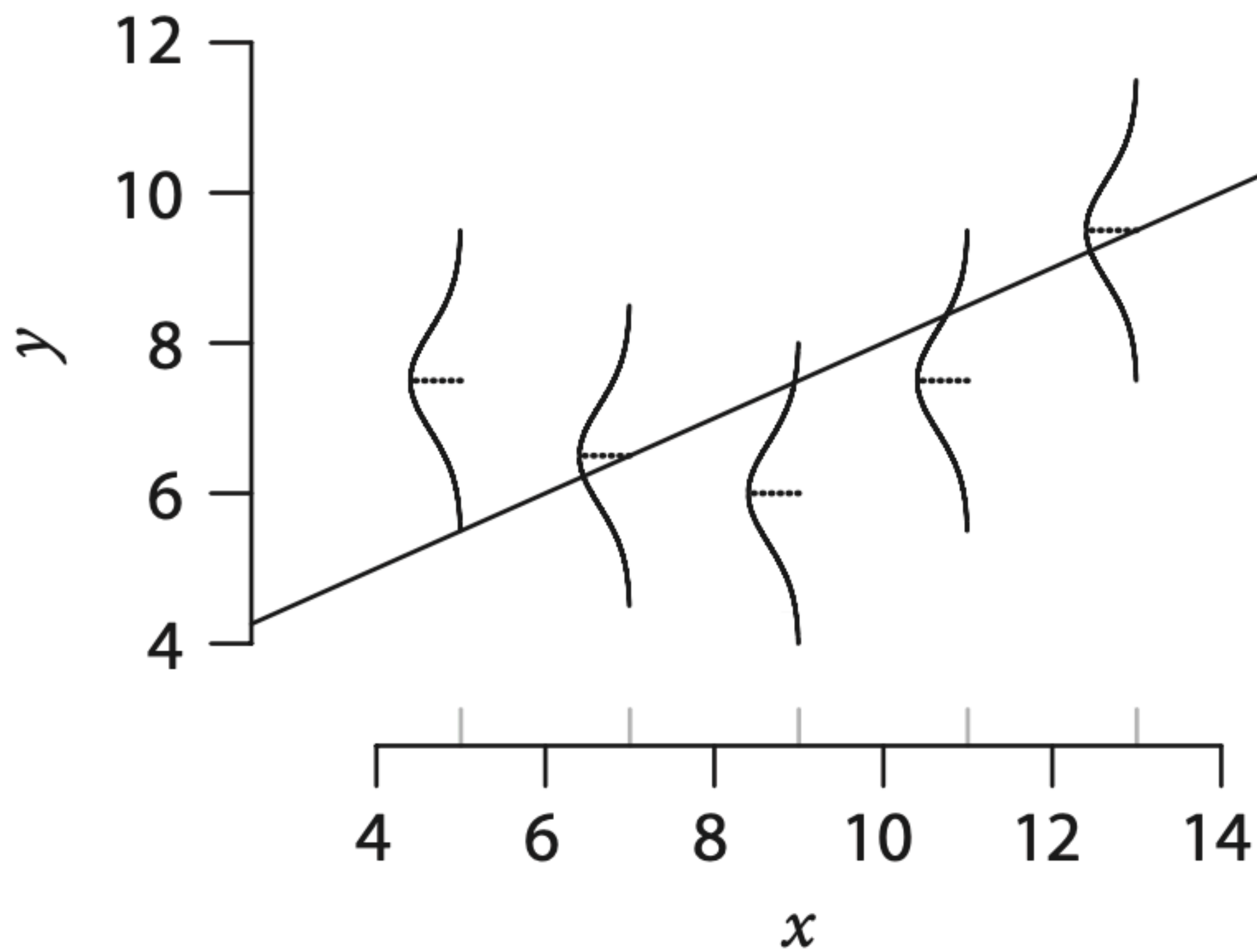
Inference vs. Prediction



Inference vs. Prediction



Linearity violated



“Homoscedasticity” assumption

$$\text{Var}[\epsilon \mid X = x] = \sigma_{\epsilon}^2$$

“The variance of the error is a constant, regardless of the value of X ”

“Homoscedasticity” assumption

$$\text{Var}[\epsilon | X = x] = \sigma_{\epsilon}^2$$

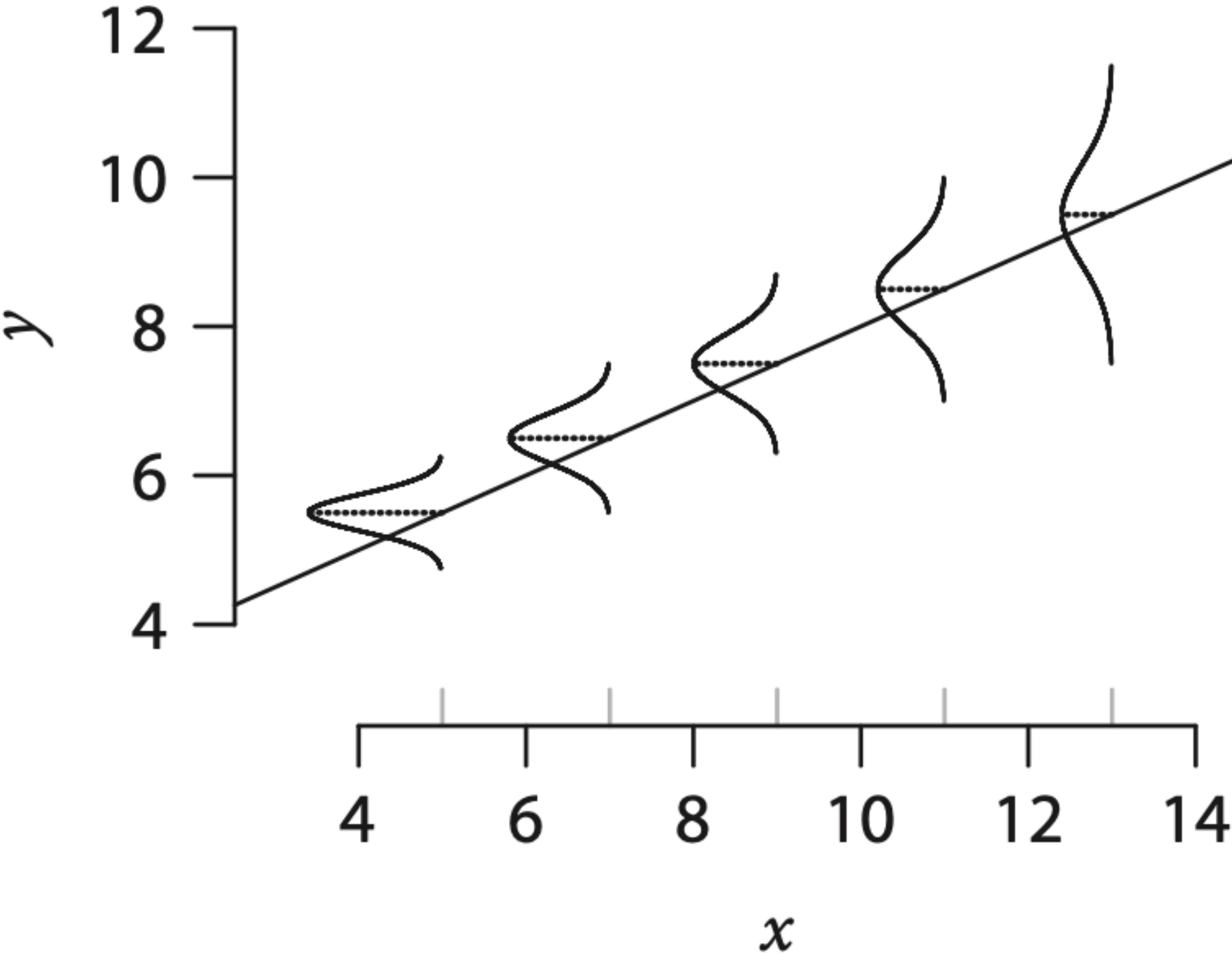
“The variance of the error is a constant, regardless of the value of X”

Then:

$$\text{Var}[Y | X = x] = \sigma_{\epsilon}^2$$

“The variance of Y is a constant (equal to the variance of the error), regardless of the value of X”

Homoscedasticity violated



“Normality” assumption

$$\epsilon_i | (X = x) \sim \text{Normal}(0, \sigma_\epsilon^2)$$

and all ϵ_i are independent of each other

“The error is normally distributed, regardless of the value of X ”

“Normality” assumption

$$\epsilon_i | (X = x) \sim \text{Normal}(0, \sigma_\epsilon^2)$$

and all ϵ_i are independent of each other

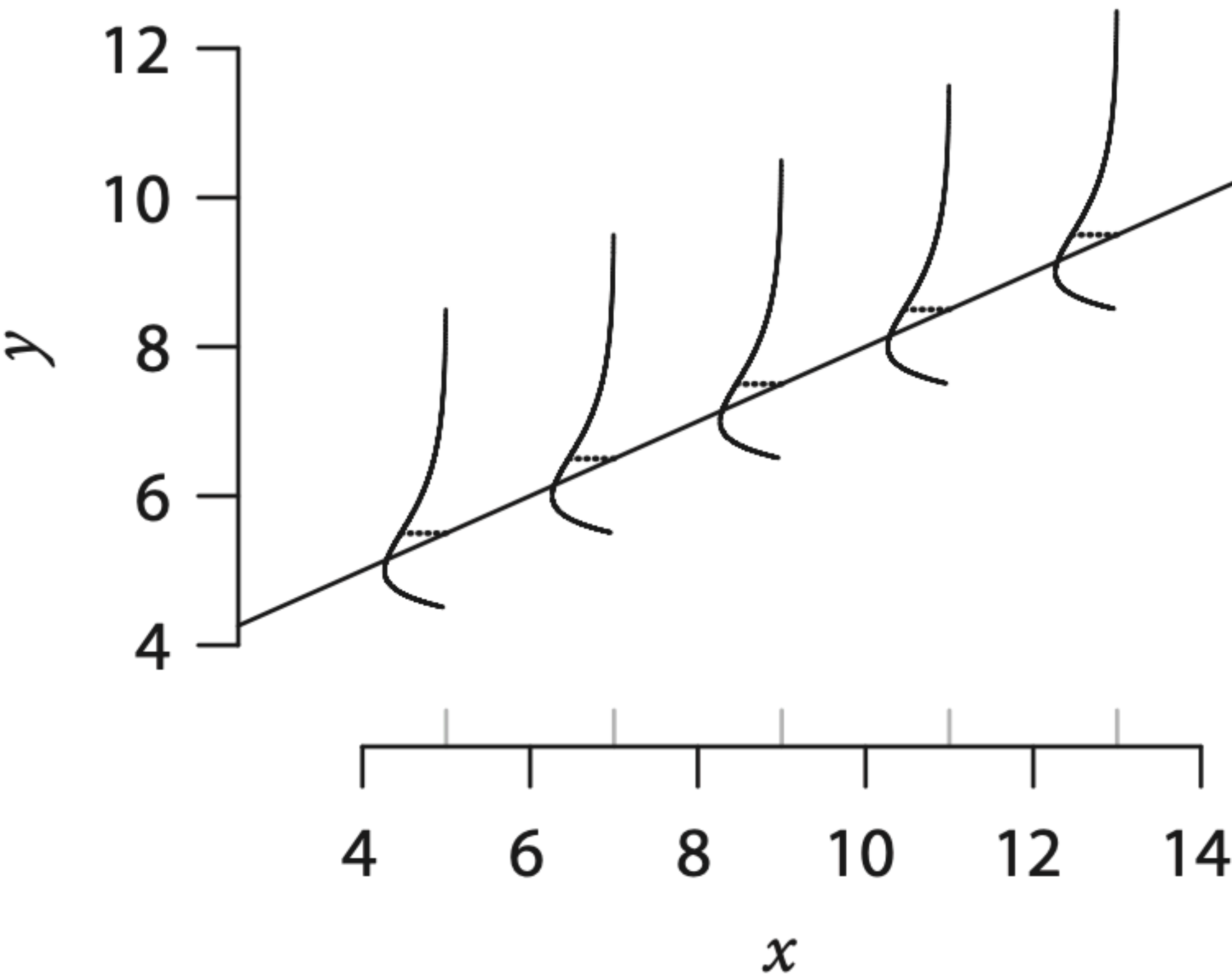
“The error is normally distributed, regardless of the value of X”

Then:

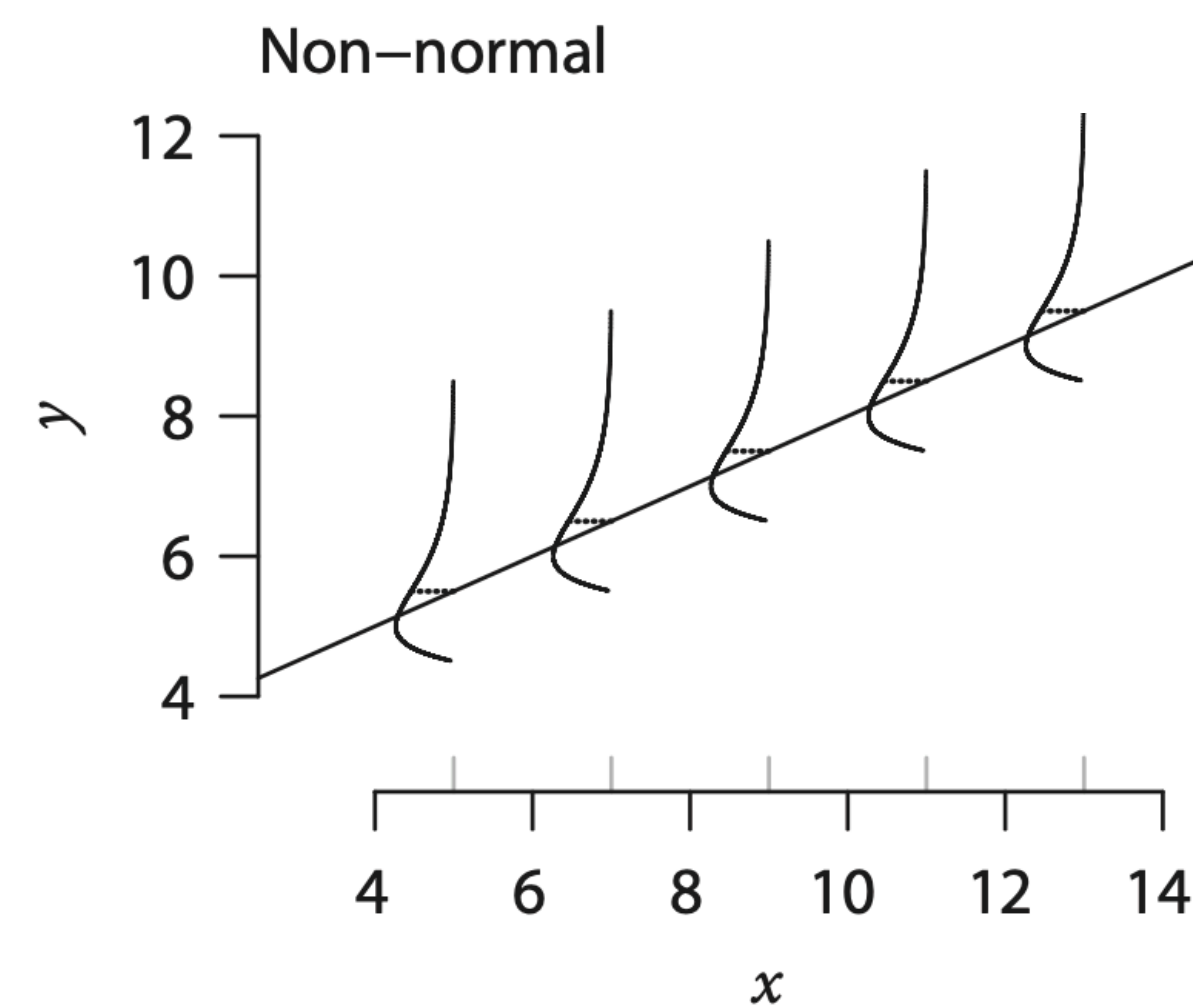
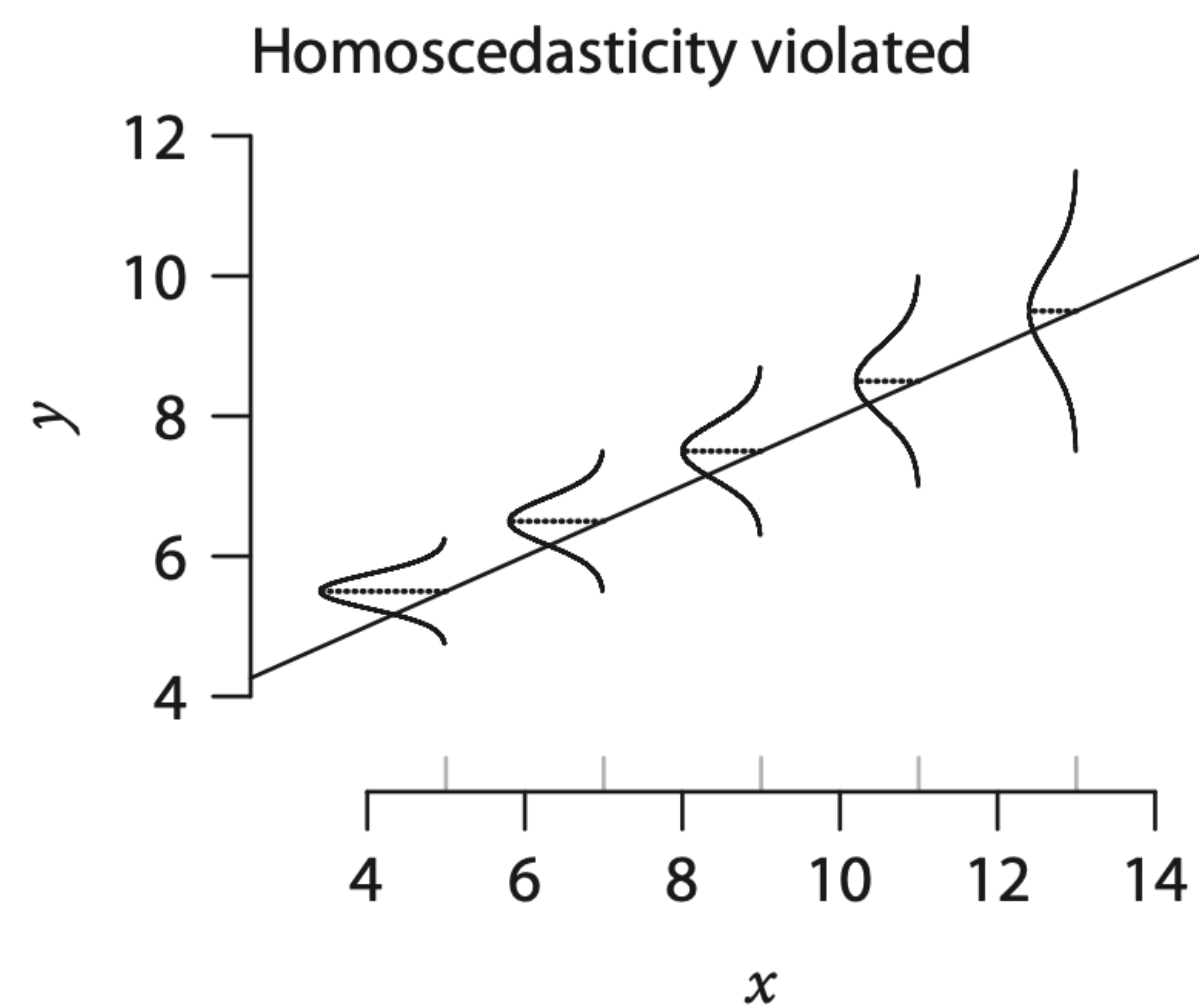
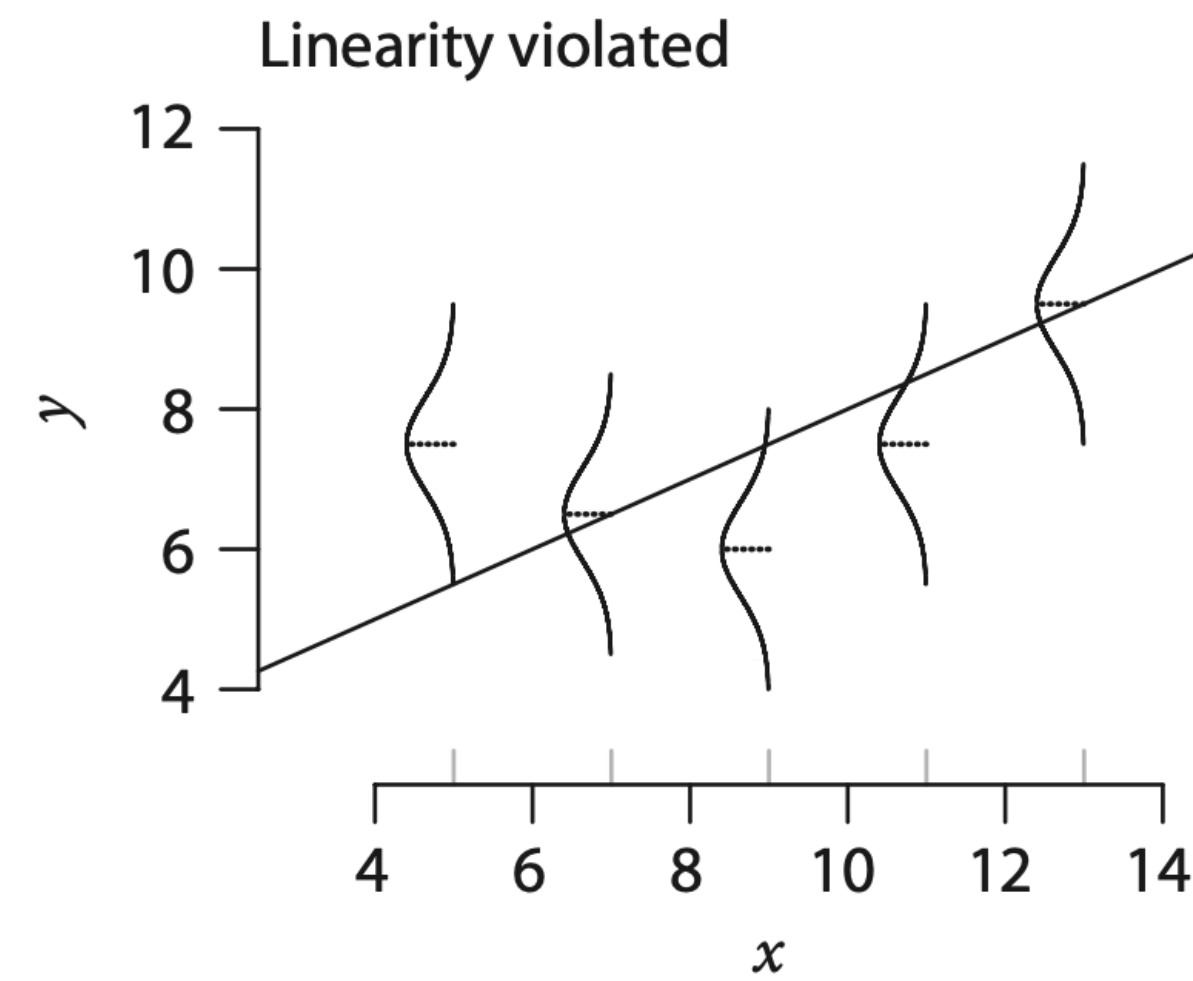
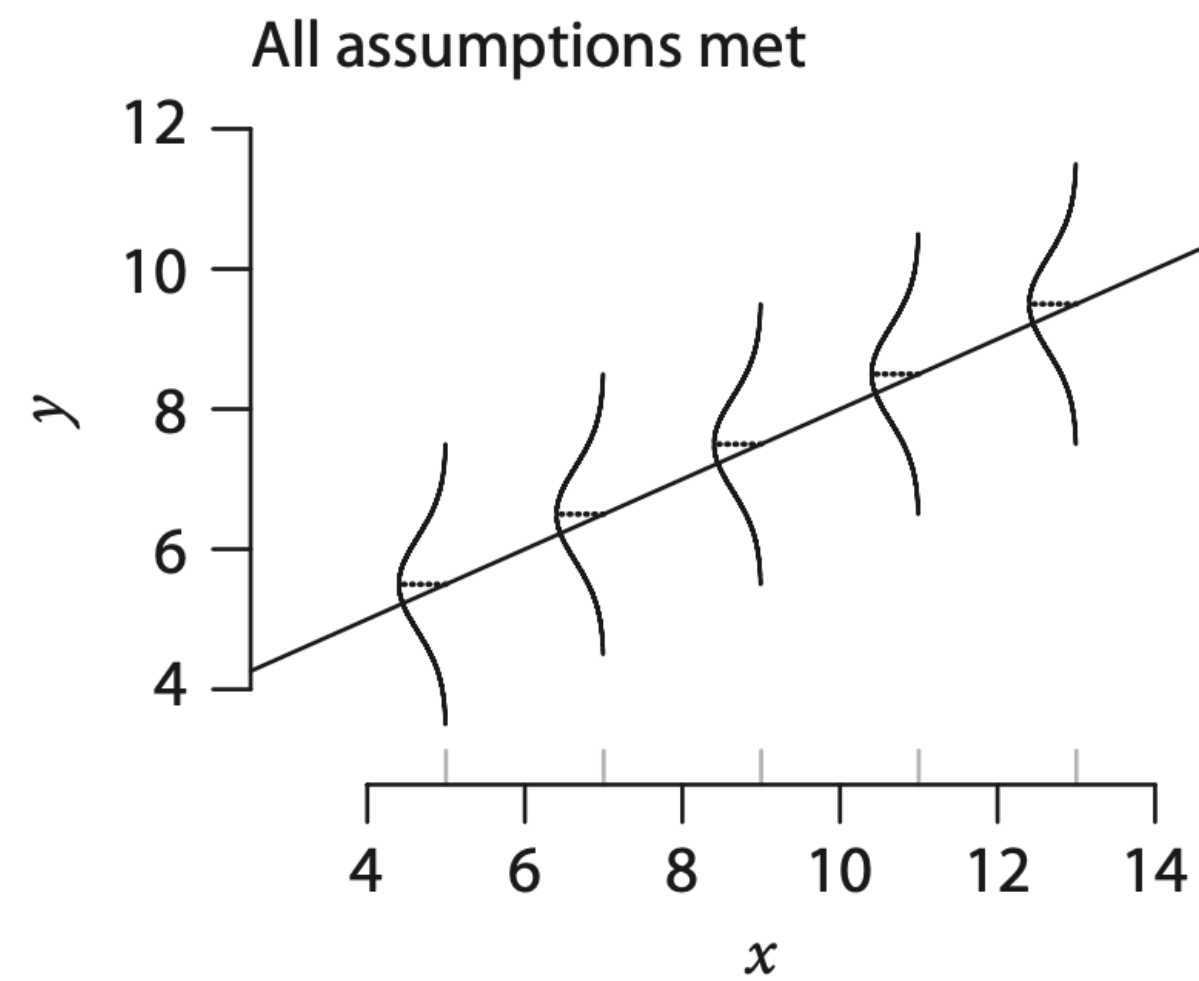
$$Y | (X = x) \sim \text{Normal}(\beta_0 + \beta_1 x, \sigma_\epsilon^2)$$

“Y is normally distributed, with mean equal to a linear function of the value of X”

Normality violated



Violations of assumptions



**That's it for simple linear
regression today, but:**

A zoo of models...

Simple linear regression

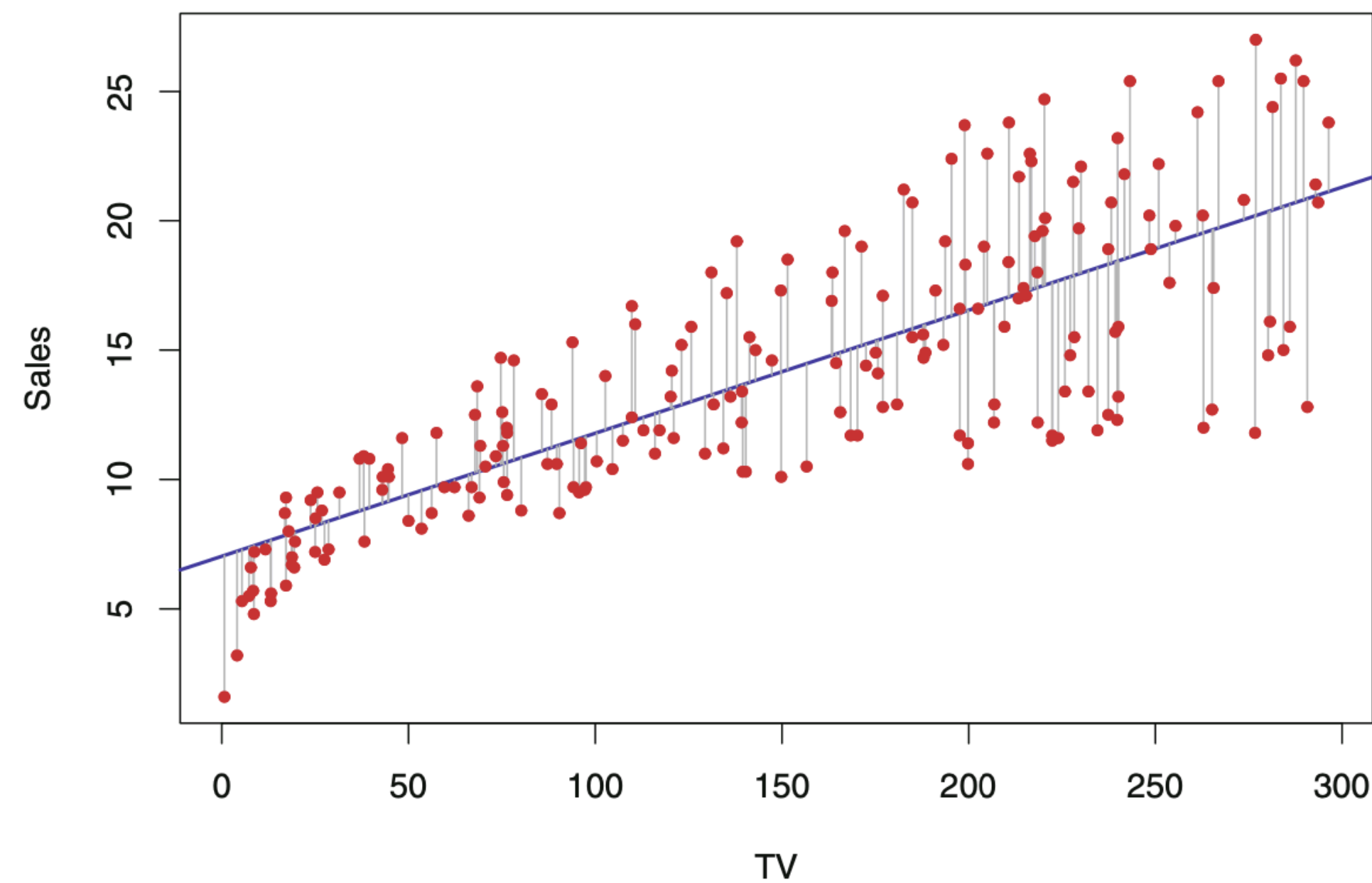
- **1** predictor variable (x)
- **1** response variable (y)

$$y = \beta_0 + \beta_1 x + \epsilon$$

Variable **y** is a linear function of **x**, plus some **noise**

Simple linear regression

- **1** predictor variable (x)
- **1** response variable (y)



Multiple linear regression

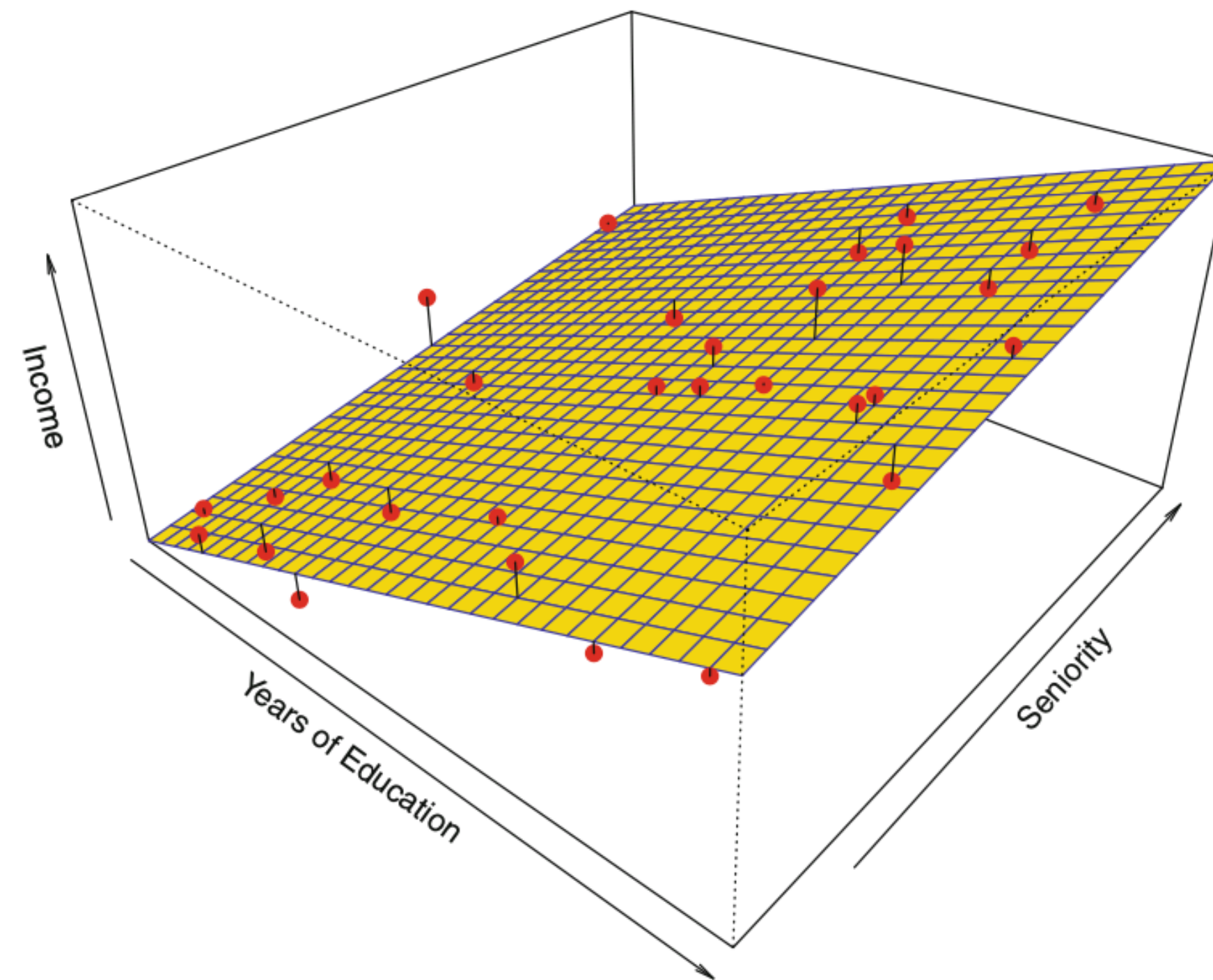
- **Several** predictor variables
- **1** response variable

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \epsilon$$

Variable **y** is a linear function of **x₁**, **x₂**, **x₃**, **etc.** plus some **noise**

Multiple linear regression

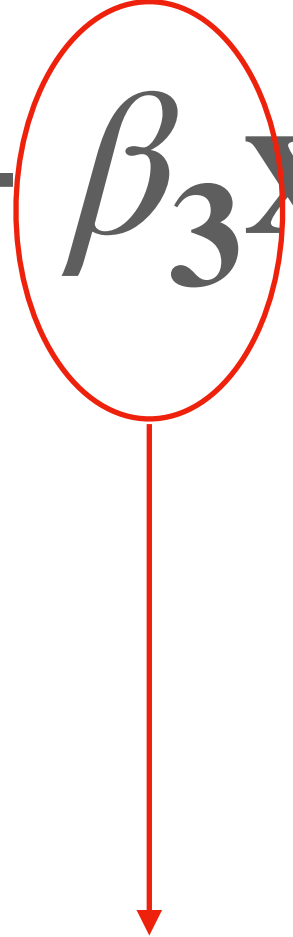
- **Several** predictor variables
- **1** response variable



Expanding the model: interaction terms

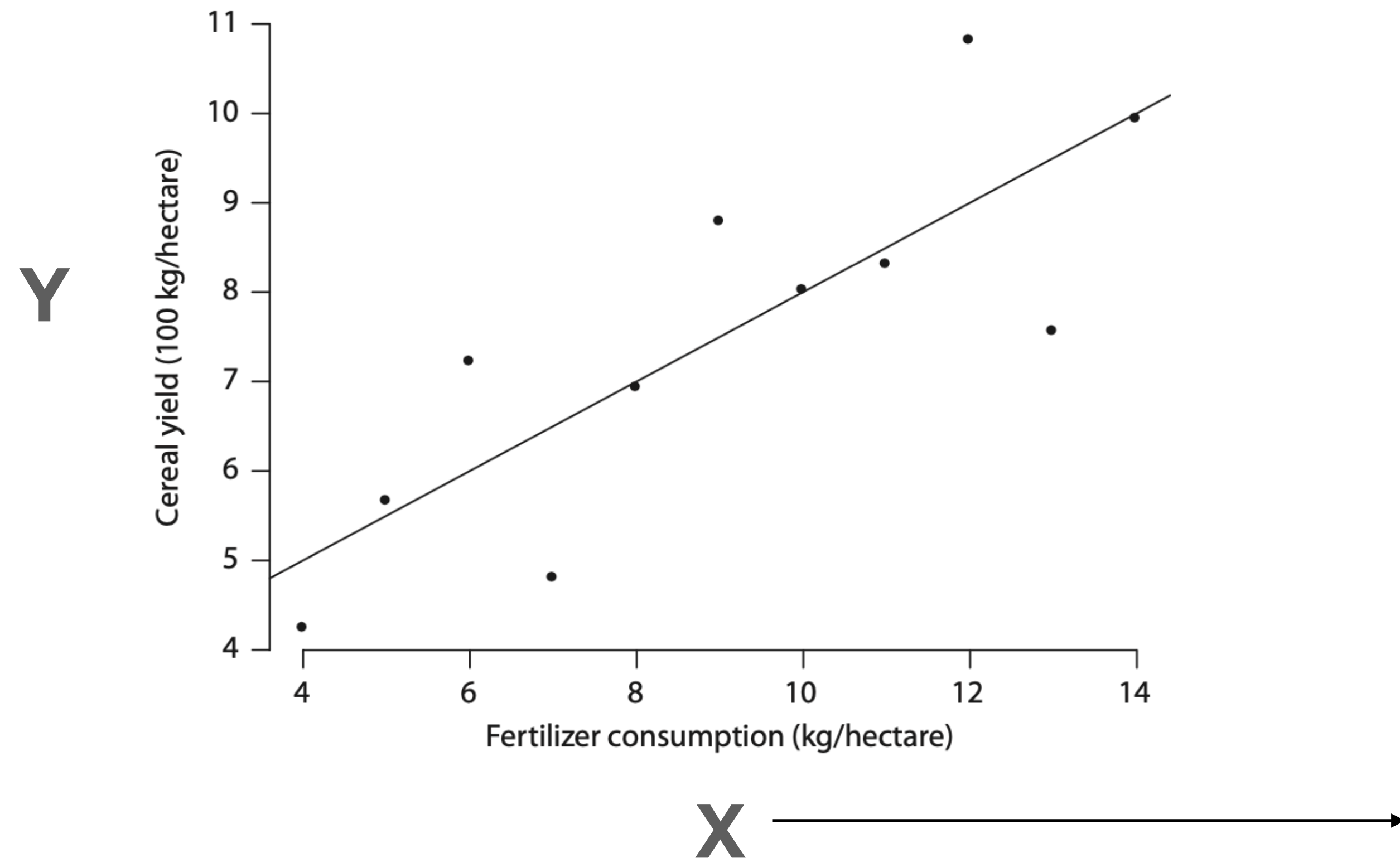
$$y = \beta_0 + \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 + \beta_3 \mathbf{x}_1 \mathbf{x}_2 + \dots + \epsilon$$

Expanding the model: interaction terms

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \dots + \epsilon$$


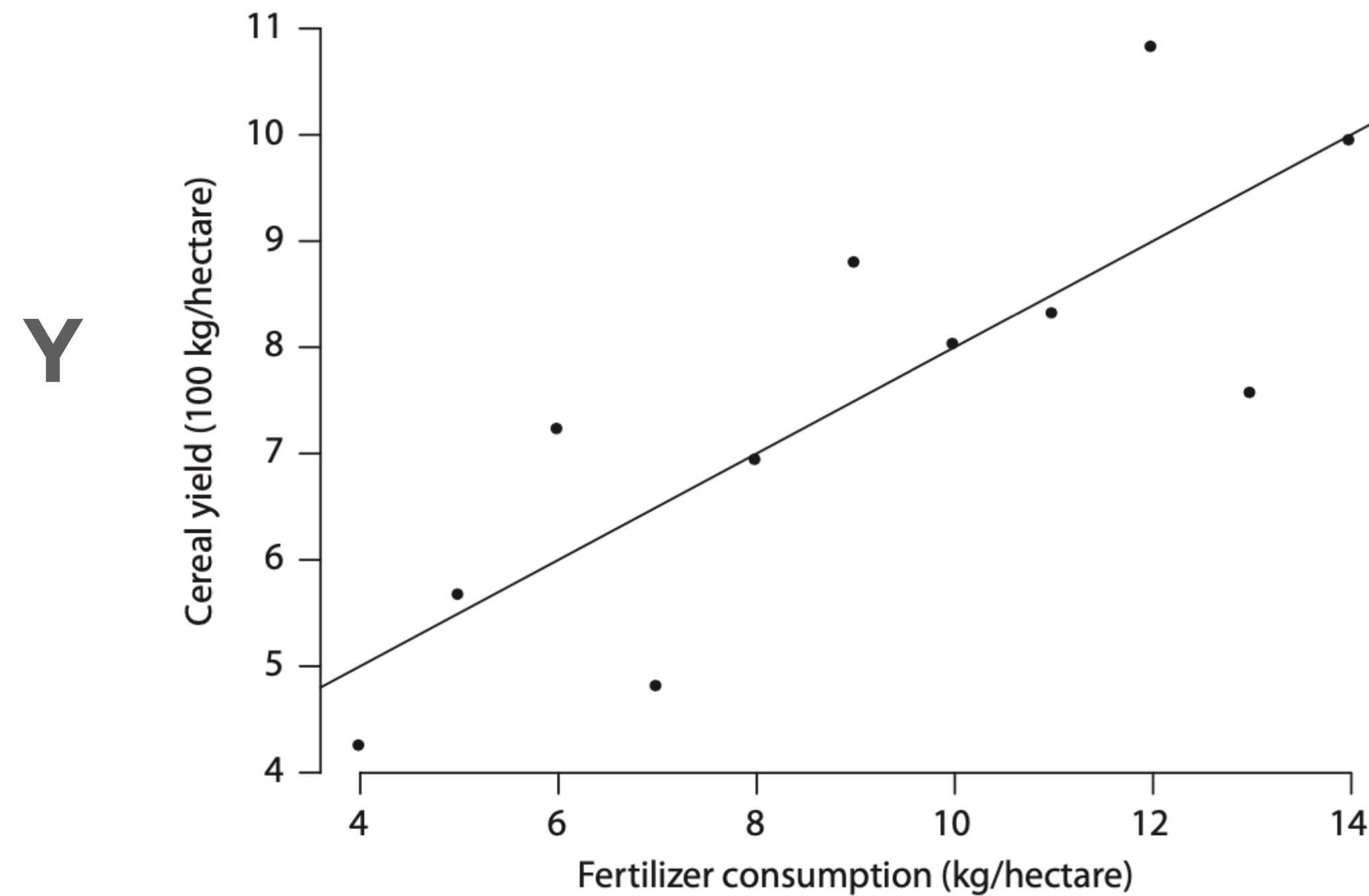
will be large when the combined behavior of x_1 and x_2 jointly influence the value of y

Multiple linear regression with categorical predictors (ANOVA)



So far, we've assumed that the value of **X** is a continuous number

Multiple linear regression with categorical predictors (ANOVA)



What if it was a **category** instead?

X

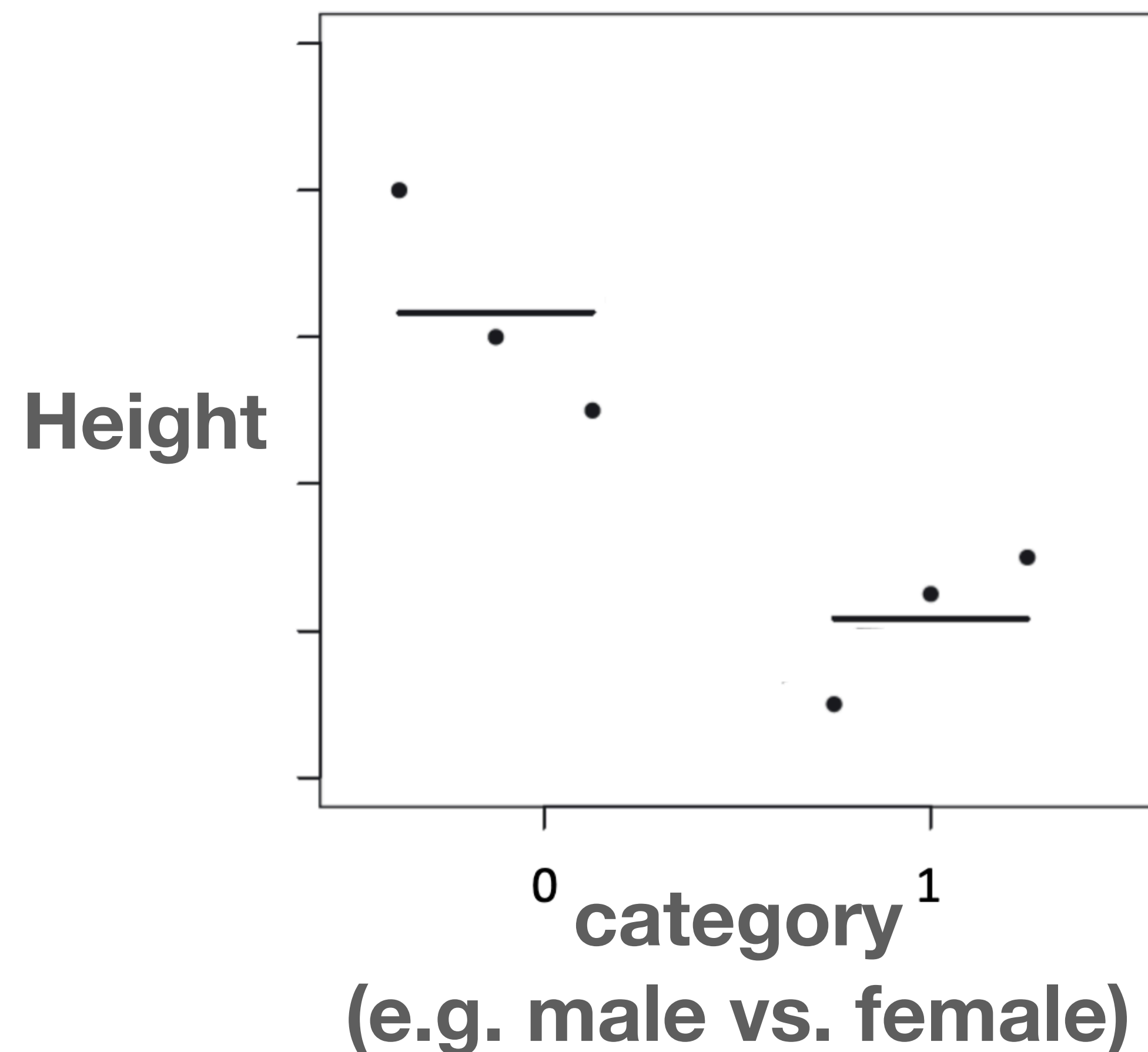


So far, we've assumed that the value of **X** is a continuous number

Multiple linear regression with categorical predictors (ANOVA)

$$y = \beta_0 + \beta_1 \mathbf{x} + \epsilon$$

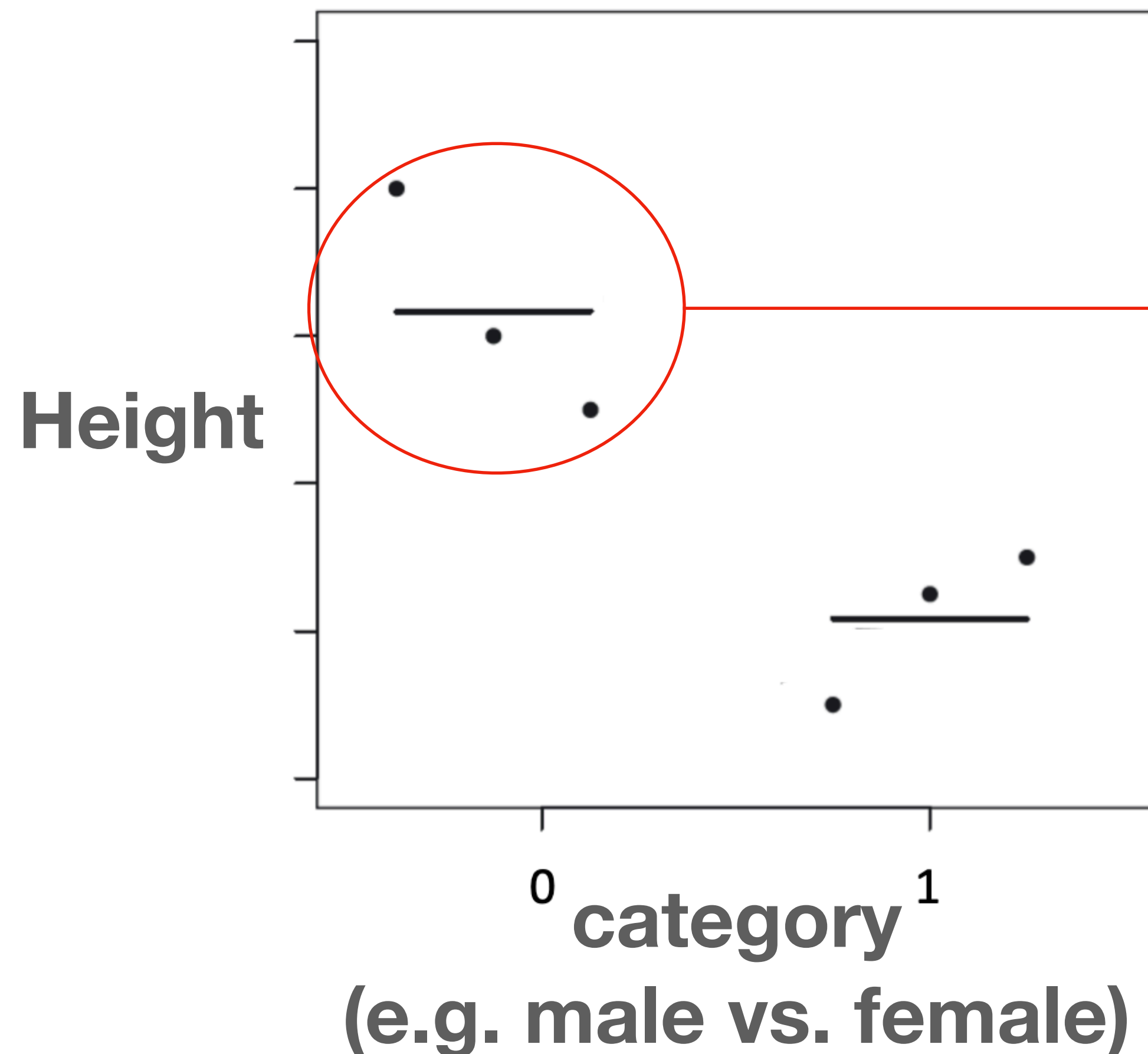
We can modify our original linear model such that $\mathbf{x}$ is now a **dummy variable**.



Multiple linear regression with categorical predictors (ANOVA)

$$y = \beta_0 + \beta_1 \mathbf{x} + \epsilon$$

We can modify our original linear model such that $\mathbf{x}$ is now a **dummy variable**.

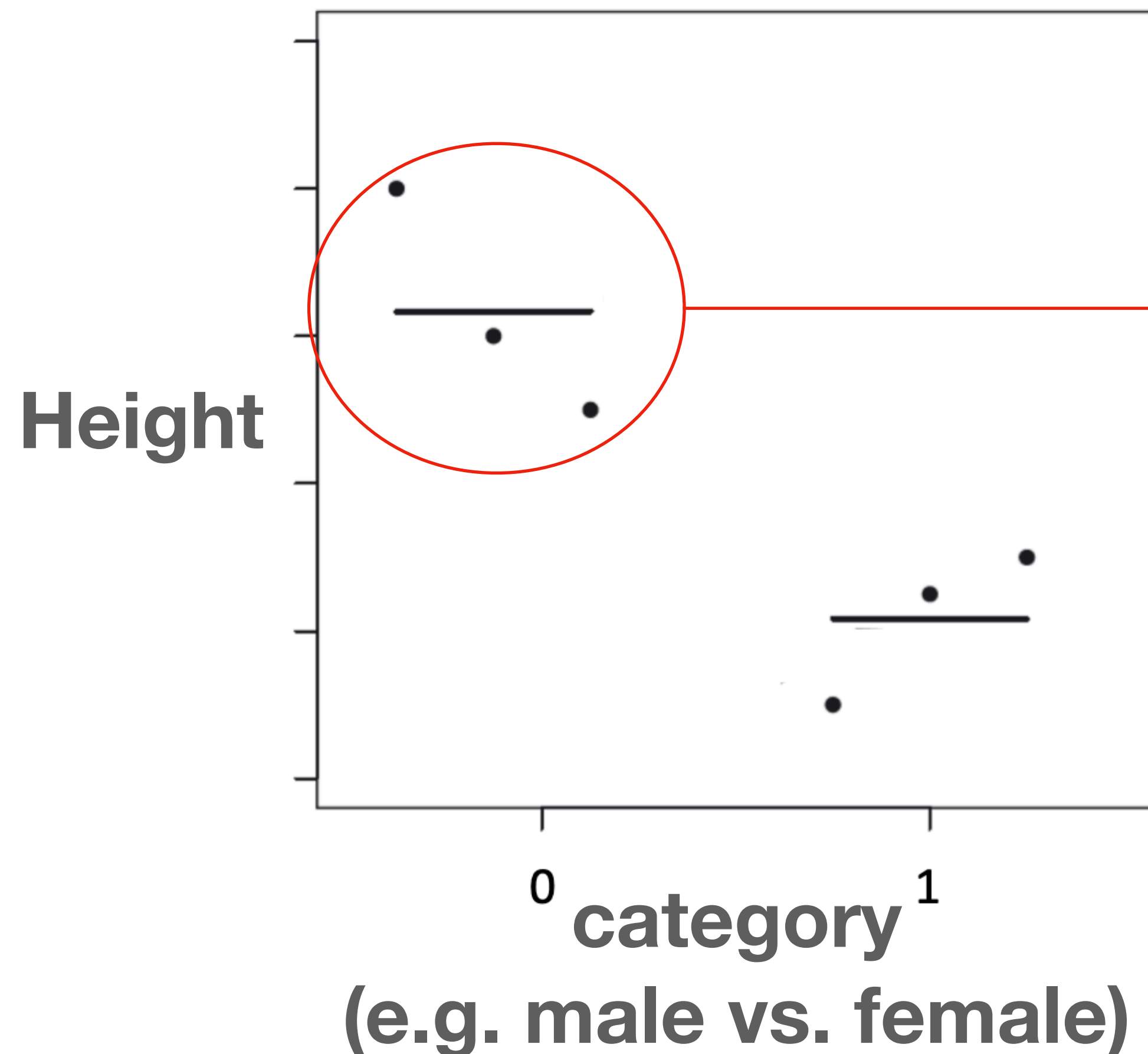


These observations get assigned $x = 0$

Multiple linear regression with categorical predictors (ANOVA)

$$y = \beta_0 + \beta_1 x + \epsilon$$

We can modify our original linear model such that x is now a **dummy variable**.



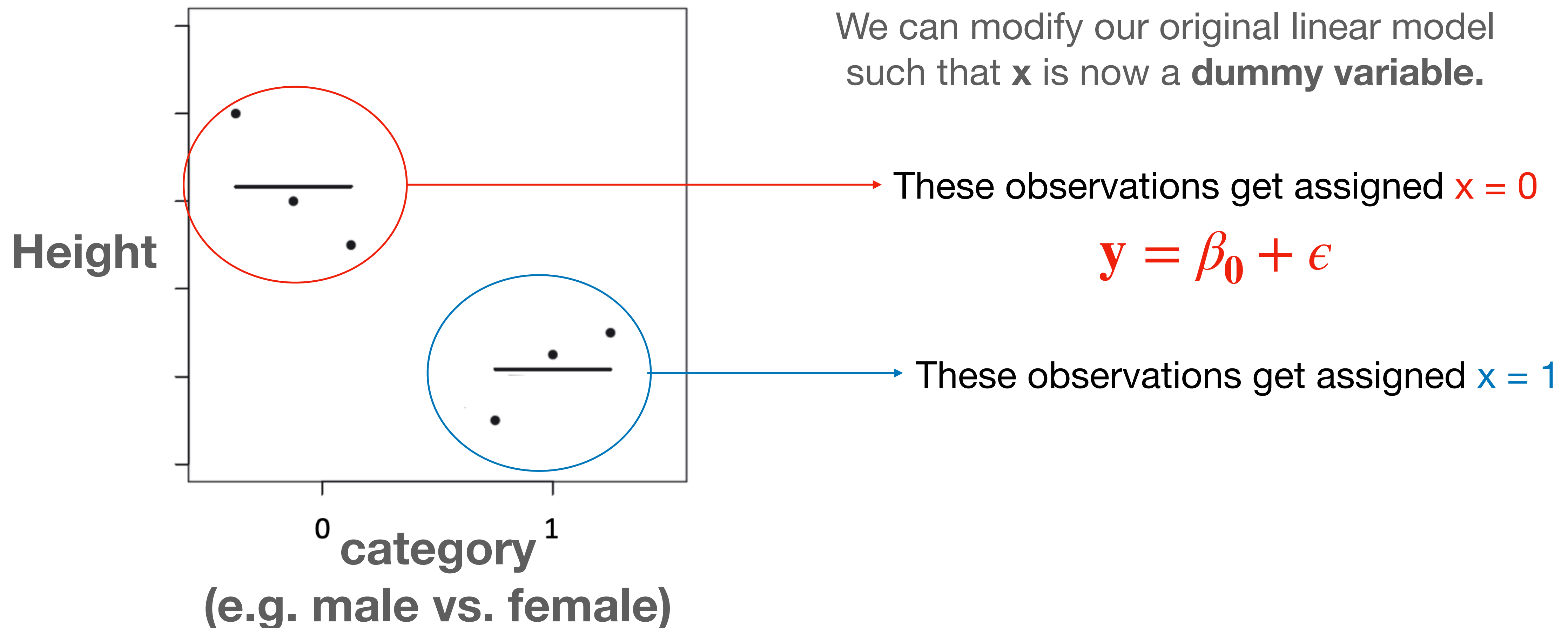
These observations get assigned $x = 0$

$$y = \beta_0 + \epsilon$$

Multiple linear regression with categorical predictors (ANOVA)

$$y = \beta_0 + \beta_1 x + \epsilon$$

We can modify our original linear model such that x is now a **dummy variable**.

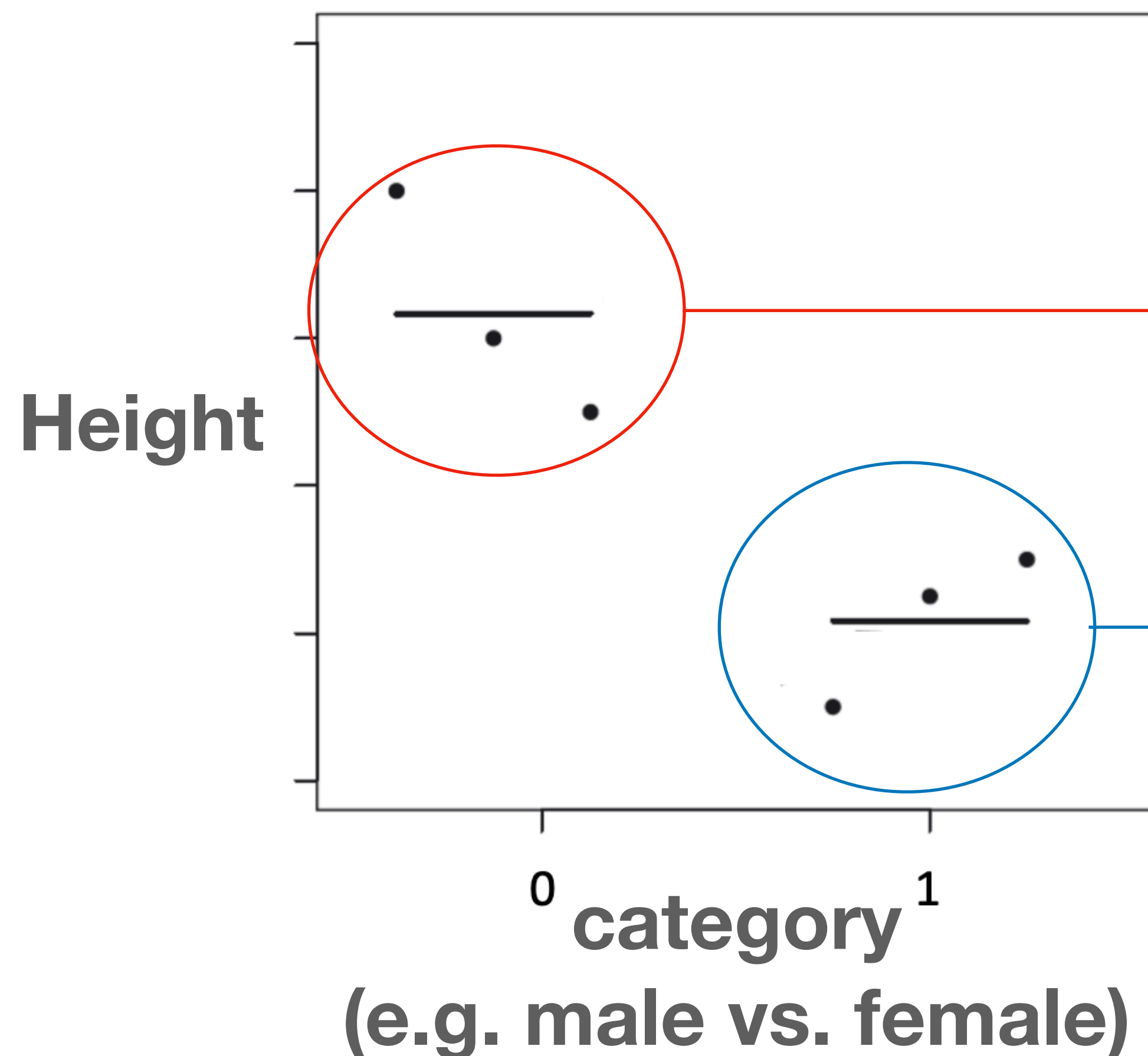


$$y = \beta_0 + \epsilon$$

Multiple linear regression with categorical predictors (ANOVA)

$$y = \beta_0 + \beta_1 x + \epsilon$$

We can modify our original linear model such that x is now a **dummy variable**.



These observations get assigned $x = 0$

$$y = \beta_0 + \epsilon$$

These observations get assigned $x = 1$

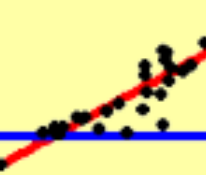
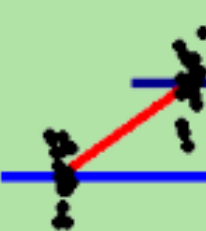
$$y = \beta_0 + \beta_1 + \epsilon$$

A little secret...

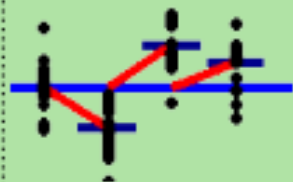
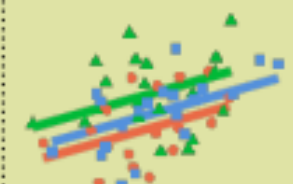
Common statistical tests are linear models

Last updated: 02 April, 2019

See worked examples and more details at the accompanying notebook: <https://lindeloev.github.io/tests-as-linear>

	Common name	Built-in function in R	Equivalent linear model in R	Exact?	The linear model in words	Icon
Simple regression: $\text{lm}(y \sim 1 + x)$	y is independent of x P: One-sample t-test N: Wilcoxon signed-rank	t.test(y) wilcox.test(y)	lm(y ~ 1) lm(signed_rank(y) ~ 1)	✓ for N > 14	One number (intercept, i.e., the mean) predicts y . - (Same, but it predicts the <i>signed rank</i> of y .)	
	P: Paired-sample t-test N: Wilcoxon matched pairs	t.test(y1, y2, paired=TRUE) wilcox.test(y1, y2, paired=TRUE)	lm(y2 - y1 ~ 1) lm(signed_rank(y2 - y1) ~ 1)	✓ for N > 14	One intercept predicts the pairwise y2-y1 differences. - (Same, but it predicts the <i>signed rank</i> of y2-y1 .)	
	y ~ continuous x P: Pearson correlation N: Spearman correlation	cor.test(x, y, method='Pearson') cor.test(x, y, method='Spearman')	lm(y ~ 1 + x) lm(rank(y) ~ 1 + rank(x))	✓ for N > 10	One intercept plus x multiplied by a number (slope) predicts y . - (Same, but with <i>ranked x</i> and y)	
	y ~ discrete x P: Two-sample t-test P: Welch's t-test N: Mann-Whitney U	t.test(y1, y2, var.equal=TRUE) t.test(y1, y2, var.equal=FALSE) wilcox.test(y1, y2)	lm(y ~ 1 + G2) ^A gls(y ~ 1 + G2, weights=... ^B) ^A lm(signed_rank(y) ~ 1 + G2) ^A	✓ ✓ for N > 11	An intercept for group 1 (plus a difference if group 2) predicts y . - (Same, but with one variance <i>per group</i> instead of one common.) - (Same, but it predicts the <i>signed rank</i> of y .)	

A little secret...

Multiple regression: $\text{lm}(y \sim 1 + x_1 + x_2 + \dots)$	P: One-way ANOVA N: Kruskal-Wallis	<code>aov(y ~ group)</code> <code>kruskal.test(y ~ group)</code>	$\text{lm}(y \sim 1 + G_2 + G_3 + \dots + G_N)^A$ $\text{lm}(\text{rank}(y) \sim 1 + G_2 + G_3 + \dots + G_N)^A$	✓ for N > 11	An intercept for group 1 (plus a difference if group $\neq 1$) predicts y . - (Same, but it predicts the <i>rank</i> of y .)	
	P: One-way ANCOVA	<code>aov(y ~ group + x)</code>	$\text{lm}(y \sim 1 + G_2 + G_3 + \dots + G_N + x)^A$	✓	- (Same, but plus a slope on x .) <i>Note: this is discrete AND continuous. ANCOVAs are ANOVAs with a continuous x.</i>	
	P: Two-way ANOVA	<code>aov(y ~ group * sex)</code>	$\text{lm}(y \sim 1 + G_2 + G_3 + \dots + G_N + S_2 + S_3 + \dots + S_K + G_2 * S_2 + G_3 * S_3 + \dots + G_N * S_K)$	✓	Interaction term: changing sex changes the y ~ group parameters. <i>Note: $G_{2 \text{ to } N}$ is an indicator (0 or 1) for each non-intercept levels of the group variable. Similarly for $S_{2 \text{ to } K}$ for sex. The first line (with G_i) is main effect of group, the second (with S_j) for sex and the third is the group * sex interaction. For two levels (e.g. male/female), line 2 would just be "S_2" and line 3 would be S_2 multiplied with each G_i.</i>	[Coming]
	Counts ~ discrete x N: Chi-square test	<code>chisq.test(groupXsex_table)</code>	Equivalent log-linear model $\text{glm}(y \sim 1 + G_2 + G_3 + \dots + G_N + S_2 + S_3 + \dots + S_K + G_2 * S_2 + G_3 * S_3 + \dots + G_N * S_K, \text{family}=\dots)^A$	✓	Interaction term: (Same as Two-way ANOVA.) <i>Note: Run glm using the following arguments: <code>glm(model, family=poisson())</code> As linear-model, the Chi-square test is $\log(y_i) = \log(N) + \log(\alpha_i) + \log(\beta_j) + \log(\alpha_i \beta_j)$ where α_i and β_j are proportions. See more info in the accompanying notebook.</i>	Same as Two-way ANOVA
	N: Goodness of fit	<code>chisq.test(y)</code>	$\text{glm}(y \sim 1 + G_2 + G_3 + \dots + G_N, \text{family}=\dots)^A$	✓	(Same as One-way ANOVA and see Chi-Square note.)	1W-ANOVA

Visit <https://lindeloev.github.io/tests-as-linear/> for more information

General multivariate regression

- **Several** predictor variables
- **Several** response variables

General multivariate regression

- **Several** predictor variables
- **Several** response variables

$$y_1 = \beta_0 + \beta_{1,1}x_1 + \beta_{2,1}x_2 + \beta_{3,1}x_3 + \dots + \epsilon_1$$

Variable **y₁** is a linear function of **x₁, x₂, x₃, etc.** plus some **noise**

General multivariate regression

- **Several** predictor variables
- **Several** response variables

$$y_1 = \beta_0 + \beta_{1,1}x_1 + \beta_{2,1}x_2 + \beta_{3,1}x_3 + \dots + \epsilon_1$$

Variable **y_1** is a linear function of **x_1 , x_2 , x_3 , etc.** plus some **noise**

$$y_2 = \beta_0 + \beta_{1,2}x_1 + \beta_{2,2}x_2 + \beta_{3,2}x_3 + \dots + \epsilon_2$$

Variable **y_2** is a linear function of **x_1 , x_2 , x_3 , etc.** plus some **noise**

General multivariate regression

- **Several** predictor variables
- **Several** response variables

$$y_1 = \beta_0 + \beta_{1,1}x_1 + \beta_{2,1}x_2 + \beta_{3,1}x_3 + \dots + \epsilon_1$$

Variable **y₁** is a linear function of **x₁, x₂, x₃, etc.** plus some **noise**

$$y_2 = \beta_0 + \beta_{1,2}x_1 + \beta_{2,2}x_2 + \beta_{3,2}x_3 + \dots + \epsilon_2$$

Variable **y₂** is a linear function of **x₁, x₂, x₃, etc.** plus some **noise**

$$y_3 = \beta_0 + \beta_{1,3}x_1 + \beta_{2,3}x_2 + \beta_{3,3}x_3 + \dots + \epsilon_3$$

Variable **y₃** is a linear function of **x₁, x₂, x₃, etc.** plus some **noise**

Generalized linear models

Generalized linear models

$$y = \beta_0 + \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 + \beta_3 \mathbf{x}_3 + \dots + \epsilon$$

Generalized linear models

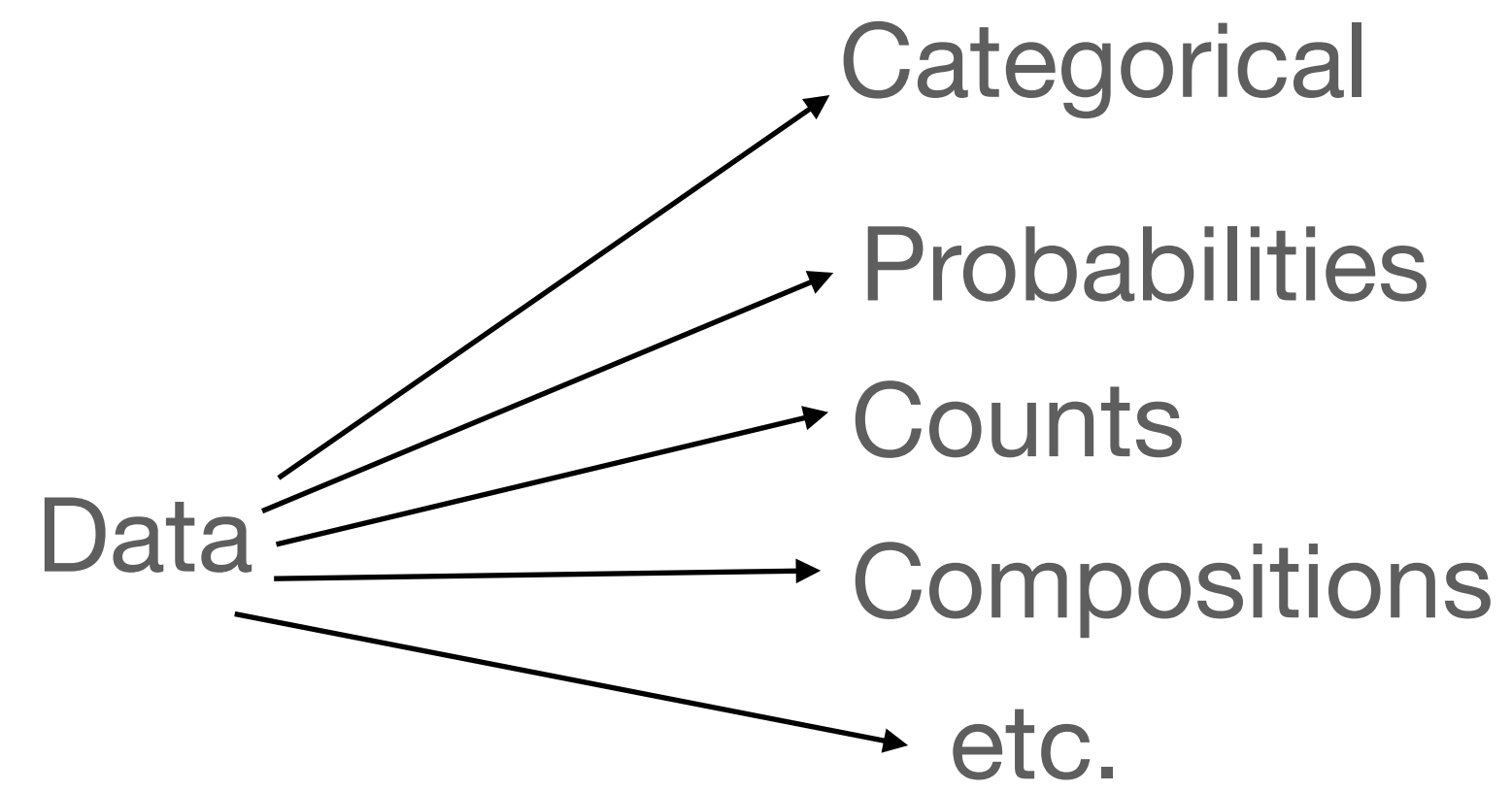
$$y = \beta_0 + \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 + \beta_3 \mathbf{x}_3 + \dots + \epsilon$$

y is continuous between -inf and inf

Generalized linear models

$$y = \beta_0 + \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 + \beta_3 \mathbf{x}_3 + \dots + \epsilon$$

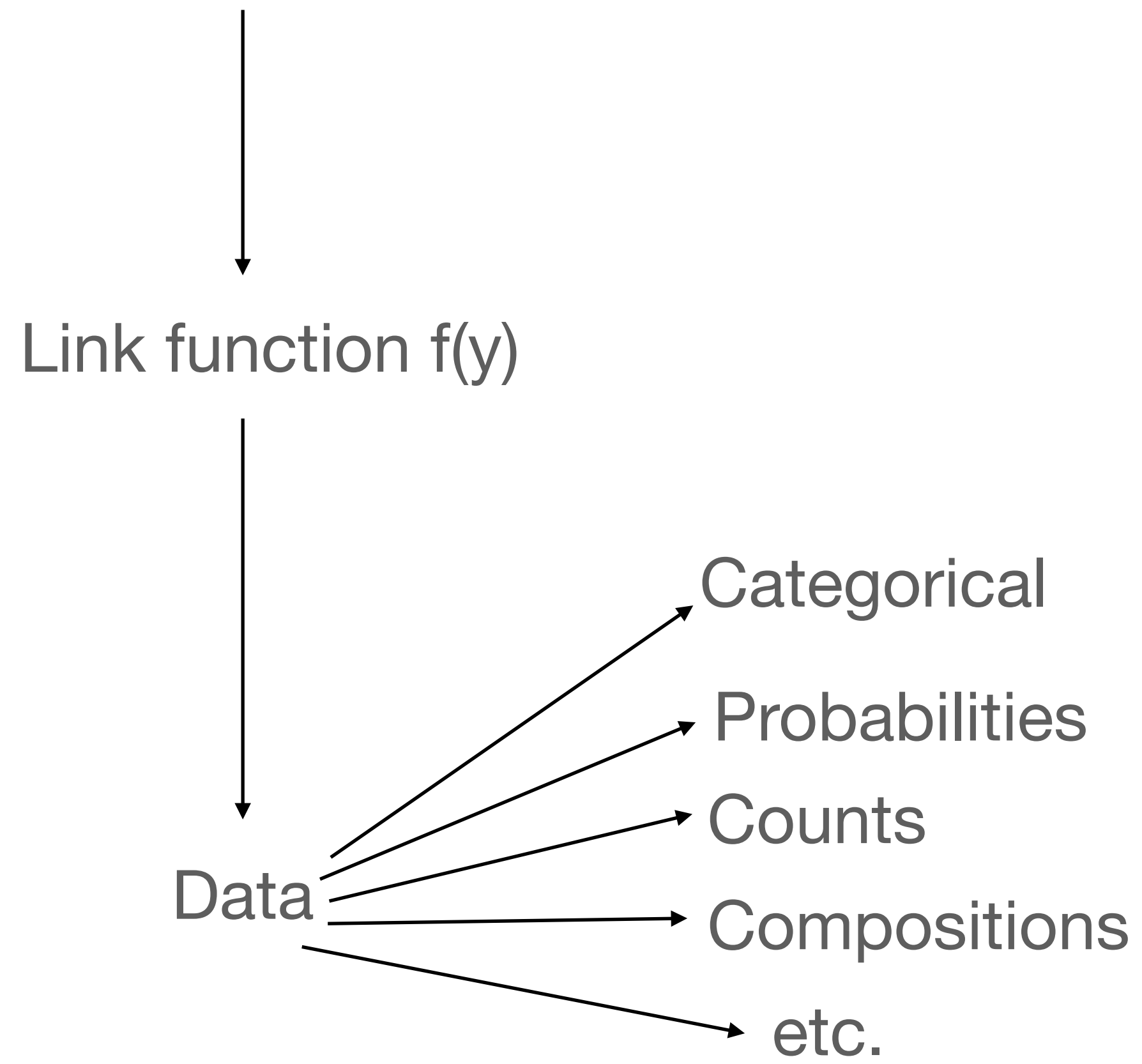
y is continuous between $-\infty$ and ∞



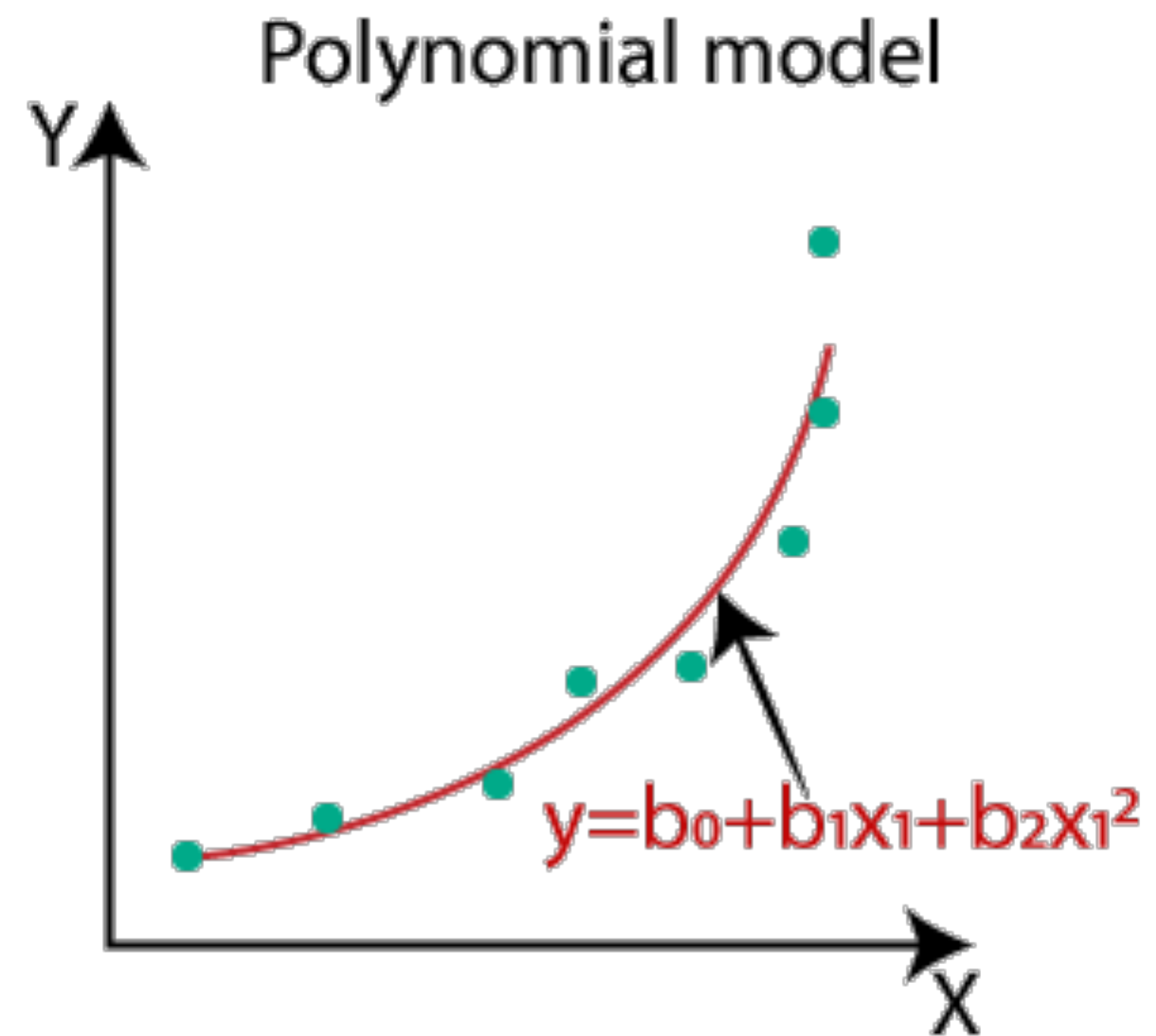
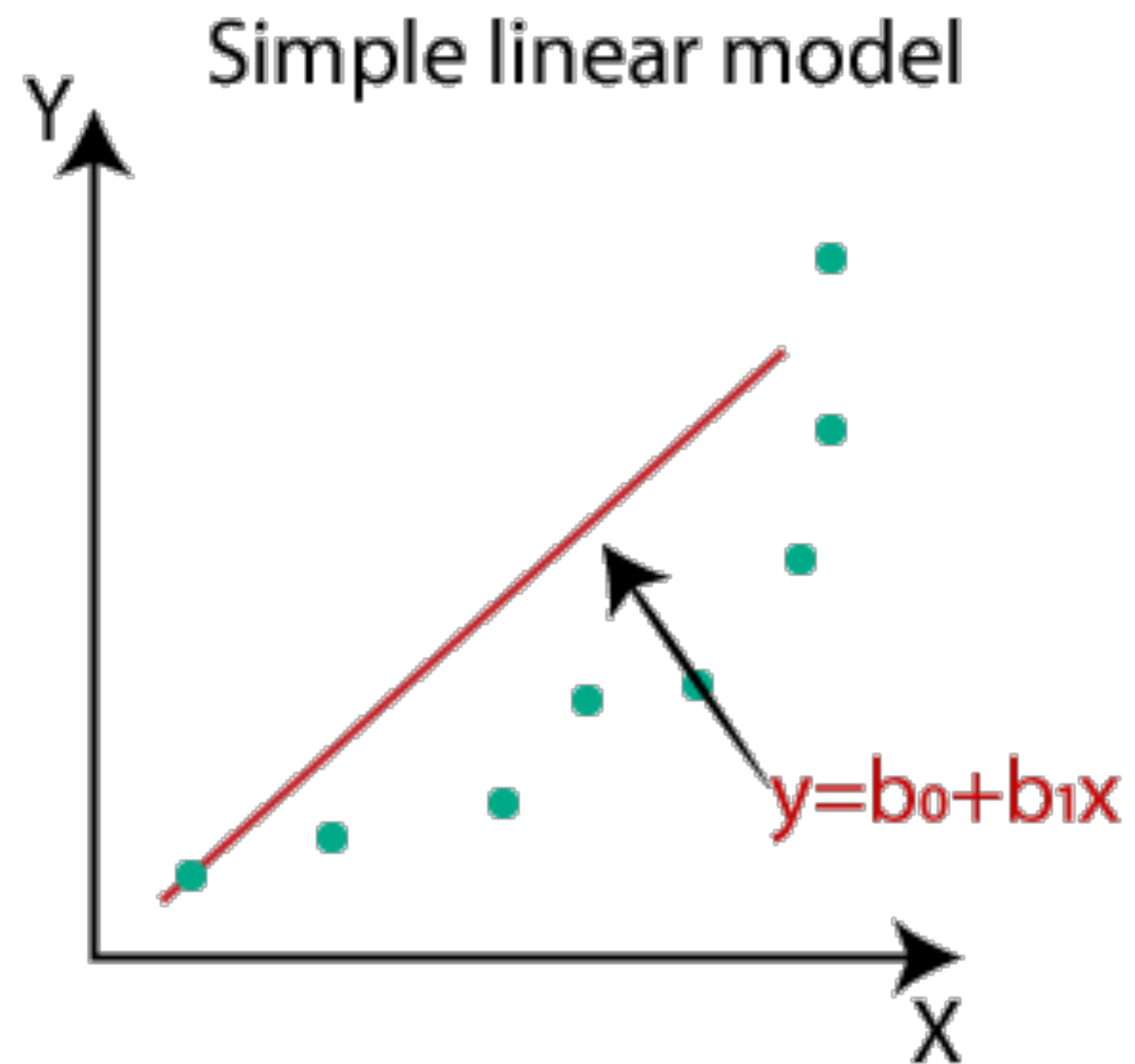
Generalized linear models

$$y = \beta_0 + \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 + \beta_3 \mathbf{x}_3 + \dots + \epsilon$$

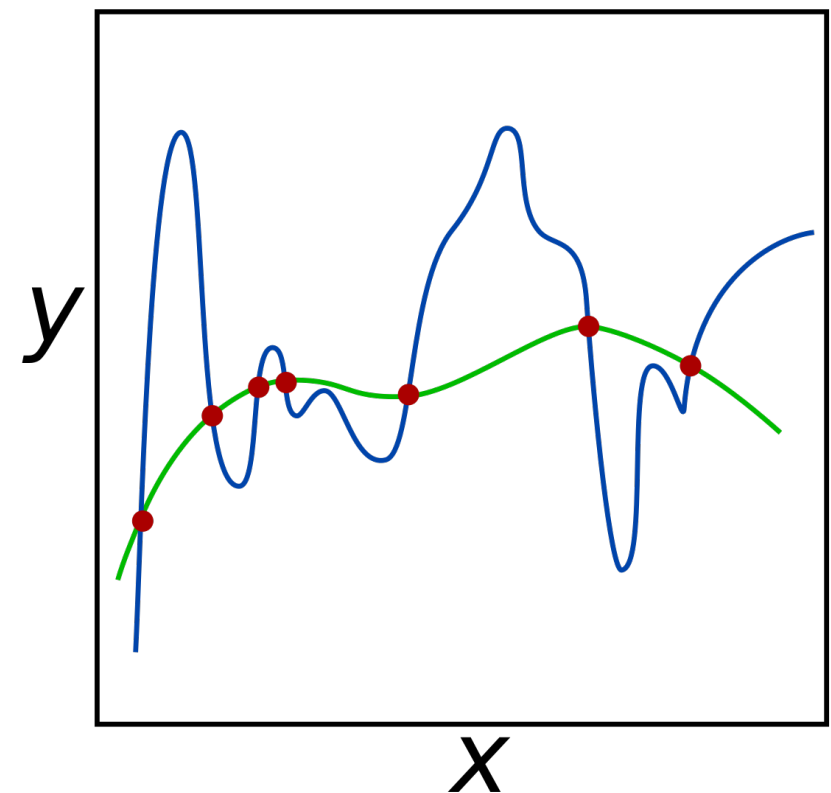
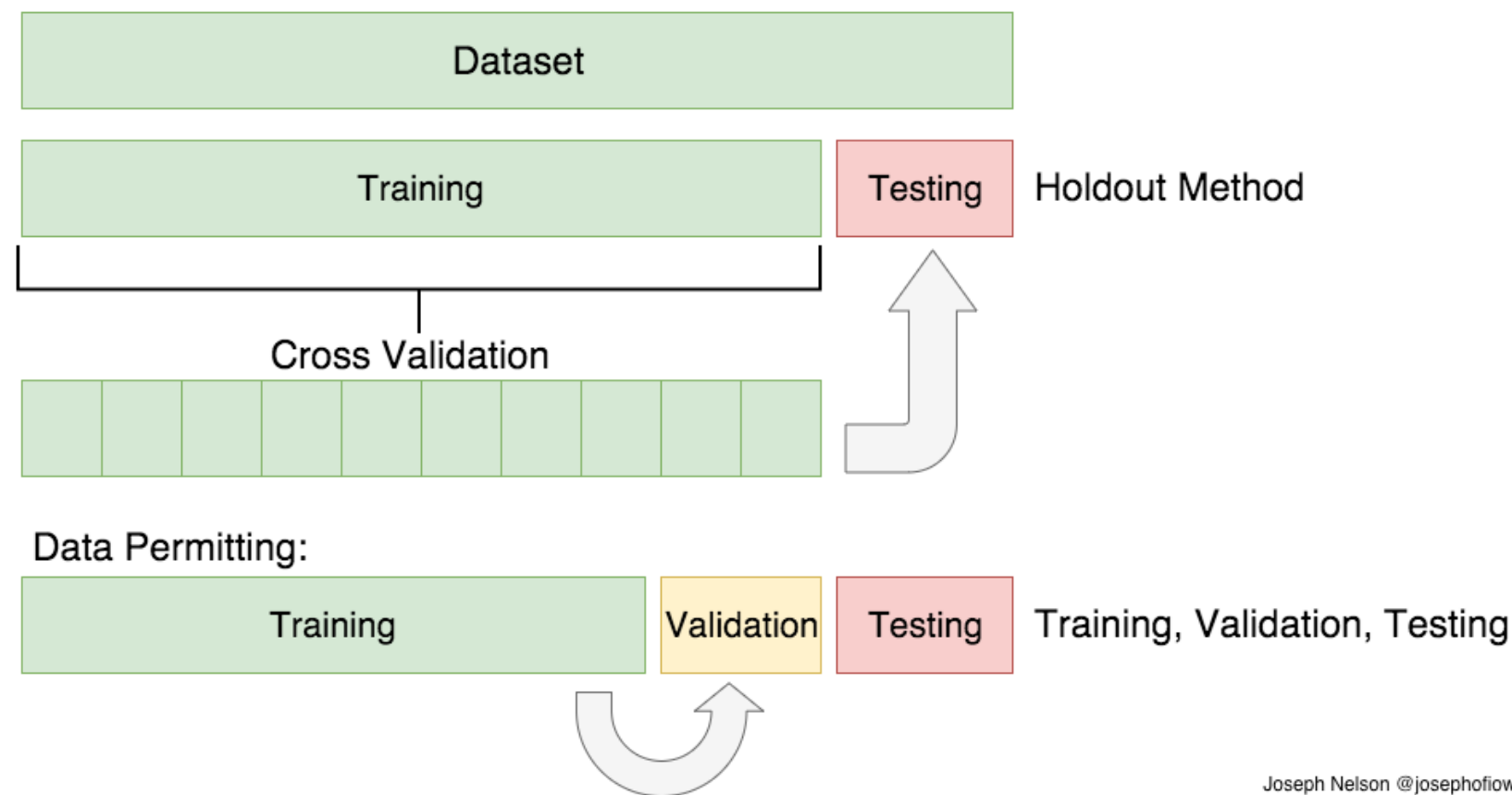
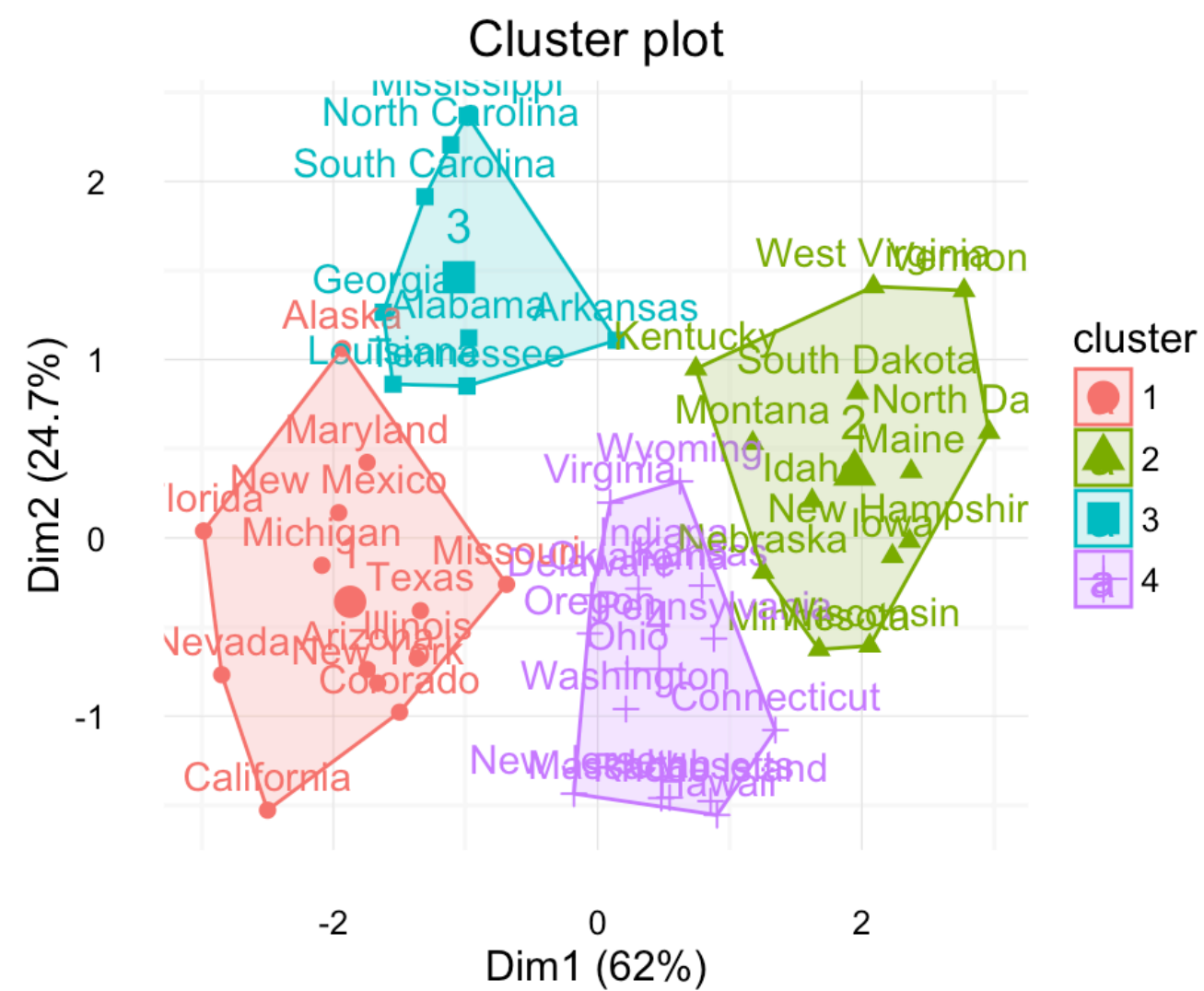
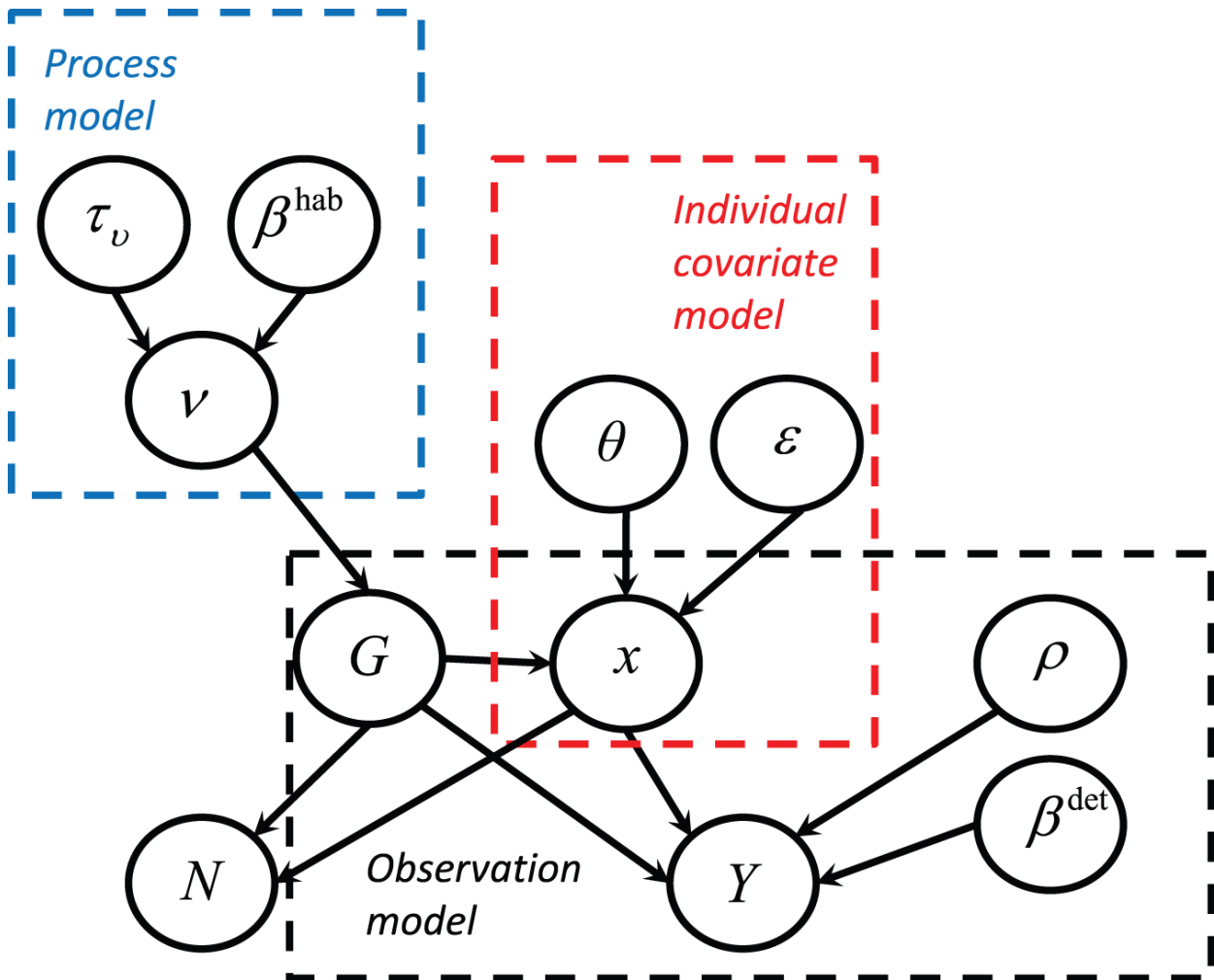
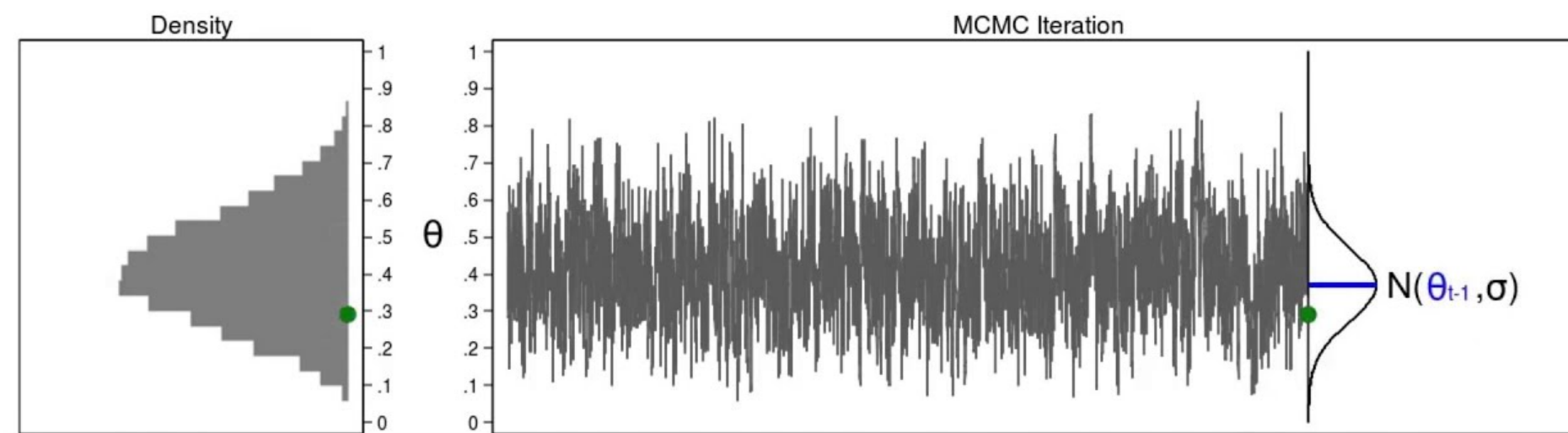
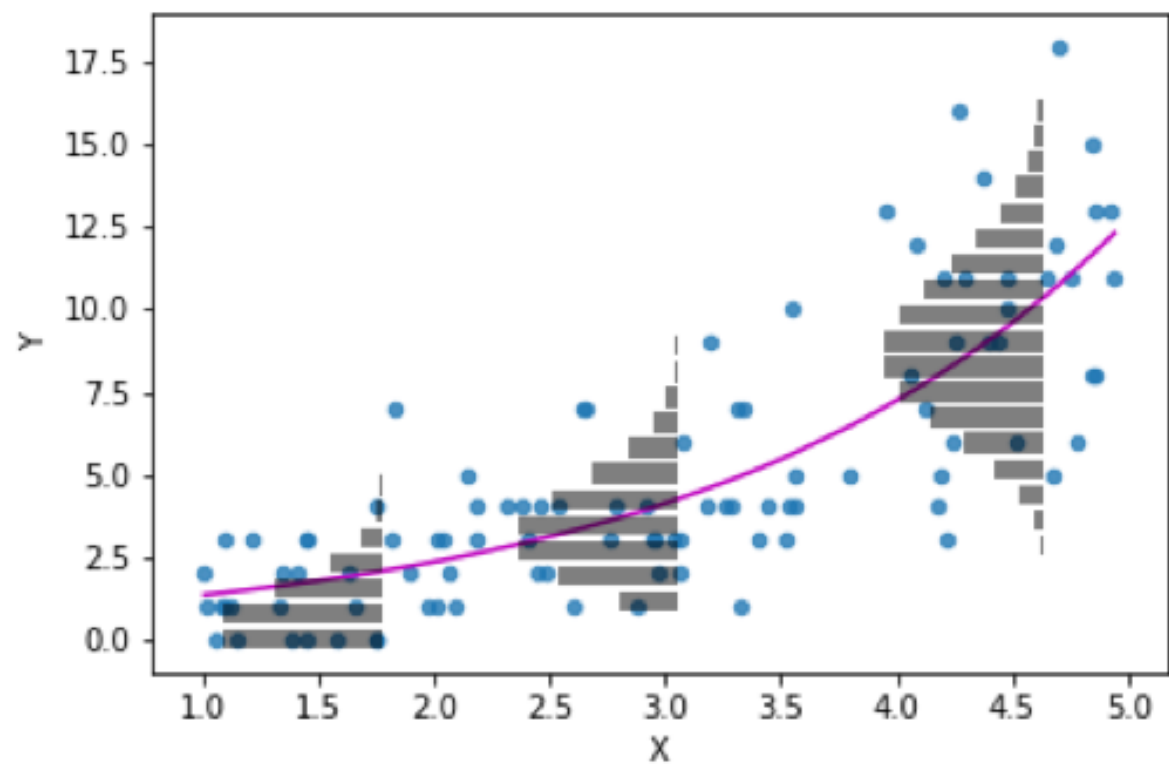
y is continuous between -inf and inf



Non-linear models



Advanced Topics in Data Analysis



- **Simulating a linear model in R**
- **Multivariate linear regression**

